



Cancer Detection Through Histopathology Images

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CERTIFICATE

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE
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Project Overview

Cancer is among the major causes of mortality in the world and its early diagnosis is vital in enhancing the survival of patients. Conventional diagnosis of cancer by use of histopathology images are physically examined by a pathologist, which is time-consuming, subjective, and highly relies on human knowledge. The complexity of tissue structures, staining variations and visual similarity between benign and malignant cells complicate proper diagnosis which in most cases lead to delayed or inconsistent diagnosis. These issues demonstrate the necessity of an effective, automated and efficient diagnostic system which is able to help medical practitioners identify tissues with cancerous tissues more precisely and faster. The title of the project is a Cancer Detection Through Histopathology Images This is an intelligent system that uses extensive artificial intelligence and image processing techniques to automatically identify the malignant cells from microscopic images. The algorithm constituted image filtering/feature detection/classification of normal vs abnormal tissue patterns using deep learning. The system is trained with a large number of histopathology images, which enables efficient detection of complex cell structures as well as prediction. A user-friendly interface allows healthcare providers to upload images, and receive a diagnostic clue without having to wait for the non-reproducibility in results to appear this would aid them make faster and more accurate assessments. The objective of this project includes image capture, preprocessing, model training, result visualization, and performance evaluation. This system reduces human errors, improves diagnostic accuracy, and minimizes errors greatly compared to the traditional manual methods of operating the system. The proposed approach harnesses new technology as compared to the current systems. It uses deep learning algorithms which show better performance in pattern recognition and classification procedures, which provide more credibility in medical diagnostics. The system helps to enhance the effectiveness of cancer detection and aid in the early diagnosis of cancer, which eventually helps healthcare professionals to make informed clinical decisions. Future improvements can encompass the extension of the model to identify various types of cancer, real time image analysis, and the implementation of the system in the information within a hospital networks to medical use on a larger scale.

Dedication

Firstly, we dedicate our project to the creator Allah Almighty and dedicate to whom the world owes its existence Muhammad (Peace Be Upon Him) and dedicate this to our beloved parents, our extremely dedicated and generous teachers and supportive friends, their prayers always pave the way to success for us.

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Chapter 1

Introduction

This project is an online cancer detection system in which users, mainly medical practitioners, can upload histopathology images of tissue samples for analysis. The system automatically preprocesses the uploaded images and applies deep learning algorithms to classify the tissue as benign or malignant. Once the analysis is complete, the results are displayed to the doctor, enabling them to make informed diagnostic decisions. Doctors can also track the history of uploaded images and view previous results for comparison or further investigation.

Pathologists and researchers registered on the system can access unknown image datasets for research purposes, improve model performance, and provide expert feedback on the accuracy of predictions. The software also provides facilities for producing reports on the results obtained after analysis that can then be saved or even printed out in patients' files. Users will receive notifications when the analysis is done.

An administrator of the system handles all the registered users who include doctors, pathologists, and researchers among others. Administrators have the capability of monitoring use of the software, managing image datasets, getting feedback regarding the model performance, and updating or retraining models. In the event of any wrong predictions made by the software, all such cases are logged in the software. The system also provides recommendations as to how to enhance image quality and annotations.

1.1 Product Purpose

In this project, there will be developed a software system aimed at simplification of the processes of data processing, report creation, and result checking. At present, companies face problems with manual procedures of work, inaccuracy of reporting, and inability to monitor processes. Such problems reduce productivity and affect decision making. This software system will provide a more effective means of solving such issues and provide users with accurate results, notifications, and analytics.

For implementing the project, there will be used certain approaches to the process of software engineering, including system analysis, system modular design, programming, debugging, testing, and integration. The major landmarks are the different stages of requirement gathering, developing modules such as report generation, result checking, notification module, logging activities module, feedback module and dashboard analysis.

Finally, after the completion of the entire process, the aim is for a software application that automates processes, makes work more accurate, and provides instant information to have been developed. The deadline for the completion of this project is expected to be July, 2026. Its effectiveness will depend on how efficient it is, how well the users like it, and the accuracy of the reports generated.

1.2 Product Scope

These are the three main groups that will directly interact with the system, i.e Admin, Doctor, and Patient. Keeping these stakeholders in mind, the scope of the project is to provide a web-based system for cancer detection using histopathology images.

Table 1.1: Abbreviation

Serial #	Abbreviation	Definition
1	CDHI	Cancer Detection through Histopathology Images
2	CNN	Convolutional Neural Network
3	DB	Database
4	AI	Artificial Intelligence
5	FL	Federated Learning
6	IDE	Integrated Development Environment
7	SDLC	Software Development Life-Cycle
8	SRS	Software Requirement Specification

Doctors can upload histopathology slides and view AI-generated diagnostic results. Patients can register, view their history, and access their reports generated by the doctor. The Admin has the authority to manage users, doctors and patients, and overall system activities including feedbacks, audit logs. The system aims to assist doctors in accurate and efficient cancer diagnosis. The abbreviations used in this document are listed in Table 1.1.

1.2.1 Existing System Description

PathAI [1]: is an artificial intelligence-based pathology platform that provides assistance to pathologists in analyzing histopathology images for cancer detection. The system uses deep learning algorithms to examine slides of tissue samples and classify them as benign or malignant. It improves diagnostic accuracy by reducing manual errors and providing automated assistance during analysis. The platform also supports continuous learning by training models on annotated pathology datasets. PathAI can be accessed through cloud-based services, allowing remote accessibility and collaboration among healthcare professionals. It is widely used in digital pathology research and clinical workflows to enhance efficiency and consistency in cancer diagnosis.

Key Features:

- AI-based analysis of histopathology images.
- Classification of tissue samples as benign and malignant categories.
- Reducing errors caused by manual pathology assessment.
- Continuous learning using labelled datasets.

- Cloud accessible platform and remote collaboration support.

Limitations:

- Requirement of centralized storage of medical data.
- Compromising patient privacy and security concerns.
- Highly dependent on cloud infrastructure.
- Expensive deployment and maintenance costs.
- Requires high computational resources.

Paige.AI [2]: is an FDA-approved artificial intelligence software used in digital pathology for cancer detection. The platform specializes in diagnosing cancers for example; prostate and breast cancer from histopathology images. It highlights doubtful regions in tissue slides to assist pathologists in identifying abnormalities efficiently, and precisely. The system integrates smoothly into existing laboratory workflows and supports both cloud and on-premise deployment options. Paige.AI improves diagnostic efficiency and helps reducing the workload of medical professionals through automation.

Key Features:

- AI pathology platform is approved by FDA.
- Prostate and breast cancers are detected accurately.
- Highlights suspicious tissue regions automatically.
- Integrates with laboratory workflows.
- Supports both cloud and on-premise deployment.

Limitations:

- Limited to specific cancer types.
- Requires histopathology slides.
- High costs of licensing and infrastructure.
- Advanced imaging equipment is needed.
- For different scanners and stains it requires calibration.

IBM Watson Health (Oncology) [3]: is an AI-powered healthcare platform developed to assist oncologists in detecting the cancers and planning how to treat it. The system analyzes histopathology slides, patient records, clinical notes, and medical literature to provide evidence-based recommendations. It supports healthcare professionals

in making informed decisions and improves collaboration among medical teams through system integration. The platform is integrated with hospital management systems and electronic health records for better clinical support. It is recognized widely for combining medical data analysis with AI-driven decision support.

Key Features:

- AI-assisted cancer diagnosis and treatment planning.
- Analyzes medical images and patient records together.
- Provides evidence-based recommendations.
- It supports integration with healthcare systems.
- Enhanced collaborations among medical professionals.

Limitations:

- For various cancer types accuracy differs.
- Centralized storage systems are used.
- Implementation and integration process is expensive.
- AI decision-making lacks complete transparency.
- It depends heavily on well-structured medical data.

Aiforia [4]: is a cloud-based digital pathology platform that assists researchers and pathologists in analyzing whole-slide images using artificial intelligence. The system provides AI-assisted image annotation, labeling, and visualization tools for pathology research and medical studies. Aiforia mainly focuses on research applications and supports collaborative pathology analysis through cloud services. It helps improve image processing and data organization for pathology workflows. Commonly used in academic and research environments for studying tissue samples and disease patterns as the images are labelled and annotated. Though it has very limited cancer detection capabilities and not efficient enough to generate complete reports.

Key Features:

- AI-assisted image annotation and labeling.
- Supports whole-slide image analysis.
- Cloud-based accessibility for researchers.
- Useful for pathology research applications.
- Provides image visualization and management tools.

Limitations:

- It has very limited cancer detection capabilities.
- Generates incomplete diagnosis reports.
- Completely dependent on cloud services.
- Requires very stable internet connection.
- Not much suitable for clinical diagnostic use.

PathPresenter [5]: is an online pathology image-sharing platform that allows users to upload, view, and discuss pathology slide images through the internet. The system is majorly made for educational, research, and collaborative purposes among oncologists and medical students. Supporting online case discussions and enables users to share online pathology data easily. PathPresenter provides web-based accessibility, making it useful for remote learning and pathology consultations. But, it doesn't provide AI-based cancer detection or automated diagnosis functionality.

Key Features:

- Online sharing of histopathology slides.
- Supports online case discussions and collaboration.
- Very useful for educational and medical purposes.
- Seamless access to the platform through web interface.
- Enables pathologist to upload and view pathology cases.

Limitations:

- It lacks AI-based cancer detection.
- Doesn't have automated image analysis features.
- Diagnostic reports can not be generated.
- Too limited privacy and security mechanisms.
- Falls short for the requirement of complete clinical diagnosis workflows.

Proscia [6]: is an AI-driven digital pathology platform developed to assist laboratories in managing and analyzing whole-slide pathology images. The system uses AI algorithms for cancer detection and workflow automation in pathology laboratories. It provides data management, collaboration tools, and integration with laboratory information systems to improve operational efficiency. Proscia supports cloud-based pathology services and enables pathologists to work collaboratively from different locations.

The platform applies advanced deep learning models to identify complex patterns in tissue samples and assists pathologists in making faster and more accurate diagnostic decisions. It also focuses on improving reproducibility in pathology analysis by significantly reducing the manual workload and routine diagnostic workflows. Proscia's solutions are designed to support scalable digital pathology adoption across healthcare organizations.

Key Features:

- AI-driven powered pathology image analysis.
- Automation of workflow for laboratories.
- Enables pathologists to collaborate.
- Seamless integration with laboratory information system networks.
- Scalable cloud-based digital management platform.

Limitations:

- High costs of licensing and infrastructure.
- Requires fully digitized histopathology slides.
- Difficult integration for small laboratories.
- Needs advanced hardware resources.
- Dependence on cloud-based systems.

DeepBio [7]: is an AI-based medical imaging system that develops deep learning algorithms for cancer diagnosis using histopathology images. The platform mainly focuses on detecting diseases such as prostate cancer through automated image analysis techniques. It helps pathologists improve diagnostic accuracy and reduce workload by automating repetitive analysis tasks. DeepBio uses high-resolution digital pathology images to train AI models for identifying cancer patterns. It is mainly designed for specific cancer detection applications in digital pathology. It is done automatically to reduce workload of pathologists. But the drawback is that it is limited to specific cancer types.

Key Features:

- Deep learning-based cancer diagnosis system.
- Automated histopathology image analysis.
- Reduces workload for pathologists.
- Improves diagnostic accuracy and speed.
- Supports AI-driven medical decision-making.

Limitations:

- Limited to specific cancer types.
- Requires high-resolution digital images.
- Difficult to generalize across datasets.
- Needs large training datasets.
- High computational requirements.

Google DeepMind Health [8]: is an artificial intelligence-based healthcare system developed to support medical professionals in disease diagnosis and medical image analysis. The platform uses advanced deep learning algorithms to analyze pathology and radiology slides for detecting abnormalities and assisting in cancer detection. It helps pathologists to improve the diagnostic accuracy by identifying patterns in tissue samples that may not be easily visible through manual observations. The system also supports integration with hospital workflows and e-medical records for efficient clinical management. DeepMind Health aims to enhance healthcare decision-making through AI-driven automation and predictive analysis. The platform is widely recognized for applying machine learning methods in medical research and diagnostics.

Key Features:

- AI-driven medical image analysis system.
- It supports cancer detection and diagnosis.
- Uses deep learning algorithms for pattern recognition.
- Integrates with healthcare management systems.
- Enhances diagnostic speed and accuracy.

Limitations:

- Large amount of training data is required.
- Inevitably raises concerns regarding patient data privacy protections.
- High costs of implementation and computational.
- Completely depends on advanced digital infrastructure.
- Less transparency in AI decision-making steps.

Tempus [9]: is an advanced AI-enabled precision medical platform that assists healthcare professionals in analyzing clinical and molecular data for cancer detection and personalized treatment planning. The system combines histopathology slides, genomic

data, and patient records to provide personalized insights for pathologists. Tempus helps in improving treatment strategies by identifying patterns of the disease and recommending specific therapies using machine learning models. It also supports data sharing and integration with healthcare systems for collaborative clinical workflows. The platform is mainly used in oncology research and precision medicine applications. Tempus aims to improve patient outcomes through data-driven healthcare solutions and AI-assisted analysis.

Key Features:

- Combines histopathology and genetic data analysis.
- It supports personalized cancer treatment planning strategies.
- Advanced AI-assisted clinical decision support systems.
- Seamless integration with healthcare systems and e-records.
- Useful for oncology research and precision medicine applications.

Limitations:

- High costs of setup and operational investments.
- Large and high-quality medical datasets are required.
- Depends on well-structured patient records.
- Accessibility is limited for small healthcare centers.
- Severe data privacy and security concern risks.

Indica Labs [10]: is a digital histopathology and image analysis platform that provides AI-powered tools for analyzing pathology slides and managing pathology workflows. The system actively assists pathologists and researchers in detecting tissue abnormalities, measuring biomarkers, and performing quantitative image analysis workflows. The system comprehensively supports whole-slide imaging and offers proper solutions for research laboratories and healthcare institute environments. Indica Labs seamlessly integrates with lab information systems and enables efficient remote collaboration through advanced pathology management tools. This platform is widely used in cancers research and diagnostic applications to improve the accuracy and workflow efficiency. It helps automate image analysis tasks and reduces manual workload from pathology professionals.

Key Features:

- Advanced AI-powered histopathology image analysis capabilities.
- It supports robust biomarker detection and quantification.
- Whole-slide image management and analysis.

- Integration with laboratory information systems.
- Improves workflow efficiency and automation.

Limitations:

- Requires advanced imaging infrastructure.
- High software and licensing costs.
- Needs trained professionals for operation.
- Dependent on digitized pathology slides.
- Limited usability in low-resource laboratories.

The following Table 1.2 presents a comprehensive comparison of the 10 existing systems based on the key features. These features include login, image upload, automatic image pre-processing, detection, federated learning support, privacy preservation, local model training, performance evaluation, report generation, visualization of detected cancer regions, feedback systems, and multi-user access. The table highlights the strengths and limitations of each application by indicating the presence or absence of these functionalities, allowing for a clear evaluation and comparison with the proposed Histo-Scan-AI system.

Table 1.2: Applications Comparison

Features	Applications										
	PathAI [1]	Paige.AI [2]	IBM Watson Health [3]	Aiforia [4]	PathPresenter [5]	Proscia [6]	DeepBio [7]	Google DeepMind Health [8]	Tempus [9]	Indica Labs [10]	Histo-Scan-AI
Login	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Upload Histopathology Image	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Automatic Image Preprocessing	×	✓	✓	✓	×	✓	✓	✓	✓	✓	✓
Cancer Cell Detection	✓	✓	✓	✓	×	✓	✓	✓	✓	✓	✓
Federated Learning Support	×	×	×	×	×	×	×	×	×	×	✓
Data Privacy Preservation	×	×	✓	✓	×	×	×	×	✓	×	✓
Local Model Training	×	×	×	×	×	×	×	×	×	×	✓
Performance Evaluation	✓	×	×	×	×	✓	✓	✓	✓	✓	✓
Download / Print Diagnostic Report	×	×	✓	×	×	×	×	×	✓	✓	✓
Visualization of Detected Cancer Regions	×	×	✓	✓	×	✓	✓	✓	✓	✓	✓
Doctor / Pathologist Feedback System	×	×	✓	×	✓	×	×	×	✓	×	✓
Multi-user Access	×	×	✓	×	✓	✓	×	✓	✓	✓	✓

Note: The comparative data was collected from the official websites and available applications of each service. Information was gathered between September 2025 and May 2026 through feature analysis of publicly available service descriptions, pricing models, and interfaces. Each entry ($\checkmark/\times$) indicates the presence or absence of a feature as described by the provider at the time of access. Source links and access dates are listed in the references.

1.2.2 Future System Usage Analysis

In the future, such a method can be widely applied to all hospitals and laboratories where pathologists detect diseases. With the help of automation technologies, doctors can work more efficiently and reliably by using such a system and not wasting time on other tasks. The process becomes shorter and well-organized compared to the previous methods.

Any specialist can send their histopathology pictures from anywhere, and thus, the process of working with this software becomes easier and more flexible. It can be convenient for doctors who cannot travel to some clinics and work remotely. Moreover, patients will be able to undergo treatment immediately after visiting a clinic and getting a result from this service.

Due to the automatic analysis and classification, there will be fewer human errors in the process of diagnosing. This system can detect tumors at the initial stage because of the objectivity and consistency of the analysis. Thus, patients have more chances to receive necessary treatment at a proper time.

This platform uses artificial intelligence, which is why it is constantly learning and growing its skills depending on the number of cases. Thus, it becomes increasingly accurate and reliable. In conclusion, it can be noted that this system provides doctors and specialists with a convenient, dependable, and affordable tool for analyzing cancer.

1.3 Objectives

Here is the list of objectives for the project:

- To design an artificial intelligence-based system for automatic and precise detection of cancer in histopathology images.
- To equip pathologists and oncologists with an instrument to lower manual effort and avoid errors.
- To facilitate the remote upload and evaluation of histopathology images, thus speeding up the diagnostic process.
- To apply deep learning algorithms that can recognize whether tissues are benign or malignant.
- To enhance the accuracy of diagnoses through continuous training with new labeled data sets.
- To create an intuitive interface that ensures convenient access to the tool by hospitals, laboratories, and research facilities.

- To facilitate the early detection of cancer and thus help in making clinical decisions.
- To give stakeholders (doctors, laboratory assistants, and scientists) accurate performance indicators and result visualization.

1.4 Problem Statement / Limitations

Cancer diagnosis using microscopic images suffers from being a tedious process and highly susceptible to human error. The shortage of expert pathologists and the high number of patients have made early detection a challenge. Moreover, differences in stains, illumination conditions, and even tissue samples complicate the process further. Therefore, there is an urgent need for a fast, efficient, and accurate process that could help in diagnosing cancer early.

1.5 Proposed Solution

The proposed system, Cancer Detection through Histopathology Images, aims to automate and enhance the process of detecting cancer from histopathology slides. Manual diagnosis is time-consuming, prone to human error, and requires highly specialized expertise. This system leverages artificial intelligence and deep learning techniques to classify tissue samples as benign or malignant, assisting pathologists in making faster and more accurate decisions. The project will be implemented as a web-based platform, accessible to hospitals, diagnostic laboratories, and patients, with secure role-based access for each user type.

The system is divided into 12 modules to handle distinct functionalities, as described below:

1.5.1 User Authentication Module:

- Secure login (Admin, Doctor, Patient)
- Role-based access control
- Password encryption
- Two-Factor Authentication
- Session Management

1.5.2 Doctor / Pathologist Module:

- Secure doctor registration with admin approval
- View assigned patients and medical history
- Add patient checkup
- Upload histopathology images for analysis
- Generate AI-assisted diagnostic reports

1.5.3 Patient Module:

- Secure patient registration and login
- View personal medical profile and history

- Access AI-generated diagnosis reports
- Download medical reports
- Receive Notifications

1.5.4 Hospital Admin Module:

- Approve or reject doctor registrations
- Add or delete patient registrations
- Monitor system usage activity
- Enforce hospital data policies
- Maintain activity and audit logs

1.5.5 Dataset Management Module

- Dataset selection and filtering
- Binary classification
- Dataset splitting
- Model training using EfficientNet-B0
- Save and export trained model weights

1.5.6 Pre-processing Module

- Load pre-trained EfficientNet-B0 model
- Pre-process uploaded histopathology images
- Perform binary cancer classification
- Generate prediction confidence score
- Return results to reporting module

1.5.7 Reporting & Analytics Module

- Generate AI-assisted diagnosis reports
- Combine patient, doctor, and AI results
- Store reports for future access
- Generate federated training improvement reports
- Display report history to patients

1.5.8 Notifications and Alerts Module

- Send alerts for new reports or updates
- Notify users about system activity
- Provide reminders for pending tasks
- Providing important system-wide announcements
- Ensure real-time updates for pathologists and patients

1.5.9 Audit and Activity Logging Module

- Log all user activities (logins, uploads, feedback, report generation, etc.)

- Maintain timestamped records for auditing and troubleshooting
- Track modifications in datasets, models, and results
- Provide admin access to view or export audit trails
- Detect unusual or unauthorized access attempts and notify the admin

1.5.10 Federated Learning Module

- Implement Flower framework
- Aggregate updates on central server
- Distribute global model back to hospitals
- Handles 03 hospital contributions
- Guarantees data privacy during federated training process

1.5.11 Dashboard and Analytics Module

- Dynamically display critical KPIs including patients, doctors, AI scans, accuracy, and user feedback ratings
- Provides comprehensive filtering by date range, specific users, affiliated hospitals and AI models
- Vividly visualize AI performance, patient, doctor, hospital, dataset, and system usage analytics
- Presenting a clear AI performance metrics
- Automatically generates insights on user activity, hospital performance, and system trends

1.5.12 Feedback and Continuous Learning Module

- Systematically collects structured feedback and ratings from patients, doctors, and administrators
- Efficiently manages feedback records with search, filtering, and builtin analytics features
- Securely stores AI prediction history along with detailed confidence scores and validation results
- Leverages doctor-confirmed diagnoses as ground truth for continuous automated model evaluation
- Precisely calculates and display AI performance metrics including overall accuracy

1.6 Intended Market of Product

The product can be used in different healthcare and scientific organizations where there is a need for histopathology image analysis and diagnosis of cancers. With the application of such an advanced combination of modules including federated learning, preprocessing, dataset management, reporting, dashboards, and continuous learning, the targeted institutions are expected to include hospitals as well as other institutions, where scalability, safety, and data-driven diagnostics are needed.

1.6.1 Primary Market

- **Medical Laboratories:** Image analysis, preprocessing, and datasets processing can help laboratories handle big amounts of tissue image data faster and more accurately.
- **Diagnostic Centers:** With quick cancer screening, report creation, and patient interactions, the system can serve perfectly well diagnostic centers that need quick workflows of diagnosing cancer.
- **Hospitals & Oncology Departments:** Pathologists and clinicians can use AI for diagnosis and get information about their patients' histories, do hospital-level analytics, receive alerts, use dashboards, and apply federated learning to improve the diagnostics quality while maintaining confidentiality of the information.

1.6.2 Secondary Market

- **Medical Research Institutions:** The team of researchers studying cancer progression, biomarkers, histopathology data and other medical phenomena can use the data storage and preprocessing modules as well as train models using scientific data.
- **Universities & Teaching Hospitals:** Educational organizations that work with students in fields of medical studies can use the system for training purposes.
- **Federated Learning Collaboratives:** Such organizations that participate in crosshospital learning can apply federated learning for training of global AI models without revealing sensitive patient data.

1.6.3 Emerging Market Opportunities

- **Health-Tech Companies:** Businesses that have developed digital pathology solutions or AI-powered diagnostic tools can seamlessly incorporate these modules (AI detection, analytics, dashboards, and compliance audit log) within their existing framework.
- **Cancer Research Consortia & NGOs:** Projects focused on early-stage cancer detection in resource constrained areas with limited resources could adopt this system because of its automation capabilities, economical implementation, and transparent audit trail.
- **Telemedicine & Remote Diagnostics Services:** With features like online reports, notifications, patient portals, and AI-assisted predictions, the product fits perfectly into remote cancer screening ecosystems.

1.6.4 Overall Market Fit

The diversity of modules ranging from user authentication and preprocessing to federated learning, analytics, instant notifications, and feedback-driven continuous learning fully enables such system to support:

- Enables efficient cancer detection processes in real-time
- Handles medical data management securely at a scale
- Facilitates cross-facility training and analysis of models
- Streamlines hospital management operations end-to-end
- Accelerates dataset annotation and AI research

Though, this solution is not merely a standard cancer detector but a versatile digital pathology and AI system, thereby perfectly qualifying it for wide-spread use across various healthcare and research fields.

1.7 Intended Users of Product

Every system is made for some users or stakeholders to help them complete their daily routine work much easier and faster. Consequently our system has meticulously developed for these three intended users:

- Pathologists
- Hospital Administrator
- Patients

1.7.1 Pathologists (Primary Users)

Pathologists are the major users of this system. They upload histopathology images for analysis and receive AI-generated diagnostic results. The system assists them in identifying cancerous patterns, validating predictions, and generating final medical reports. Pathologists also provide feedback to improve AI accuracy and can review patient history, previous reports, and analytics to support better clinical decision-making.

1.7.2 Hospital Administrators (Secondary Users)

The hospital administrators oversee the entire process within the system. They perform tasks like creating accounts for doctors, allocating privileges, auditing the system activities, and maintaining adherence to hospital rules. They have access to the system's dashboards, which provide information related to system performance, audit trail, and analytics at the hospital level. Administrators maintain the appropriate levels of data confidentiality and security within the system.

1.7.3 Patients (End Users)

Patients interface with the system to get their diagnostic reports and monitor the status of their samples that they had submitted to the system. They have access to detailed reports of their diagnostics, downloading capabilities, notifications, and an opportunity to maintain records of their health.

1.8 Software Process Model

Choosing a methodology for a system is very important and a crucial part of any project to make sure it ends successfully. The software process model we are choosing for our product is scrum and we will divide the work as iterations as it's in scrum development. In scrum we will work on a small cycles called sprints. In scrum each sprints have 2-4 weeks.

1.8.1 Process Model Introduction

To develop the proposed system we will follow Scrum methodology which is an agile framework for development of software product. It is one of the approaches that is used primarily for development of software projects with the goal of delivering new functionality or capability every sprint. A sprint is a time boxed period and is usually of 2 to 4 weeks but can be of one week as well. Scrum is the most popular agile methodology and is being used widely by software development teams to develop high quality software faster.

1.8.2 Justification of Proposing the process model

The Scrum methodology adheres to the values and principles of agile, which combines incremental and iterative process models. We choose it since it is a flexible model and meets our needs. change continues even after progress has started. Additionally, it offers incremental delivery for each interval and is appropriate for projects with fewer team members. Higher productivity, better products, greater stakeholder satisfaction, and shortened time to market are further advantages of implementing Scrum approach.

1.8.3 Steps of Process Model

Federated learning offers a transformative approach to collaborative medical AI development by enabling model training without centralized data sharing. This project documents the development of such a system for histopathology image classification, employing Agile Scrum methodology over 9 monthly sprints. Table 1.3 outlines the fundamental characteristics that establish the framework for this privacy-preserving healthcare AI solution.

The development process in Scrum methodology comprises a series of iterative stages referred to as sprints, where during each sprint, the aim is to produce certain product elements or enhancements. Sprint planning takes place at the start of each sprint, when objectives are determined and tasks are selected from the product backlog. Regular daily scrum meetings are organized to assess progress, discuss obstacles that may have occurred, and facilitate coordination between team members.

At the completion of each sprint, a sprint review occurs, during which the outcomes are assessed and feedback is gathered from all stakeholders. Finally, a sprint retrospective is performed, allowing the team to analyze their performance.

Table 1.3: Project Overview and Methodology

Details	Section
Project Goal	Develop a secure, privacy-preserving FL platform for pathology slides categorization that facilitates collective AI model training across all hospitals without sharing fully decentralized raw patient data, with features for image upload, AI-based diagnosis, model aggregation, and performance monitoring.
Methodology	Agile – Scrum Framework
Sprint Duration	1 Month per Sprint
Total Duration	9 Months (11 Sprints)
Team Roles	Product Owner: Defines and prioritizes features. Scrum Master: Facilitates process and removes blockers. Development Team: Developers (AI/ML engineers, Back-end/Frontend developers), Testers, Security specialists

In Table 1.4, the decomposition of all activities involved in the project from its initiation to deployment is given, which acts as a map for executing the project. The sprint-wise project plan outlines the development schedule of CDHI system using the Agile Scrum methodology. The project is divided into monthly sprints, where each sprint focuses on a specific objective and delivers measurable progress. User stories are used to capture stakeholder requirements, while tasks define the development, testing, and integration activities performed during each sprint.

Table 1.4: Work Breakdown Structure for CDHI Project

Level	Task	Description
1.0	CDHI Project	Overall development and deployment of the Cancer Detection through Histopathology Images system.
1.1	Initiation	Initial phase focused on understanding project goals, feasibility, and stakeholder expectations.
1.1.1	Requirement Gathering	Collecting functional and non-functional requirements from stakeholders and domain experts.
1.1.2	Elicitation	Extracting detailed requirements through interviews, discussions, and observations.
1.1.3	Questionnaire	Designing and distributing questionnaires to gather structured feedback from users.
1.1.4	Negotiation	Resolving conflicts and prioritizing requirements based on feasibility and scope.
1.1.5	Requirement Management	Documenting, tracking, and managing changes in requirements throughout the projects life cycle.

Level	Task	Description
1.2	Planning & Estimation	Planning project activities, timelines, resources, and cost estimation.
1.2.1	Design Project Planning	Defining project schedule, milestones, and development methodology.
1.2.2	Plan Resources	Identifying human, hardware, and software resources required for the project.
1.2.3	Plan Budgets	Estimating overall project cost including tools, hosting, and development expenses.
1.2.4	Products Backlog	Creating a prioritized list of crucial features and system requirements.
1.2.5	Sprint Backlog	Selecting tasks from the product backlog for each development sprint.
1.2.6	Design and Plan Risk Management Processes	Identifying potential risks and defining mitigation strategies.
1.2.7	Design and Plan Requirements	Structuring and organizing requirements for implementation.
1.2.8	Designing UML Diagrams	Creating UML diagrams to represent system architecture and workflows.
1.3	Implementation	Actual development of system components and functionalities.
1.3.1	Develop Systems Design	Designing systems architecture and overall technical structures.
1.3.2	Develop Interface Mockup	Creating visual prototypes of system interfaces.
1.3.3	Develop Admin Interface	Implementing functionalities specific to the admin role.
1.3.4	Develop Doctor Interface	Developing features that allow doctors to upload images and view results.
1.3.5	Develop Patient Interface	Creating patient-side functionalities for accessing reports and information.
1.3.6	Develop Database and Integrated System	Designing database schema and integrating backend with frontend.
1.3.7	Manage Changes	Handling updates and modifications during development.
1.4	Reviewing/Testing	Verifying system correctness, performance, and reliability.

Level	Task	Description
1.4.1	Test Case	Designing test case to validate systems requirements.
1.4.2	Unit Testing Checklists	Testing individual modules for correctness.
1.4.3	Systems Testing Checklists	Testing complete system as the whole.
1.4.4	User Acceptance Test	Validating system functionality with end users.
1.5	Releasing	Final deployment and availability of the system for use.
1.5.1	Setup Purchase Domain and Hosting	Configuring domain name and hosting environment.
1.5.2	Deploy System on Web	Deploying the final system on a live web server.

The FL platform follows an Agile Scrum methodology, dividing the complete project life cycle precisely into eight monthly sprints to ensure well-structured and incremental progress where each sprint is designed with specific objectives to achieve, user stories, tasks, and deliverables to support systematic development and continuous evaluation. While this sprint-wise plan provides a clear-cut road map from initial requirement gathering to final deployment and feedback integration, ensuring the in-time completion and effective coordination across all development phases as presented in the detailed breakdown of table 1.5.

Table 1.5: Sprint-wise Project Plan

Sprint	Weeks	Goal	User Stories	Tasks	Deliverables
1	1–4	Requirements Gathering	Sign Up, Login, Update Profile	Collect stakeholder requirements, define workflows, list roles, create initial mockups	Requirements Document, User Story List
2	5–8	System & UI Design	View Assigned Patients History, Upload Histopathology Images, Annotate	Design UI screens, create dashboards, plan image upload and annotation workflows	UI Mockups, UML Diagrams, Design Documents

Sprint	Weeks	Goal	User Stories	Tasks	Deliverables
3	9–12	AI Model + Preprocessing	Request AI Analysis, AI Classification, Generate Diagnostic Report	Develop preprocessing pipeline, CNN model training, AI integration, report generation logic	Trained AI Module, Pre-processing Pipeline, Base Diagnostic Report Template
4	13–16	Federated Learning Setup	Patient Downloads Report, Track Historical Reports	Implement federated learning nodes, secure aggregation, multi-user access	Working FL System, Multi-node Training Setup
5	17–20	Admin & Dataset Management	Admin Management & Monitoring, & Management	Admin dashboard setup, dataset upload and organization, approval workflows	Functional Admin Panel, Access Control Setup
6	21–24	Notifications + Logging	Notifications & Alerts, Auditing & Logging	Develop notification system, alert triggers, system logging, auditing framework	Notification Module, Logs, Auditing Reports
7	25–28	Feedback-driven Learning	Feedback-driven Continuous Learning	Implement feedback collection, incremental AI retraining, validation of improvements	Feedback Collection System, Updated AI Model
8	29–32	Analytics & Final Improvements	Analytics & Dashboards	Develop analytics module, generate reports, finalize dashboards, optimize system performance	Analytics Dashboard, Final System Release, Documentation

The CDHI system plan is laid out in a way that makes sense. It is being developed using the Agile Scrum methodology. This plan for the Cancer Detection through Histopathology Images system is divided into eight parts that will take thirty-two weeks to complete. Each part of the CDHI system plan has its goal.

Chapter 2

SOFTWARE REQUIREMENTS SPECIFICATION

The technical term requirement means a detailed, definition of the system functions and core constraints. An SRS is basically the fundamental blueprint for defining the requirements. If the requirements are done well and validated early, the subsequent software design flows logically and smoothly across the entire life cycle. Conversely, if the requirements are done poorly and lack clarity, the resulting architectural design is awkward and the coding is more difficult.

2.1 Introduction

The following comprehensive SRS contains the detailed perspective of the system captured from multiple stakeholders. It also provides a thorough general description of behavioral characteristics of the end-user, the product's hardware, and the functional and non-functional requirements of the product. The purpose of the SRS document is to collect and analyze the architectural ideas that have come up to clearly define the system and its requirements w.r.t primary consumers of the product.

This comprehensive SRS will also provide a broad overview of our software product, its operational parameters and strategic goals to be achieved. This technical document describes the projects target audience and its necessary hardware and software requirements.

2.1.1 Document Scope

This SRS is specifically aimed at specifying the technical requirements of software to be developed. The purpose of this formal document is to present a highly detailed description of the system to be developed.

2.1.2 Audience

This documentation is targeted for all the stakeholders of the project and the creators of the system but primarily it is written for the developers and describes the details of the functionality of the product in technical terms.

Testers of the project can also use this documentation to assist them in testing phase as some errors are easy to find with the help of the document. By this method testing will become more systematic and less time consuming.

- Development Team
- Administration
- Pathologists

- Patients
- Maintenance Team
- Technical Writers

2.1.3 Functional Requirements

Functional requirements [11] specify the features and functions that a product must have. It describes a specific system outcome when a user action is taken on them. The procedures, data, and interactions that the client wants developed as well as the interactions between the system and its surroundings are included in these requirements. Here are the Functional Requirements of our project.

FR1: User Authentication Module

Provides encrypted login process for all system users based on the access roles.

FR2: Doctor/Pathologist Module

Enables the doctors to control the medical imaging, verify AI-based predictions, and view the history of diagnoses made.

FR3: Patient Module

Facilitates the patients in accessing their reports, notifications, and medical records.

FR4: Hospital Admin Module

Enables the administrators in controlling doctor accounts and tracking system operations.

FR5: Dataset Management Module

Enables the upload, label, and organize datasets required for training and validation of the artificial intelligence algorithm.

FR6: Preprocessing Module

Carries out image enhancement operations, creates image patches, and pre-processes the images for use in machine learning algorithms.

FR7: Federated Learning Module

Enables the training of AI models across multiple hospitals through secure aggregation.

FR8: Reporting & Analytics Module

Produces summary of diagnostics, trend graphs, and performance analysis of the model.

FR9: Notifications and Alerts Module

Furnishes notifications and alerts about any relevant event to system and diagnosis.

FR10: Audit and Activity Logging Module

Traces and logs actions taken in the system for audit purposes.

FR11: Dashboard and Analytics Module

Enables the visualization of key performance indicators of the AI-based system.

FR12: Feedback & Continuous Learning Module

Captures corrections from the doctors on the predictions and retrain the global model.

Below is the table 2.1 that shows the description and working of each functional requirement one by one. That how it works and what is it doing in the website.

Table 2.1: Functional Requirements

Serial No.	Functional Requirement	Description
FR1	User Authentication Module	This module manages all processes related to system access. It verifies user identity using credentials and role-based authorization.
FR2	Doctor/Pathologist Module	The current module is used as the main working environment for the doctors. The histopathology images can be uploaded in this module, pre-processing begins here, and it gives the predictions of the artificial intelligence. Tumors can be marked here by the doctors, giving feedback and approving the diagnosis. The module also provides full medical records of the patients.
FR3	Patient Module	In this module, users can have a secure access point to view their diagnostics results. It guarantees access to the medical report, downloading facility, and also their health report in a chronological format. This module also helps in getting notifications, appointments updates, and information regarding past conversations with the AI system.
FR4	Hospital Admin Module	The hospital administrator uses this module to control all operational aspects of the platform within a hospital environment. It includes managing doctor registrations, customizing role privileges, approving or revoking access, and enforcing hospital data governance rules. The module also provides tools to monitor system activity and view usage statistics.
FR5	Dataset Management Module	This module centralizes dataset operations such as uploading, labeling, annotating, and deleting histopathology images. It maintains metadata like class distribution, image resolution, and dataset size. It supports collaborative labeling workflows, prevents unauthorized dataset modifications, and ensures that properly structured datasets are available for pre processing and model training. Versioning of datasets and validation of dataset quality are also included.

Serial No.	Functional Requirement	Description
FR6	Pre processing Module	This module performs all transformations required to prepare medical images for deep learning. It includes patch extraction, resizing, normalization, stain normalization, noise reduction, artifact removal, and image augmentation. The module also manages splitting of datasets into training, validation, and testing subsets, ensures balanced class distribution, and maintains reproducibility. It initiates CNN training, saves extracted features and model weights, computes evaluation metrics such as accuracy, recall, precision, and F1-score, and logs pre processing and training steps for performance tracking. The module supports batch processing, GPU acceleration, and integration with federated learning pipelines.
FR7	Federated Learning Module	This module enables privacy-preserving collaborative model training across hospitals. Local models are trained on hospital-specific data, and only weight updates (not raw patient data) are shared with the central server. The module consolidate these updates using federated averaging, monitors contribution quality, handles model versioning, distributes the improved global model back to participants, and ensures secure, encrypted transmission channels for data privacy and compliance.
FR8	Reporting & Analytics Module	This module creates diagnostic reports using AI predictions, doctor's validation, and patient history. It gives visual analytics such as hospital-wise reports, trend of detections accuracy, confusion matrices, class wise distribution, and comparative analysis of models. These reports are exported in formats like PDF along with timestamps for audit and also allows drilling down facilities.
FR9	Notifications and Alerts Module	This module acts as a communication layer between all modules of the system. In this module, automated alerts get generated based on various events such as diagnosis completion, update of models, auditing process, malicious login detection, and request for feedbacks. Alerts are generated based on role and context awareness and are also prioritized.

Serial No.	Functional Requirement	Description
FR10	Audit & Activity Logging Module	This module keeps track of each major actions in the system, including logins, uploading of images, creation of reports, model training, dataset updates, and administrative activities. It keeps timestamped records of all actions for accountability and compliance with medical data regulations. This feature will enable administrators to search, filter, and export audit logs.
FR11	Dashboard and Analytics Module	This module visualizes KPIs like the percentage of accurate predictions, total number of uploads, workloads, number of datasets created, federated learning iterations, and statistics about doctors' and patients' activities in real time. It provides graphical representations that can be filtered according to hospitals, dates, and user categories. Drill-down is an important functionality provided by this dashboard.
FR12	Feedback & Continuous Learning Module	This module enables doctors to give feedback about AI's predictions. It helps record corrections made by the experts and adds them to the feedback dataset for model training. The module evaluates the impact of the re-trained AI models on performance parameters to calculate the gains in accuracy.

The system caters to both physicians and patients by ensuring the protection of all personal information related to users, medical images, and the AI-based diagnostic service. The system offers services such as data processing and federated learning that can assist hospitals in training an AI model without exposing their data to any potential risks. The system also enables reporting, dashboard creation, and alerts. With each update, the performance of the AI model is improved.

2.2 Non-Functional Requirements

Non-functional requirements are those requirements that determine fundamental properties of system [12]. The non-functional requirement describes behavior of a system and sets the limits on its functions. The set of requirements that cannot be covered with functional ones is called non-functional requirements.

- **Performance:**

- The system should work efficiently analyzing histopathology pictures in a couple of seconds for the standard picture size.
- The performance of the system is inherently constrained and may be somewhat affected by underlying server capabilities and the speed of network connections.

- **Reliability:**
 - The system must possess the robust ability to seamlessly replicate databases for ensuring patient data backup and storing them off-site for disaster recovery.
 - The system must consistently deliver stable and highly reliable AI predictions under all normal operational conditions.
- **Maintainability:**
 - The system architecture must be fully modifiable to accomodate seamless updates for AI models, image processing algorithm updates, and interface changes.
 - To maintain operational reliability there should be an error logging and monitoring tool for rapid system analysis and debugging.
 - Proper documentation should be available for further developers for efficient modification and updating.
- **Security:**
 - Patient, pathologist, and admin passwords shall always be encrypted and never displayed in plain text.
 - Only authenticated users shall have access to their respective data.
 - Admin-level servers shall be accessible only to authorized personnel.
 - Sensitive medical data shall be transmitted securely using encrypted protocols.
- **Usability:**
 - The system will provide a uniform and self-explanatory interface for patients, pathologists, and admins.
 - Standard icons and toolbars shall be used to improve usability and reduce learning time.

2.2.1 Other Non-Functional Requirements

- **Hardware Interface:** Since the application is web based and runs over the internet, so all the hardwares that are required to connect with the internet will be called hardware interface.
- **User Interface:** The user interfaces of the softwares should be optimized to any web-browser such as Internet Explorer, Chrome, through which user will access the dashboard.
- **Legal and Ethical Compliance:** The system shall ensure secure handling of patient medical data. All records will be confidential and protected under data privacy standards similar to HIPAA compliance. Proper access control, encryption, and ethical use of AI results will also be maintained to ensure long-term system reliability and trustworthiness.
- **Software Tools:** The system will employ C-sharp in the backend programming, TensorFlow framework in the AI model creation, and structured SQL in the database programming. Where these technologies will help ensure efficient communication between modules, secure data handling, and long-term maintainability of the system.

2.3 Requirement Gathering Techniques Used

We have used three techniques for requirement gathering. Below are discussed the techniques of brain storming, sample questions and observation method.

2.3.1 Brainstorming:

The brainstorming technique [13] was used to derive the necessary requirements of our system. The brainstorming sessions involved the creation of various ideas concerning different modules and perspectives by our team members. This was carried out in an atmosphere where all participants were free to contribute without any limitations.

We conducted six brainstorming sessions, which lasted between 30 and 60 minutes. Each of the brainstorming sessions had a specific focus in the system and gave us the required information.

Brainstorming about system scope definition, module identification, technical requirements, UX requirements, security and data handling requirements and lastly continuous learning requirement was done.

Outcomes from Each Brainstorming Session:

- **Session 1: System Scope Definition:** We clarified the overall objective of creating an AI-powered system to classify cancer as either positive or negative using the images of histopathology. We specified the three roles that can use the system, namely the Admin, Doctor, and Patient.
- **Session 2: Module Identification:** The major modules of the system include authentication, doctors, patient, admin, uploading images, pre-processing, reporting, and notification were identified.
- **Session 3: Technical Requirements:** Requirements related to the datasets, pre-processing, model training, federated learning, and cloud storage were defined.
- **Session 4: UX Requirements:** Ideas such as simple UI, drag-and-drop file upload feature, loading indicators, report creation, visualization capabilities, and dashboards for each user type were defined.
- **Session 5: Security and Data Handling Requirements:** It was decided that there should be security requirements such as encryption, data protection, role-based access control, audit log recording, etc.
- **Session 6: Continuous Learning Requirement:** A continuous learning cycle where feedback is provided by doctors was finalized. Corrections made by doctors would be taken to improve AI performance.

2.3.2 Sample Questions

In order to avoid conducting one-on-one interviews with users like pathologists, patients, or system administrator, we created some sample questions based on what can be asked during user interviews [14].

Example Sample Questions:

- What problems may arise during histopathology images analysis?
- What current tools or methods do you use for tracking diagnostic results?
- What features would be valuable for an AI-based cancer detection system?
- Is it critical for you that patient information was securely stored and protected?
- Do you prefer to get visual images with detailed explanations or simple diagnostic results?
- What frequency should be used to update the machine learning model in your opinion?
- Is there a necessity in the system integration with the existing hospital/laboratory software?

Outcomes of Sample Questions:

- **Outcome for Q1:** The most common problems pointed by the respondents included time-consuming analysis, difficulties with cancer detection, low quality of images, and expert-level required interpretation of images.
- **Outcome for Q2:** Respondents stated that their current ways to track diagnostic results include written notes, Excel spreadsheets, and hospital-specific software that doesn't have automation features.
- **Outcome for Q3:** In their opinion, most valuable features of the proposed system include reliable diagnostic results, easy uploading of images, informative visual indicators, ability to produce reports, and opportunity to correct predictions.
- **Outcome for Q4:** All participants agreed that data privacy and security were highly important aspects of the product since it deals with sensitive patient information.
- **Outcome for Q5:** A majority of respondents chose the option to receive both images with visualization of cancer cells locations and summaries.
- **Outcome for Q6:** Respondents claimed that periodic updating of the AI model, preferably once per month or upon collecting new verified dataset, would be most suitable.
- **Outcome for Q7:** All participants agreed that integration with hospital/laboratory software could be advantageous, although it's not a prerequisite for the product success.

This approach allowed us to define clear and structured system requirements while maintaining the perspective of potential end-users.

2.3.3 Observation Method:

The observation technique [15] was utilized in order to study the relation between pathologists, patients and hospital administrators with currently made system and its workflow. Instead of solely depending on verbal answers, our group analyzed first-hand processes like image analysis, documentation, reporting, and flow of communication within a medical context.

These observations took place in brief sessions each lasting 20-30 minutes on separate days. Where the idea was to observe the real user behavior, inefficiencies in the current workflow, and potential places where the application of artificial intelligence could be of help and by this method, we could detect unspoken work flow requirements that users themselves may have failed to consider while being interviewed.

Sample Questions for Observation Method:

- How do pathologists analyze and interpret images of histopathology?
- What tools/software do doctors and hospital administrators use in the process?
- What processes take the most amount of time in the whole diagnosis process?
- What kind of mistakes do users commit while performing manual reporting?
- How do users respond to complicated/repetitive tasks?
- How do doctors and administrators store/retrieve/share data among doctor, patient, and administrator?

Outcomes of Observation Method Questions:

- **Result of Question 1:** Most pathologists use manual microscopes and simple viewers that slow down the process and increase the chances of human fatigue.
- **Result of Question 2:** Doctors and hospital administrators widely use tools like email, Excel spreadsheet, hospital management system, and stand-alone image viewer, yet no software is specialized in histopathology workflows.
- **Result of Question 3:** Image interpretation and writing reports are the most time-consuming stages in the entire process, particularly for large batches of images.
- **Result of Question 4:** Minor inaccuracies were discovered during the manual report generation stage, typically caused by lack of time, mislabeling, and human error.
- **Result of Question 5:** Users became visibly frustrated at repetitive tasks, including searching for old records, renaming files, uploading images, and manually creating reports.
- **Result of Question 6:** Data storage is distributed across computer disks, USB drives, and various software, causing delays in data retrieval and making users susceptible to data loss. The centralized data access was highly needed.

2.4 Time Frame

Definition Of Time Frame [16] is defined as period of time that relates to any activity or a project. Time frame Inception requires 5 days. Process of elicitation may be done in 7 days. Process of elaboration may be done in 4 days. Process of negotiation may be done in 7 days. Process of specification requires 5 days. Process of validation may be done in 4 days. Requirements management process may take 8 days. Table 2.1 demonstrates the time frame of requirement phase. Time Frame / Duration of requirement phase.

Table 2.2: Time Frame

Serial #	Phase	Duration
1	Inception	5 days
2	Elicitation	7 days
3	Elaboration	4 days
4	Negotiation	7 days
5	Specification	5 days
6	Validation	4 days
7	Requirement Management	8 days

This table shows how much time will be spent on each stage of the requirement engineering process. All stages of the process are done sequentially to make sure that all requirements are defined, analyzed, and managed properly before starting the development process. The first step is inception, which takes 5 days. In this stage, we define what the problem is, what we want to achieve, and who are our stakeholders. The second step is elicitation, which requires 7 days. We need to gather requirements from our stakeholders by doing interviews, surveys, or observations. Then elaboration, which takes 4 days. In this step, we analyze all gathered requirements, eliminate any ambiguities, and figure out what relationships exist between different requirements.

Next comes the negotiation stage, which requires 7 days. In this step, we need to solve any conflicts that arise between different requirements or stakeholders. After that comes the specification stage, which takes 5 days. In this step, we document all requirements in a structured manner. They usually put everything in an SRS. Then comes the validation stage, which requires 4 days. Here, we check whether all documented requirements are correct, complete, and meet the requirements of stakeholders. The last stage is requirement management and it takes 8 days. Here, we manage changes in requirements throughout the life cycle of our project because requirements can change during the development process.

Chapter 3

SOFTWARE PROJECT PLAN

A plan of software development serves as a structured approach to transforming project requirements into a fully functional application. It encompasses various phases, including comprehensive project planning, detailed conceptualization, software development, documentation, system deployment, official launch, and ongoing maintenance. Where explicitly this document ensures a systematic workflow, allowing efficient resource utilization and proactive risk mitigation throughout the entire life cycle of the project.

For proposed system, the plan of software development acts as a strategic blueprint, offering a clear road map for all the involved stakeholders, such as project managers, developers, designers, and end-users furthermore. This structured guide facilitates smooth execution by defining key objectives, technical requirements, development processes, and deployment strategies. So that by following a well-defined plan, the engineering team can ensure timely completion of the project while maintaining high software quality and operational efficiency.

3.1 Deliverables of the Project

This project aims at the development of a web-based AI application designed to automatically detect cancerous cells from histopathology images. It is a web-based universal platform through which pathologists can analyze and detect cancers from uploaded histopathology images while administrators are responsible for the management of the system's functions. Some of the major deliverables include:

1. **AI-Based Web Application:**

This is an interactive website where users will be able to upload their histopathology images, analyze them through pathologists' assistance, and get the result based on AI models training.

2. **SRS Document:**

The SRS document includes all types of functional and non-functional requirements of the system. All relevant information regarding system scope, roles of users, and other important details is included.

3. **Design Document:**

This document discusses the architecture of the system. Detailed information regarding the database, DFDs (Data Flow Diagrams), and module description is included for the developer's perspective.

4. **Project Plan:**

The project plan outline the schedules, resource allocation, and milestones. It defines phases like requirement gathering, design, implementations, testings, and deployment.

5. Product Backlog:

A highlighted list of system features, its enhancements, and user requirements created during the initial Agile phase. It serves as a roadmap for development and iteration.

6. Sprint Backlog:

A document containing detailed user stories and tasks to be completed in each sprint, ensuring focused progress toward the final deliverable.

7. Interface Mockups:

The system interface will be created in interactive mockup forms for all 03 stakeholders; admin, doctors and patients. Where subsequently the mockups will be rigorously checked by stakeholders for usability and strategic design enhancements.

8. Prototypes:

The initial functional prototype will be developed in incremental mode to examine each functional software module separately which includes:

- **Patient Portal Interface:** The portal precisely helps patients to see their final diagnostic reports and monitor the real-time analysis progress using a highly intuitive dashboard.
- **Pathologist Diagnostic Interface:** A dedicated specialist interface that helps pathologists access high resolution image uploads, analyze AI-assisted detection metrics, and prepare and authenticate official diagnostic reports.
- **Admin Management Interface:** This interface allows admin to control all user credentials, closely monitor the overall performance of the system, and maintain records.

These prototypes will be used later in one web-based application.

9. Integrated System and Module Completion:

After integrating the user interface and backend services, modules for patient, pathologists, and admin will work together. Images will be processed and AI detection will be performed in order to prepare and analyze reports.

10. Test Report:

The system will pass through several testing modes like unit testings, integration testings, and system testings and after that, a detailed test report will be prepared.

11. Final Deployed Application:

The finalized version of the web-based Histopathology Image Detection System will be launched to perform the AI-based cancer cell detection job efficiently and securely.

3.2 Software Project Management Plan

Software Project Management [17] Plan aims at making sure that all team members get acquainted with their duties and requirements related to their performance. In order

to coordinate team members' actions and make them understand their project responsibilities in full, a software project management plan is now developed. Purpose of its implementation is to provide guidelines on project activity management.

Project failure can be associated with inefficient planning, which can be considered one of the major causes of the inability of successful software development. The lack of project planning prevents identification of the extent of the work, since it makes difficult determination of the scope of project activities. Also, the absence of proper planning makes it problematic to monitor the materials and development approaches required to ensure the achievement of the set project goals.

As is evident, the proper planning is crucial for efficient implementation of a project within the assigned time frame and the limited budget. Since the current project is being implemented by a team of three developers and should be carried out in an approximate period of ten months, a detailed project management plan is needed. This will contribute to more even distribution of tasks among team members and more stable progress of development process.

The table below presents the detailed plan of project development with the description of all tools, technologies and other resources to be used for project deliverables' implementation. At the same time, it reflects documentation planning and resource allocation strategies as well as contains the data related to the development of a project Gantt chart that would allow visualization of the timeline.

In addition, the developed project plan implies communication planning that ensures frequent meetings and status reporting. In this way, all participants will be informed about the current status of activities and prevent any misunderstandings.

The Gantt chart is a kind of a timeline for the entire project. There, one will be able to see dependencies between tasks, milestones, and deadlines. The instrument will help in continuous monitoring of the process and, in case of some delays, adjust activity.

Lastly, the issue of documentation must be taken into account. It is imperative to document all information pertaining to requirements, design, development, and testing outcomes. This is not just a necessity in terms of making the entire process transparent and understandable but also because documentation is very important in the future maintenance or enhancement of the product.

3.2.1 Project Planning

Poor planning is constantly being identified as one of the major reasons for project failure. Without developing suitable project plan at the beginning, determining project scope becomes practically impossible. Without a project plan it is actually difficult to keep track of resources, tool and techniques that are required to achieve the goals. Therefore, a good project plan has to be developed in order to successfully complete the project within the given schedule and budget.

In order to deliver the above mentioned project deliverables we require sufficient human resources, time, and budget. As we are three group members who will be working on the project and we need to deliver the project in approximately 10 months, so we have to make a plan accordingly.

Our Project Plan is given below in table 3.1, which shows all the tools, technologies and other resources that will be utilized for each of the deliverable, followed by Documentation Plan, Resource Plan and a Gantt Chart of our project.

Table 3.1: Project Planning

Serial No.	Deliverables	Abdul Rafay	Aman Siddiqui	Rutaba Sajid	Tools	Language	Time Weeks
1	SRS Document	✓	✓	✓	OverLeaf	LaTeX	1
2	Design Document	✓	✓	✓	OverLeaf	LaTeX	2
3	Project Plan	✓	✓	✓	OverLeaf	LaTeX	2
4	Product Backlog	✓	✓	✓	OverLeaf	LaTeX	2
5	Sprint Backlog	✓	✓	✓	Draw.io	–	2
6	Interface Mockup	✓	✓	✓	VS	HTML CSS, JS	18
7	Prototypes	✓	✓	✓	SQL	–	3
8	Database	✓	✓	✓	SQL SSMS	HTML CSS, JS	4
9	Integrated System	✓	✓	✓	VS	HTML CSS, JS	3
10	Test Report	✓	✓	✓	OverLeaf	LaTeX	2
11	Final Deployed Application	✓	✓	✓	VS	HTML CSS, JS	4

In this table it displays the complete project planning and assignment of tasks among the three members of the group who are Abdul Rafay, Aman Siddiqui, and Rutaba Sajid. All the project outputs are described along with the respective team members who have been allotted the task of completing these outputs, the technology that will be used in the process, the tools needed, and the number of weeks it will take to complete it.

3.2.1.1 Milestone Plan

The milestone chart highlights the important stages of the project, ranging from requirements collection through implementation up to final evaluation. Milestones indicate

important deadlines and expected results in order to ensure proper project implementation in stages. Table 3.2 shows the milestones together with their durations and goals. The milestone chart acts as an important timeline which guides project progress, ensuring consistency and timely completion of each phase of software development.

Table 3.2: Milestones Plan

Serial No.	Milestones	Milestone Goals	Duration (Weeks)
1	Requirement Engineering	Collect requirements from all users, understand diagnostic needs; validate requirements with experts.	2
2	AI Model Development	Design cancer-detection AI model; pre-process histopathology images;	5
3	Web Application Design	Design dashboards for all users, create responsive, user-friendly interface layouts.	3
4	Database Design	Create SQL schema; store medical records, diagnosis reports, and user data securely.	2
5	Model Integration	Integrate trained AI model with web interface; enable real-time detection and reporting.	4
6	Federated Learning Integration	Implement local training at each hospital; aggregate weight updates using FedAvg without sharing data.	3
7	Testing & Validation	Conduct unit and system testing; validate AI output with test datasets; document results.	3
8	Report Module	Develop automated report system; add visual result display and formatted diagnostic output.	2
9	Deployment	Deploy system to server; ensure secure, stable performance and user access.	2
10	Documentation & Final Review	Prepare SRS, design, and testing documents; final review and refinements before submission.	1

The process is broken down into different stages ranging from requirement engineering

up to deployment and documentation. Each milestone entails specific objectives including requirement acquisition, AI model creation, designing the user interface, database creation, model integration, testing, report writing, and deployment of the system. In addition, a specific time frame for each stage in weeks is provided to facilitate time management. Using the milestones will make it easy to gauge the progress of the project and to ensure that one activity is completed before moving on to the other. The milestones will facilitate better coordination when undertaking development activities.

3.2.1.2 Documentation Plan

The following table represents the documentation strategy of the project, providing information about the necessary documents that should be generated throughout the development process. Table below 3.3 shows the document plan of our project.

Table 3.3: Documentation Plan

Serial #	Document Name	Documenter Name	Duration
1	Project Proposal	Rafay, Aman, Rutaba	2 weeks
2	Chap 1	Rafay, Aman, Rutaba	1 week
3	Chap 2	Rafay, Aman, Rutaba	1 week
4	Chap 3	Rafay, Aman, Rutaba	1 week
5	Chap 4	Rafay, Aman, Rutaba	1 week
6	Chap 5	Rafay, Aman, Rutaba	1 week
7	Chap 6	Rafay, Aman, Rutaba	1 week
8	Chap 7	Rafay, Aman, Rutaba	1 week
9	Chap 8	Rafay, Aman, Rutaba	1 week

This includes the documentations of project proposals and each individual chapter. In addition, it mentions people who will perform the task and the time required for each. This is done to create proper documentation as per the academic requirements.

3.2.1.3 Resources Plan

The resource planning of the project is shown, specifying the primary activities, staff involved, budget estimates, and duration of each activity. The plan includes all significant project phases such as requirement gathering, document creation, design process, implementation of AI model, database creation, testing process, and deployment. Table below 3.4 show the resource plan of our project.

The resource plan at the same time helps assign tasks to appropriate team members without exceeding realistic line and budgetary constraints over time. Most phases involve the whole team to encourage efficient communication and balanced workload throughout development. Estimates for time have been distributed based on degree of difficulty

and relevance, as implementation module respectively will take the most effort. Family budgets are devoted mainly to system construction and web deployment items, while documents writing or planning tasks use existing resources. It helps control project progress, maximize productivity and achieve completion of the CDHI system in a successful timeline.

Table 3.4: Resource Plan

Serial No.	Resource Name	Task	Budget	Time (Weeks)
1	Rafay, Aman, Rutaba	Requirement Gathering	-	1
2	Rafay, Aman, Rutaba	SRS Document	-	2
3	Rafay, Aman, Rutaba	Develop Design Document	-	3
4	Rafay, Aman, Rutaba	Develop Project Plan	-	2
5	Rafay, Aman, Rutaba	Develop Product Backlog	-	3
6	Rafay, Aman, Rutaba	Develop Sprint Backlog	-	2
7	Rafay, Aman, Rutaba	Develop Interface Mockup	-	2
8	Rafay, Aman, Rutaba	Build Prototype	-	2
9	Rafay, Aman, Rutaba	Develop Database	-	2
10	Rafay, Aman, Rutaba	Implementing AI Model	-	3
11	Rafay, Aman, Rutaba	Completion of all modules	15,000	12
12	Rafay, Aman, Rutaba	Perform Testing	-	2
13	Rafay, Aman, Rutaba	Develop User Manual	-	1
14	Rafay, Aman, Rutaba	Develop Test Report	-	1
15	Rafay, Aman, Rutaba	Deploy on web	15,000	1

Each activity is assigned to the same set of staff to encourage teamwork and effective allocation of tasks. The budget cell specifies the financial inputs required for important milestones like completing modules and deploying them. Overall, the plan enables proper management of time, manpower, and financial investments.

3.2.1.4 Quality Plan

Quality management plan serves as one of the core aspects of the software development process aimed at making sure that the end product complies with certain criteria and standards. The emphasis is put on the provision of consistent and efficient development process. In the case of the current project, it is essential to mention that the quality activities are implemented during every stage, which enables addressing potential problems beforehand and avoiding bugs in the system.

The quality management plan defines a number of characteristics including scalability,

usability, security, and performance which are necessary for measuring the effectiveness of the system. Testing of each characteristic will be carried out in accordance with the specific time frame in order to ensure systematic approach towards the validation of system behavior.

It is evident that this strategy helps to achieve balance between the functionality of the system and its quality criteria.

The main objectives of this quality management plan are:

1. Apply strict quality control process.
2. Make sure that the quality of the project output meets the end-user expectations.

Table 3.5 given below describes our quality plan of our project.

Table 3.5: Quality Plan

Serial No.	Quality Attribute	Start Date	End Date	Description
1	Scalability	2 Jan 2026	9 Jan 2026	In this test we check our system is scalable according to usage environment. It supports data of large or small audience.
2	Usability	10 Jan 2026	13 Jan 2026	In this test we check how our system interface responds to user's input. Is UI user-friendly or not?
3	Security	13 Jan 2026	17 Jan 2026	In this test we check if user information is secure or not. User's data remains private.
4	Performance	18 Jan 2026	20 Jan 2026	In this test we check that our system performs tasks quickly and responds fast for any user.

In fact, this plan allows conducting system testing systematically in order to examine all the essential aspects of a particular system in specific time frames. Testing will be conducted at each stage in order to increase the reliability of the system and find out its vulnerabilities. As a result, such an approach guarantees high standards regarding the quality of software.

3.2.2 Project Scheduling

Project scheduling defines the timeline for each deliverable and ensures that development activities are organized and completed on time. It clarifies task sequences, durations,

and dependencies, helping the team plan work efficiently.

The schedule sets clear start and end dates for all major phases, from documentation and planning to design, development, testing, and final delivery. Through assigning the expected duration, it becomes possible to monitor the progress of the process and identify any delay that may be occurring during the work.

Table 3.6 shows the comprehensive schedule of major deliverables together with their expected duration. This method of scheduling allows for better resource allocation, effective distribution of workload among the team members, and maintenance of uniformity during the entire life cycle of the project. Moreover, following the schedule becomes crucial in completing the project within the expected time span while fulfilling all the required specifications. The deliverables listed in the above table include srs document, design document, project plan, product backlog, sprint backlog, inference mock up, prototype, database, complete user interface, test report, and final product.

Table 3.6: Project Scheduling

Serial #	Deliverables	Start Date	End Date
1	SRS Document	1 Oct 2025	15 Oct 2025
2	Design Document	16 Oct 2025	22 Oct 2025
3	Project Plan	23 Oct 2025	24 Nov 2025
4	Product Backlog	25 Nov 2025	28 Nov 2025
5	Sprint Backlog	28 Nov 2025	30 Nov 2025
6	Interface Mockup	30 Nov 2025	5 Dec 2025
7	Prototype	6 Dec 2025	25 Dec 2025
8	DataBase	26 Dec 2025	1 Jan 2026
9	Complete User Interface	2 Jan 2026	7 March 2026
10	Test Report	8 March 2026	20 March 2026
11	Final Product	21 March 2026	2 June 2026

Project schedule table is the complete time frame of deliverables expected during the software development process. This starts with SRS Document (1 October 2025–15 October 2025), where functional and nonfunctional requirements of the system will be collected and stated to lay a strong foundation of the project. Then comes the Design Document (16 October 2025–22 October 2025), which defines the architecture, database schema and technical design of the product. Next, the Project Plan (23 October 2025–24 November 2025) defines tasks and strategies for the correct organization of the project. After that, Product Backlog (25 November 2025–28 November 2025) describes what features will be included in the product, and after that, Sprint Backlog (28 November 2025–30 November 2025) breaks down the task into smaller chunks.

Next is the Interface Mockup (30 November 2025–5 December 2025), which will focus

on the user interface to enhance usability and user experience. Then comes the Prototype phase (6 December 2025–25 December 2025), in which a prototype of the product will be created to test the basic functionality. Database phase (26 December 2025–1 January 2026) defines designing and creating the database schema to store the data efficiently. After that, Complete User Interface phase (2 January 2026–7 March 2026) involves building up the complete front-end application with back-end functionalities. Next is the Test Report phase (8 March 2026–20 March 2026) that helps in finding defects in the developed software. Finally, there comes the Final Product phase (21 March 2026–2 June 2026).

3.2.2.1 Gantt Chart

The project schedule for developing a web application over a span of nine months, October to June, is provided in Figure 3.1. The plan is structured in terms of several stages, starting from documentation and design efforts, moving onto development sprints and system integration, and then finally to testing and deploying the developed application. The major milestones have been identified such that each stage occurs after the completion of its preceding dependent task. The last two milestones of system integration and the developed application are planned towards the later part of the schedule [18].

Tasks	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
SRS Document									
Design Document									
Project Plan									
Product Backlog									
Sprint Backlog									
Interface Mockup									
Prototypes									
Database									
Integrated System									
Test Report									
Final Application									

Figure 3.1: Gantt Chart

In this project, there is a life cycle of nine months between October to June, in which the project will go through a phase of foundation planning that involves setting the

requirements and architecture using the SRS Document, Design Document, and Project Plan in the first two months. In the next phase of Agile Preparation, November to December months will be spent developing the product backlog, sprint backlog, interface mock ups, prototypes, and database structure.

This is followed by a phase called Implementation where integrated system development will be done within a span of four months, which would be the maximum duration in this phase, denoting coding and module development.

3.2.2.2 Work Breakdown Structure

The figure 3.2 represents the Work Breakdown Structure (WBS) for the current project, which acts as a hierarchy of all deliverables and activities required for the accomplishment of the software development life cycle. The WBS categorizes the project into smaller parts through the decomposition of higher level goals into specific activities that could also be further divided into sub-activities if necessary. All important deliverables, whether they are documentation-related artifacts such as SRS and design documents, or development-related stages such as database development and integrated system development, are aligned against time-specific activities over a period of nine months [19].

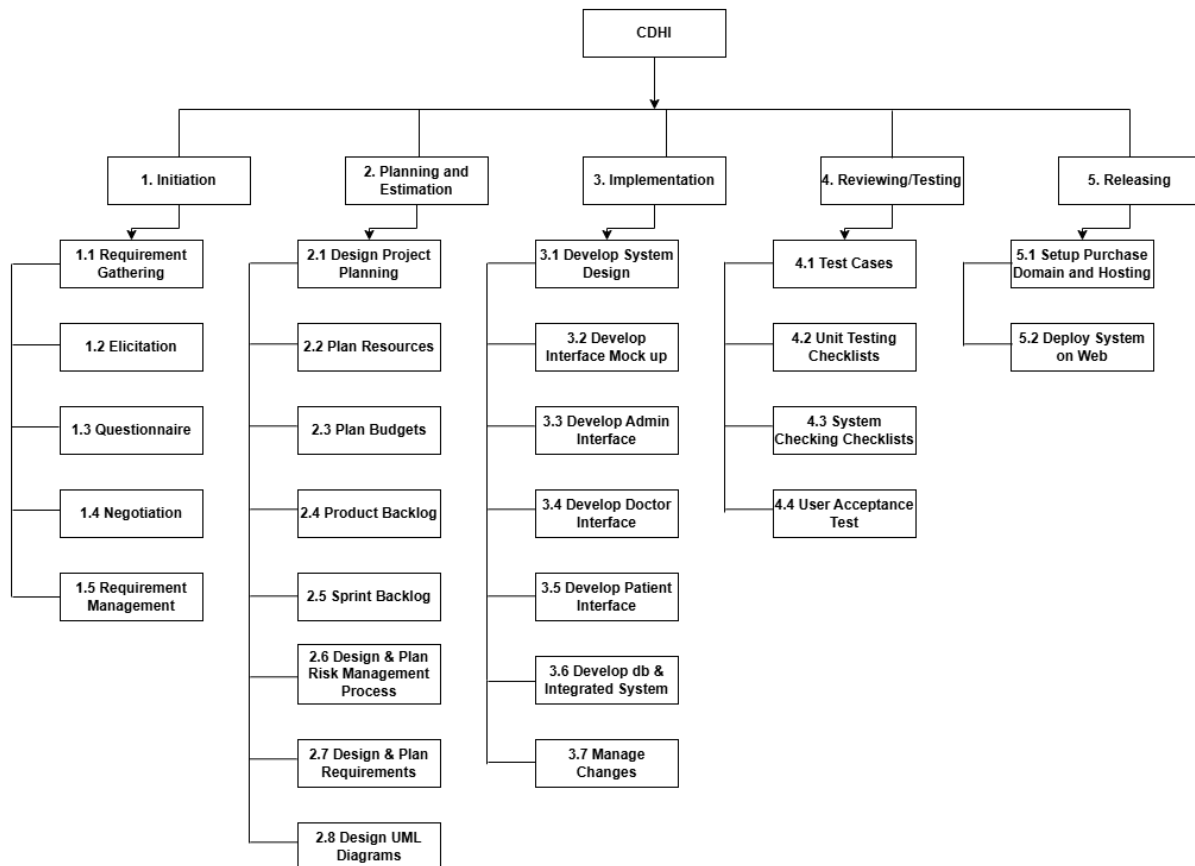


Figure 3.2: Work Breakdown Structure

Work Breakdown Structure for the CDHI Project shows the detailed hierarchical breakdown of a healthcare information system into five significant stages: Initiation,

Planning & Estimation, Implementation, Testing/Reviewing, and Releasing. In the Initiation stage, the basic requirements are set up using such activities as gathering, elicitation, questionnaires, negotiations, and management. The Planning & Estimation stage includes the longest number of tasks (eight) involved in planning, estimation, design of the project, resource and budget planning, Agile backlog setting, risk management procedures, and UML diagrams. In the Implementation stage, the plans become deliverable items in the form of system design, interface mock ups, three unique interfaces for administrators, physicians, and users, database building, and full system integration including changes management. Testing/Reviewing ensures quality assurance by developing test cases, conducting unit and system testing checklists, and user acceptance testing. The process of releasing the system will cover acquisition of domain name, hosting arrangements, and website publishing. Thus, this approach represents the hybrid Agile/Waterfall methodology with 26 work packages that allow precise resource management and control over the entire nine-month development period.

3.2.2.3 Critical Path Method

Figure 3.3 shows the critical path method [15] of the project. The Critical path is the longest sequence of dependent tasks in a project. By identifying the critical path, we can more easily prioritize tasks, build realistic schedules, keep projects within budget, and reduce or offset delays [20].

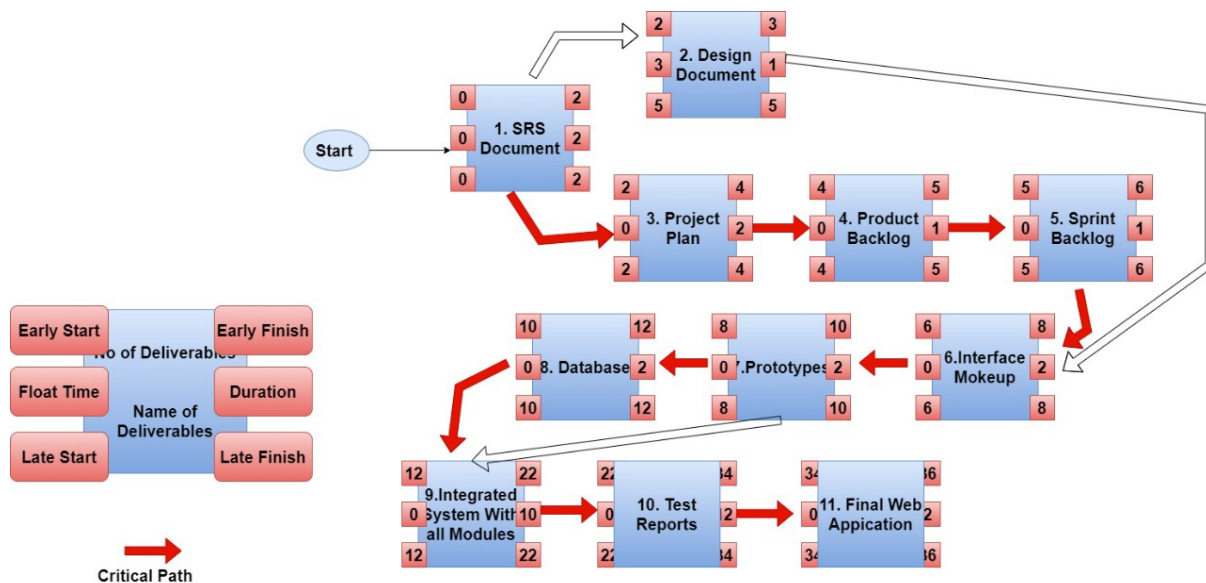


Figure 3.3: Critical Path Method

The above is a network representation of the Critical Path Method of a project for software development that comprises 11 activities. Each node contains the data of Early Start and Finish, Late Start and Finish, Duration, and Float Time. The project begins by breaking down the SRS Document into two separate paths where one path includes Design Document (non-critical path) while the other path is shown in red color arrows and represents the critical path. It begins from the SRS Document to the Final Web Application and takes up 36 time units.

The activities on this critical path have no float; hence, any delays result in the

ultimate delay in delivery. The legend helps us understand the position of each of the nodes used; Early Start at top left, Early Finish at top right, Late Start at bottom left, and Late Finish at bottom right, while Duration and Float time are along the sides. The thin arrows represent non-critical activities that can be delayed in scheduling since they have some float. Critical path diagrams are used by project managers in determining which activities require close supervision and the slack period that can be utilized for reassignment of resources. It also helps in determining the minimum time the project will take.

3.3 Managerial Process

Management is an organized approach towards planning, coordinating, and controlling activities inside an enterprise in order to accomplish certain objectives in the most efficient way possible. An important part of the management process is project management, which involves the utilization of certain knowledge, techniques, and skills in ensuring that a particular project is completed successfully. A project can be defined as an endeavor which aims at developing something unique, either tangible or intangible.

Project management is vital in ensuring that the tasks involved in the project are completed within the stipulated time frame, budget, and other quality parameters. Through project management techniques, the chances of success can be increased as well as the satisfaction of stakeholders and quality outputs can be achieved through the proper utilization of resources.

3.3.1 Management Objectives and Priorities

Project management is considered to be a key factor in the success of a software project that is why a project manager needs to have all the sufficient knowledge along with enough managerial experience to efficiently carry out all the managerial process [21].

1. **Planning:** The first objective in project management is planning, which involves establishing the project initial scope and allocating all essential resources, and dividing larger deliverables into smaller ones. determining the tasks required to achieve the deliverables, estimating how long each activity will take, identifying all potential risks and their effects, and creating a strategy to reduce or manage those risks.
2. **Execution:** After the planning phase, it is crucial to carry out all the duties outlined at this stage successfully. The project manager must take part in this process and monitor the efficient completion of all project tasks.
3. **Observe and Manage:** To ensure that project stays on track during execution phase, it is crucial to monitor all project tasks. Changes in scope, schedule, and money must be inquired and controlled by the project manager to ensure that they do not conflict with customer requirements. Additionally, keep an eye on and manage any risks that arise so they do not have an impact on the project.
4. **Closing:** The managerial process also includes ensuring that project is delivered on time and that its goals and objectives have been met. Making sure that the project deliverables have been completed according to schedule is another focus of this process.

3.3.2 Assumptions and Constraints

Constraints and project assumptions [22] are listed at the start of the project. They will be improved upon and updated over the duration of the project. Assumptions are expected future occurrences or circumstances over the life of your overall project. By using your existing knowledge or the information currently at hand, you develop these assumptions. Project constraints are restrictions placed on the project, such as the money, schedule, or available resources. The general constraints, dependencies and assumptions imposed on the implementation of project.

Objectives:

1. The project's tasks and subtasks being carried out successfully.
2. Effective communication inside and between the project team or teams.
3. Comply with all project requirements within the allotted time.
4. The project's resource allocation has been optimized.
5. Hand off a project that complies with the customer's wants and expectations.

Priorities:

1. Ensures that the project quality objectives are then met at all times.
2. The project deliverables are a must to be finished according to schedule.
3. The project is supposed to be completed on schedule and within the allotted spending limit.
4. Neither the project's schedule nor its scope should be compromised.
5. Keep track of every project milestone, deliverable, and change request.

Assumptions:

Following are a few assumptions that were made from a project management standpoint:

1. We have all the materials needed to complete and submit the project on schedule.
2. Team members are fully equipped with the skills and expertise required to create a web-based solution.
3. The system can be developed and deployed on schedule with just three members of the group.
4. Throughout the project, the project's scope won't change.
5. Domain and cloud hosting costs won't change throughout the project.

Constraints:

In our AI-based web page, the primary goal is to develop an AI-driven image-processing model for analyzing histopathology slides. Because the system depends on deep-learning techniques and real-time diagnostic analysis, we built the backend and system logic in C-sharp, avoiding platforms such as WordPress or PHP frameworks.

Implementing an AI model for medical image analysis is a complex, computation-intensive task that requires expertise in data preprocessing, model integration, and visualization. However, despite the need to invest extra effort in learning and implementing the method, it provides greater control and flexibility for the system as a whole.

3.4 Project Risk Management

Risk refers to the chances of uncertainty and novelty occurring which will impact the scheduling, quality, and efficiency of the software being created. The process of project risk management involves identifying potential hazards that could arise throughout the life cycle of project and developing a plan for how to handle them in order to keep the project on track and to achieve its intended purpose.

3.4.1 Risk Management Plan

The risk management plan [23] contains an investigation of probable risks with both high and low impact, and also some mitigation strategies to assist the project avoid being delayed. A risk management plan makes us to see further possible risks and decrease their negative impact. Table 3.7 show the project risk management plan. It describes the risks within the project and category of each risk. It also describe the probability of occurrence of each risk.

Table 3.7: Project Risk Management

Serial #	Risk	Category	Probability
1	Fail to meet the requirements	Project Risk	30 %
2	All members have sufficient development but lack experience	Known Risk	30 %
3	Insufficient training to use latest tools	Known Risk	20 %
4	Non responsive unsupportable software	Technical Risk	20 %
5	Change in requirements may occur	Project Risk	40 %
6	AI model integration or accuracy issues	Technical Risk	5 %
7	Mastering python libraries and flower for FL which is a new technology may require alot of time	Technical Risk	5 %
8	Shortage of financial resources	Business Risk	10 %
9	Less reuse than planned	Project Risk	40 %
10	Possible minor budget overrun	Project Risk	20 %

There are 10 risks in four main risk categories: Project Risks, Known Risks, Technical Risks, and Business Risks. The high probability risks of 40% include change in requirements and less reuse than planned, which are both Project Risks, signaling the uncertainty in the scope. Both failing to meet the requirements and team member lack of experience are estimated to have 30% probability, emphasizing the difficulties of both the deliverables and the team resources.

AI model integration problems and learning federated learning libraries (e.g., Flower) are technical risks with 5% probability, showing that these risks are not very concerning for the team. Insufficient tool training and software non-responsiveness have the same probability of 20%, and there may be a small budget overflow. The sole business risk that has 10% probability is insufficient financial resources.

3.4.1.1 Purpose

The purpose of risk management is to recognize possible issues before they arise or, in the case of opportunities, to attempt to take advantage of it in order to make it happen. Risk-handling methods may be used at any time over the course of the project. A risk management plan has the following important functions:

1. Finding possible risks that might have an impact on project, such as adjustments to the scope, budget, or timeline, is helpful.
2. Prioritizing risks according to their likelihood and impact allows resources to be concentrated on the greatest hazards.
3. It offers a methodical way for coming up with and putting into action measures to reduce or get rid of risks that have been recognized, like making backup plans or assigning more resources.
4. By doing this, it is possible to make sure that the project team is informed about potential hazards and equipped to handle them should they arise.
5. It offers a structure for continual risk monitoring and management throughout the project, enabling necessary corrective action to be implemented.
6. Risk management plan overall goal is to maximize likelihood that project will succeed by identifying and mitigating potential risks before they have an adverse effect on project.

3.4.1.2 Roles and Responsibilities

Delegation of certain tasks and responsibilities assigned to individuals working in an organization or team is known as role and responsibility. It includes tasks associated with a certain role or position and even responsibilities which can be imposed upon all team members.

Managing project management, conducting meetings, decision making, and teamwork with fellow team members are some of the tasks and responsibilities. For individuals to

play their role in meeting organizational objectives, they should be well aware of their roles and responsibilities.

Table 3.8 depicts roles and responsibilities as requirement gathering, developing SRS document, developing project plan, creating interface mock ups, completing all the 12 modules and then deploying the system .

Table 3.8: Roles and Responsibilities

Resource Name	Title	Email	Project Roles	Responsibilities
Abdul Rafay	Group Member	2280101@szabist-isb.pk	Requirement Engineer, Designer, Developer and Tester	Requirement Gathering, Develop SRS Document, Develop Project Plan, Develop Interface Mockups, Complete all modules, Deploy system.
Aman Siddiqui	Group Member	2280104@szabist-isb.pk	Requirement Engineer, Designer, Developer and Tester	Requirement Gathering, Develop SRS Document, Develop Project Plan, Develop Interface Mockups, Complete all modules, Deploy system.
Rutaba Sajid	Group Member	2280123@szabist-isb.pk	Requirement Engineer, Designer, Developer and Tester	Requirement Gathering, Develop SRS Document, Develop Project Plan, Develop Interface Mockups, Complete all modules, Deploy system.

The team comprises three individuals who enjoy an equal status within the team with neither any leadership nor any hierarchical structure. Each of the three individuals has been assigned similar roles, comprising all of software development, such that they are not only Requirement Engineers but also Designers, Developers, and Testers.

3.4.2 Risk Management Activities

The processes used to detect, evaluate, and mitigate risks that could possibly have an impact on a success of project are known as risk management activities. These actions are usually taken as part of a risk management strategy, and they could include the following:

1. **Identifying risks:** This involves locating potential dangers that might have an effect on this project, like adjustments to scope, budget, or timeline. Meetings could be held with engineering team to brainstorm ideas with the project team, study of project materials, or stakeholder consultation.
2. **Assessing risks:** It involves ranking them according to the statistical likelihood and severity of the impact of each detected risks and their possible detrimental effect on overall life cycle of project. To assess the likelihood and impact of hazards, might use different quantitative analysis methods or generate a risk assessment matrix.
3. **Creating risk response plans:** This involves dealing with ways to reduce up coming risks or eliminate already recognized risks, such putting together a backup plans or dedicating more resources to the project.
4. **Putting risk response plans into action:** This execution phase involves actively carrying out the risk response plans created in the previous phase.
5. **Monitoring and managing risks:** This governance loop involves continuously keeping track of and managing any emerging hazard during the entirety of the project and taking necessary corrective actions. This could further involve conducting routine risk analyses, revising the risk management strategy, or putting new risk response techniques into place.

3.4.2.1 Risk Identification The analytical process of identifying prospective hazards that might have an impact on a project's success is formally known as risk identification. As it enables the project team to proactively address risks before they have an impact on the project, it is undeniably crucial step in the risk management process. Risk identification can be done in a number of ways, including:

1. **Interviewing:** You can interview the stakeholders, project participants or experts to identify risks. In contrast with the collaborative nature of brainstorming, since you leverage specific direct questions about the project and usually conduct the interviews one-on-one to extract insights.
2. **Brainstorming:** This process involves talking about and collectively identifying potential hazards with the entire project team and key stakeholders.
3. **Examining project documents:** The analytical approach helps detect potential hazards include looking over project plans, timelines, budgets, and other project documents to find hidden structures.
4. **Consulting with specialists:** This involves asking for subject-matter experts' opinions or those of other industry professionals who may be able to spot potential problems.
5. **Using risk assessment tools:** A risk assessment matrix or a SWOT analysis are just a couple of the structured tools and methods that may be used to detect risks systematically. Once possible risks have been discovered, they can be evaluated, given a priority ranking based on likelihood and impact, and concrete strategic mitigation measures to reduce or eliminate them can be stepped up and put into immediate action.

3.4.2.2 Risk Analysis The systematic process of assessing the statistical likelihood and impact of operational prospective hazards that could have an impact on a project's success is formally known as risk analysis. The project team can use it to priorities risks and concentrate on the most relevant ones, making it a crucial stage in the overarching risk management process. Consequently effective risk analysis can be done using a variety of techniques, such as:

1. **Bow tie analysis:** Bow tie analysis [24] is a highly visual risk analysis method used to manage and reduce risk proactively. You begin this process by observing a potential risk and splitting it into two distinct categories: one that includes all the potential contributing factors and one that lists all the potential consequences. From there, you can create a new mitigation practice that directly address each of the potential causes and consequences.
2. **SWIFT analysis:** The SWIFT approach [25], or "structured what-if technique," includes a team-based approach executed in a workshop environment. This process involves critically assessing how changes from a newly implemented plan or design may affect a potential project life cycle. Using speculative "what if" questions, teams can determine how changes may affect a project or initiative down the road.
3. **Probability/consequence matrix:** The probability matrix method [26] is a prominent method of assessing risk severity when conducting a qualitative risk analysis, and it can assess risks at all levels of the company infrastructure. The probability matrix is a practical method of ranking the severity of various risks by calculating the risk's potential impact mathematically. Not only does this allow companies to better determine the overall severity of a risk, but it also allows you to determine the main contributing factors to each potential risk so resource can be allocated.

3.4.2.3 Risk Response Planning The process of creating choices and then choosing courses of the actions to increase the opportunities and lower the threats to the project's goals is known as risk response planning. It comprises systematically identifying and assigning parties or individuals to be in charge of each agreed-upon risk response. This accountability procedure makes sure that hazards are identified and adequately managed throughout the project life cycle. Risk response planning can be done in a number of ways, including:

1. **Accepting the risk:** In this case, the identified danger is accepted and no further action is taken by the project team. For low-impact threats that are statistically unlikely to arise, this passive strategy might be suitable.
2. **Reducing the risk:** This involves taking drastic operational steps to do so, such as abandoning the project or modifying its scope fundamentally. For hazards with a high impact that cannot be addressed by standard mitigation, this method might be suitable.
3. **Passing the risk:** This entails transferring the risk to a different party, for example, by using contracts or insurance.

4. **Mitigating the risk:** This is taking measures to lessen the impact or possibility of the risk, such as putting emergency plans into place or dedicating more resources.
5. **Risk Acceptance with Backup Planning:** while taking efforts to reduce it, such as putting in place backup plans or dedicating more resources, is known as accepting the risk while reducing it.

3.4.2.4 Rating Risk Likelihood and Impact

Risk management is an essential part of software project planning that helps in identifying, analyzing, and controlling potential issues that may affect project success. Each risk is evaluated based on two main factors: probability and severity.

Table 3.9 presents a structured assessment of identified project risks, enabling better planning.

Table 3.9: Rating Risk Likelihood and Impact

Serial #	Risk	Category	Probability	Severity	Impact
1	Fail to meet the requirements	Project Risk	30 %	Catastrophic	Project success
2	All members have sufficient development but lack experience	Known Risk	30 %	Critical	Project and cost
3	Insufficient training to use latest tools	Known Risk	20 %	Marginal	Project Scheduled
4	Non responsive unsupportable software	Technical Risk	20 %	Catastrophic	Project Success
5	Change in requirements may occur	Project Risk	40 %	Critical	Project Scope
6	AI model may give inaccurate predictions due to low-quality or unclear histopathology images.	Technical Risk	5 %	Marginal	Project Success

Serial #	Risk	Category	Probability	Severity	Impact
7	Shortage of financial resources	Business Risk	10 %	Critical	Project Success
8	New technology may require extra time	Technical Risk	5 %	Negligible	Project Scheduled
9	Possible minor budget over-run	Business Risk	20 %	Negligible	Project Success

Risk factors are provided with their respective categories, likelihoods, severity, and effects on the project. Both technical and project risks have been taken into account during the analysis process. These risks include not meeting the user needs, lacking the required knowledge, changes in requirements, and constraints in the system. The probability of occurrence and severity of each risk factor have been analyzed. Severity levels vary from insignificant to catastrophic. How each risk would affect the project, cost, duration, and scope have also been presented.

3.4.2.5 Risk Monitoring and Control

Risk monitoring and control [27] is process of identifying, analyzing, and then planning for newly discovered risks and managing it. Throughout the process, we have to track identified risks, reveal new risks, implement response plans, and measure effectiveness. The key point is only throughout this phase constant monitoring and due diligence is key to its success. Continuous evaluation ensures that risks remain within acceptable limits and do not negatively impact project objectives. It also helps in early detection of deviations so that corrective actions can be taken promptly. Effective risk control improves project stability, resource utilization, and overall project success rate.

1. **Risk Management Plan:** It gives details how to approach and manage the project risk efficiently. The plan describes that the how and when for monitoring risks provides guidance around budgeting and timing for risk-related activities, thresholds, reportings, and tracking.
2. **Approved Change Requests:** They are the necessary adjustments to work methods, all contracts, project scopes, and schedule. Changes can drastically impact existing risk and give rise to new risk. Approved change requests are need to be reviewed whether they will affect risk ratings and responses or if a new risks is a result.
3. **Performance Reports:** It paint a picture of the project's performance w.r.t cost, scope, schedules, resources, quality, and risks. Comparing actual performance against baseline plans may reveal risks which may cause issues in the future. It uses charts, tables, and histograms, to organize and summarize data such as earned value analysis and project work progress.

3.4.2.6 Risk Assessment

Software risk assessment [28] is a process of identifying, analyzing, and highlighting all the possible risks that may occur. As our project is a medium scale project so there is a chance that it will be influenced by multiple risks. During the risk assessment process, a solution or plan of action should be developed for all possible risks. Table 3.10 shows the probabilities, severity and impact of all the possible risk that may occur during the project.

Table 3.10: Risk Assessment

Serial No.	Risk	Category	Prob.	Severity	Impact Value	RM MM
1	Fail to meet the requirements	Project Risk	30 %	Catastrophic	1	Risk Monitoring
2	Non responsive unsupportable software	Technical Risk	20 %	Catastrophic	1	Risk Monitoring
3	AI model accuracy may drop if histopathology images are low-quality or poorly scanned	Technical Risk	5 %	Marginal	1	Risk Monitoring
4	All members have sufficient development but lack experience	Known Risk	30 %	Critical	2	Risk Mitigation
5	Change in requirements may occur	Project Risk	40 %	Critical	2	Risk Management
6	Insufficient training to use latest tools	Known Risk	20 %	Marginal	3	Risk Mitigation
7	New technology may require extra time	Technical Risk	10 %	Marginal	4	Risk Management

Serial No.	Risk	Category	Prob.	Severity	Impact Value	RM MM
8	Probability of Shortage of financial resources	Technical Risk	10 %	Negligible	4	Risk Management
9	Possible minor budget over-run	Project Risk	20 %	Negligible	4	Risk Management

The risk assessment table not only identifies the possible risks in the project; it also serves as a method for project management control. In addition, the use of the risk assessment table ensures that there is systematic analysis of risks based on standardized criteria such as probability, severity, impact value, and mitigation approach. As such, this tool enables the management team to differentiate between minor risks, which might cause distraction during the process of project implementation, and significant risks, which have the potential of derailing the entire project.

The advantage of the table is that it enables the identification of risks in terms of their priorities. For instance, risks that are highly likely and have catastrophic effects are classified as priority number one; therefore, they must be addressed immediately. On the other hand, low-probability and low-severity risks are also identified within the table; however, they are addressed using routine monitoring.

The inclusion of impact value provides a further level of analysis through transforming the risks from qualitative analysis to semi-quantification. This will help the team be able to analyze each risk without resorting to subjective analysis. The combination of all these factors can enable better forecasting of stability and resource needs for the project. In fact, even when the probability is low, risks can still have significant impact value.

The risk assessment process also facilitates proactive decision making throughout the project lifecycle. Early identification of risks enables the development team to plan contingency measures and use resources more efficiently.

Additionally, a formal framework for managing uncertainty is provided by the chosen risk management techniques, which include risk monitoring, mitigation, and management. These tactics aid in reducing the effects of unfavorable occurrences while upholding stakeholder expectations and project goals. Increased confidence in the successful implementation of the Cancer Detection via Histopathology Images system, enhanced project reliability, and better resource utilization are all facilitated by effective risk management.

Chapter 4

FUNCTIONAL ANALYSIS AND MODELING

4.1 Use Case Modeling

A helpful approach for gathering requirements is use case modeling [29]. It offers a graphic representation of a software system's requirements. The use case model is simple in both concept and presentation, making it reasonably simple to discuss its accuracy with non-technical staff (such as clients). A model that describes how the user interacts with the system to address an error is known as a "use-case model". As a result, the use case model comprises the end-users goal, the user and system communication, and the system behavior necessary to achieve these objectives.

4.1.1 User Stories

User stories are a fundamental cornerstone of Agile software development methodologies, providing a clear, concise and structured way to describe core system functionality directly from the user's perspective. They focus heavily on capturing what an end-user needs to achieve and the underlying business purpose behind it, offering valuable strategic insight into user expectations and guiding engineering development efforts toward real-world utility and tangible operational values. These stories are written in natural, accessible language, making them seamlessly understandable to both core technical teams and non-technical project stakeholders.

4.1.1.1 Admin User Stories

User and Role Management To strictly control system access and maintain robust data security, the admin manages users and assigns granular roles so that only authorized users can access critical system functionalities.

Acceptance Criteria:

- The system allows the admin to create, update, and delete users dynamically.
- User roles (Admin, Doctor, Patient) are easily accessible.
- Unauthorized access is strictly restricted.

Dataset Management To properly maintain high quality training and validation data for the AI classification model, the admin manages complex histopathology datasets.

Acceptance Criteria:

- The system allows seamless dataset upload, update, and deletion.

- Only the authenticated admin has access to dataset management module.
- Overall structural integrity of dataset is fully preserved.

Federated Learning Management To significantly enhance the cancer detection model and maintain strict patient privacy, the admin oversees federated learning where the global model is trained at multiple hospitals without exchanging the actual raw data.

Acceptance Criteria:

- A robust FL framework is provided by the system architecture.
- Model training is done locally using hospital data stores.
- Encrypted model updates are communicated securely to the central server.

System Monitoring and Audit Logs The admin regularly monitors system activities and detailed audit logs to ensure complete operational transparency and accountability.

Acceptance Criteria:

- The system automatically logs critical user action and system events.
- Logs include user identity, specific action, and timestamp.
- Audit logs is accessible only to authorized admins.

Dashboard and Analytics To closely monitor system performance and make informed strategic decisions, the admin views real-time analytics and KPIs across patients, doctors, hospitals, and AI model activity.

Acceptance Criteria:

- The dashboard displays dynamic KPIs including total patients, doctors, AI scans, detection accuracy metrix, and active users.
- Admin can filter analytics by date range, doctor, hospital, cancerous or non-cancerous classification and specific AI model versions.
- Charts are shown for performance, patient demographics, doctor trends.

Feedback Management To accurately evaluate user satisfaction and overall system quality, the admin reviews and manages feedback submitted by admins, doctors, and patients.

Acceptance Criteria:

- Doctors can submit feedback rating the AI result clarity, diagnostic workflow, efficiency and overall system usability.

- Patients can submit feedback rating the report quality, ease of access, and overall user experience.
- Admin can filter, search, and delete feedback records by role, keyword, and date range.
- Analytics display per-role satisfaction scores, per-question averages, and overall rating trends.

Notification Module To keep users informed of the system events, the system sends automated email notification alerts triggered by the specific wrong actions.

Acceptance Criteria:

- A 6-digit OTP is emailed to doctors and patients during registration and to doctors also during login for identity verification, expiring after 5 minutes.
- After signing up for doctor or patient, admin receives an email to either approve or reject them.
- Patients receive an email notification when their AI diagnostic report is ready for download.
- Notifications are sent via SMTP using TLS on port 587 and are skipped silently if the recipient email is invalid.

4.1.1.2 Doctor User Stories

Authentication and Secure Access To access diagnostic features securely, doctors authenticate themselves using valid credentials.

Acceptance Criteria:

- Doctors are authenticated using valid credentials.
- Access is role-based.
- Unauthorized access is denied.

Histopathology Image Upload To analyze tissue samples for cancer detection, doctors upload histopathology images for patients.

Acceptance Criteria:

- Doctors can upload images in supported formats.
- Each image is associated with a patient record.
- Uploaded images are securely stored.

Image Preprocessing To ensure accurate analysis, the system automatically preprocesses uploaded images.

Acceptance Criteria:

- Image resizing and normalization are performed.
- Noise reduction techniques are applied.
- Preprocessed images are used for analysis.

Cancer Detection Using AI Model To determine whether a tissue sample is cancerous or non-cancerous, the system analyzes images using an AI model.

Acceptance Criteria:

- Binary classification is performed.
- Prediction results are generated after processing.
- Acceptable response time is maintained.

Diagnostic Report Generation To document diagnosis results, the system generates diagnostic reports.

Acceptance Criteria:

- Reports include prediction results and patient details.
- Reports are viewable and downloadable.
- Access is limited to authorized users.

4.1.1.3 Patient User Stories

Secure Login To access medical information, patients securely log in to the system.

Acceptance Criteria:

- Patients are authenticated using valid credentials.
- Only patient can access their data.

View Diagnostic Results To stay informed about the health status, and patients view their diagnostic reports.

Acceptance Criteria:

- Patients view their own reports only.
- Results are displayed clearly and can be downloaded.
- Data privacy is maintained.

4.1.2 Individual Actor Use Cases

These cases presents the individual use cases of the HistoScanAI system. Each use case represents a major interaction between a specific actor and the system. The actors involved include Doctor, Patient, and Admin. For each actor, multiple use cases are defined to describe system behavior, functional responsibilities, and expected outcomes.

These structured tables enhance clarity, traceability, and validation of system requirements. A formal tabular representation of use cases, known as a use case table, is widely employed in system analysis, software design, and requirement gathering to document the behavior of system from the side of every actor.

4.1.2.1 High Level Usecases

The following Table 4.1 illustrates the use-case for doctor signup in the HistoScanAI system. It shows the process that allow a doctor to create an account by entering personal and professional information, which remains inactive until approved by the admin.

Table 4.1: Doctor Sign Up

Doctor Sign Up	
Use Case #	UC01
Use Case	Doctor Sign Up
Actors	Doctor
Type	Primary
Description	Allows a doctor to create an account by entering all credentials. The account remains inactive until reviewed by the admin.

Table 4.2 describes the actions admin can take to approve or reject a doctor's account, which either activates the account or requires resubmission.

Table 4.2: Admin Approves/Rejects Doctor Registration

Admin Approves/Rejects Doctor Registration	
Use Case #	UC02
Use Case	Admin Accepts or Deletes Doctor Registration
Actors	Admin
Type	Primary
Description	Enables the admin to review pending doctor registration requests and either approve or reject them. Approval activates the doctor's account; rejection requires resubmission.

In Table 4.3, the use case of doctor login is shown. It explains how doctors can access their dashboards using approved credentials.

Table 4.3: Doctor Login, Update Profile

Doctor Login, Update Profile	
Use Case #	UC03
Use Case	Doctor Login, Update Profile
Actors	Doctor
Type	Primary, Secondary
Description	Allows doctors to log in using approved credentials to access the dashboard.

Table 4.4 describes the use case for doctors to view assigned patients and access their history; reviewing previously uploaded reports and patient records.

Table 4.4: View Assigned Patients, Patient History

View Assigned Patients, Patient History	
Use Case #	UC04
Use Case	View Assigned Patients, Patient History
Actors	Doctor
Type	Primary
Description	Doctor views list of assigned patients and opens patient records to review previous reports, uploaded images, and history.

The use case presented in Table 4.5 details the workflow for uploading histopathology images and requesting AI-based analysis.

Table 4.5: Upload Histopathology Image, Request AI Analysis, AI Classification

Upload Histopathology Image, Request AI Analysis, AI Classification	
Use Case #	UC05
Use Case	Upload Histopathology Image, Request AI Analysis, AI Classification
Actors	Doctor, AI Model
Type	Primary
Description	Doctor uploads patient histopathology images. The AI model preprocesses images, performs classification, returns predictions with confidence scores, and stores results for doctor review.

Table 4.6 highlights the process of generating and uploading diagnostic reports. After reviewing AI outputs, doctors add clinical interpretation and recommendations, generating a final report that is uploaded to the patient portal.

Table 4.6: Generate, Upload Diagnostic Report

Generate, Upload Diagnostic Report	
Use Case #	UC06
Use Case	Generate, Upload Diagnostic Report
Actors	Doctor
Type	Primary
Description	Doctor reviews AI output, adds clinical interpretation and recommendations, generates final diagnosis report, and uploads it to the patient portal.

In Table 4.7, the patient-side use case for logging in and accessing reports is described. It shows how patients can view, download, and review current or historical diagnostic reports through the system.

Table 4.7: Patient Login, Views, Downloads Reports

Patient Login, Views, Downloads Reports	
Use Case #	UC07
Use Case	Patient Login, Views, Downloads Reports
Actors	Patient
Type	Primary, Secondary
Description	Patient logs in, views uploaded diagnosis reports, downloads current or final reports, and reviews report history.

Table 4.8 illustrates the use case of patient account management by the admin. It details the creation, assignment, updating, and deletion of patient records to maintain proper system organization.

Table 4.8: Admin Creates Patient Accounts, Manages Records

Admin Creates Patient Accounts, Manages Records	
Use Case #	UC08
Use Case	Admin Creates Patient Accounts, Manages Records
Actors	Admin
Type	Primary, Secondary
Description	Admin creates patient accounts, assigns them to doctors, updates patient profiles, and deletes records when necessary.

The use case in Table 4.9 covers dataset management by the admin. Includes uploading, removing, and organizing datasets required for AI model training or retraining purposes.

Table 4.9: Admin Dataset Management

Admin Dataset Management	
Use Case #	UC09
Use Case	Admin Dataset Management
Actors	Admin
Type	Primary
Description	Admin uploads, removes, or organizes datasets used for AI model training or retraining.

Table 4.10 provides the application of monitoring and logging by the administrator. It explains how the process is done for tracking the AI logs, doctor activity, and the patients' interaction.

Table 4.10: Admin Monitoring & Logs

Admin Monitoring & Logs	
Use Case #	UC10
Use Case	Admin Monitoring & Logs
Actors	Admin
Type	Secondary
Description	Admin monitors system activity, AI logs, doctor activity, and patient interactions to ensure transparency and compliance.

The use case in Table 4.11 explains how feedback is important to improve AI performance from doctors and patients. The table details the collection of feedback and its role in incremental retraining of the AI model.

Table 4.11: Feedback-driven AI Learning

Feedback-driven AI Learning	
Use Case #	UC11
Use Case	Feedback-driven AI Learning
Actors	Doctor, Patient, AI Model
Type	Secondary
Description	Doctors and patients provide feedback on AI predictions. System stores feedback and uses it to improve AI performance via incremental retraining.

Table 4.12 describes the system-generated notifications and alerts for doctors and patients. It highlights how doctors are informed when AI analyses are complete and patients are notified when new reports are uploaded.

Table 4.12: Notifications & Alerts

Notifications & Alerts	
Use Case #	UC12
Use Case	Notifications & Alerts
Actors	System, Doctor, Patient
Type	Secondary
Description	System generates notifications for doctors when AI analysis completes or reports are ready; patients receive alerts when new reports are uploaded.

Table 4.13 describes the Federated Learning module, highlighting how the admin initiates training rounds, hospitals contribute local model updates, and the central server aggregates them without exchanging any patient data.

Table 4.13: Federated Learning

Federated Learning	
Use Case #	UC13
Use Case	Federated Learning
Actors	Admin, System, Participating Hospitals
Type	Secondary
Description	Admin initiates a federated learning round; each hospital trains the local model on its own patient info and uploads only encrypted weight updates to the central server, which aggregates them using FedAvg without sharing any patient data.

Table 4.14 describes the Dashboard and Analytics module, highlighting how the admin monitors real-time KPIs, filters system-wide statistics, and views interactive charts across AI performance, patients, doctors, hospitals, and usage.

Table 4.14: Dashboard & Analytics

Dashboard & Analytics	
Use Case #	UC14
Use Case	Dashboard & Analytics
Actors	Admin, System
Type	Secondary
Description	Admin monitors real-time KPIs and views interactive analytics charts across AI performance, patients, doctors, hospitals, usage, and dataset growth, with filters by date range, doctor, hospital, and either cancerous or non-cancerous.

4.1.3 Use Case Diagram

Use-case diagram [30] describes the high-level functions and scope of the system and identifies the interaction between the system and its actors also describes what the system does and how the actors use it. Figure 4.1 shows the use case diagram of the whole system.

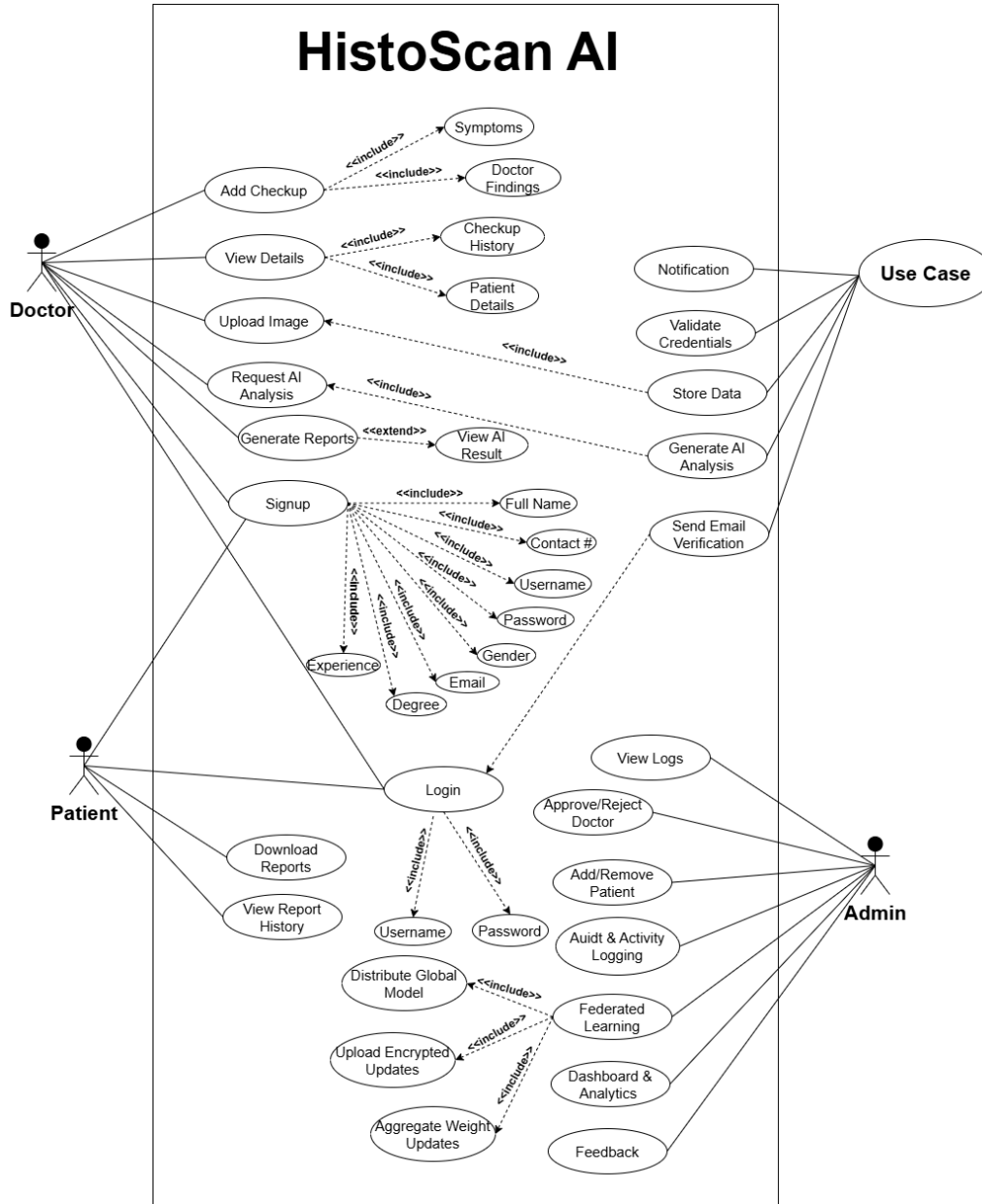


Figure 4.1: Use Case Diagram

4.2 Functional Modeling

Functional modeling [31] provides an overview of the system's intended functionality as well as an object-oriented analysis model's process perspective. Data Flow Diagrams (DFDs), Entity Relationship Diagram and Relational Schema are used to define the aim of the internal processes of system.

4.2.1 Entity Relationship Diagram

Figure 4.2 and 4.3 shows the entity relationship diagram [32] of the system. It depicts connections among people, objects, places, concepts or events . It describes the relationships between the entities. It also describes the attributes of each entity.

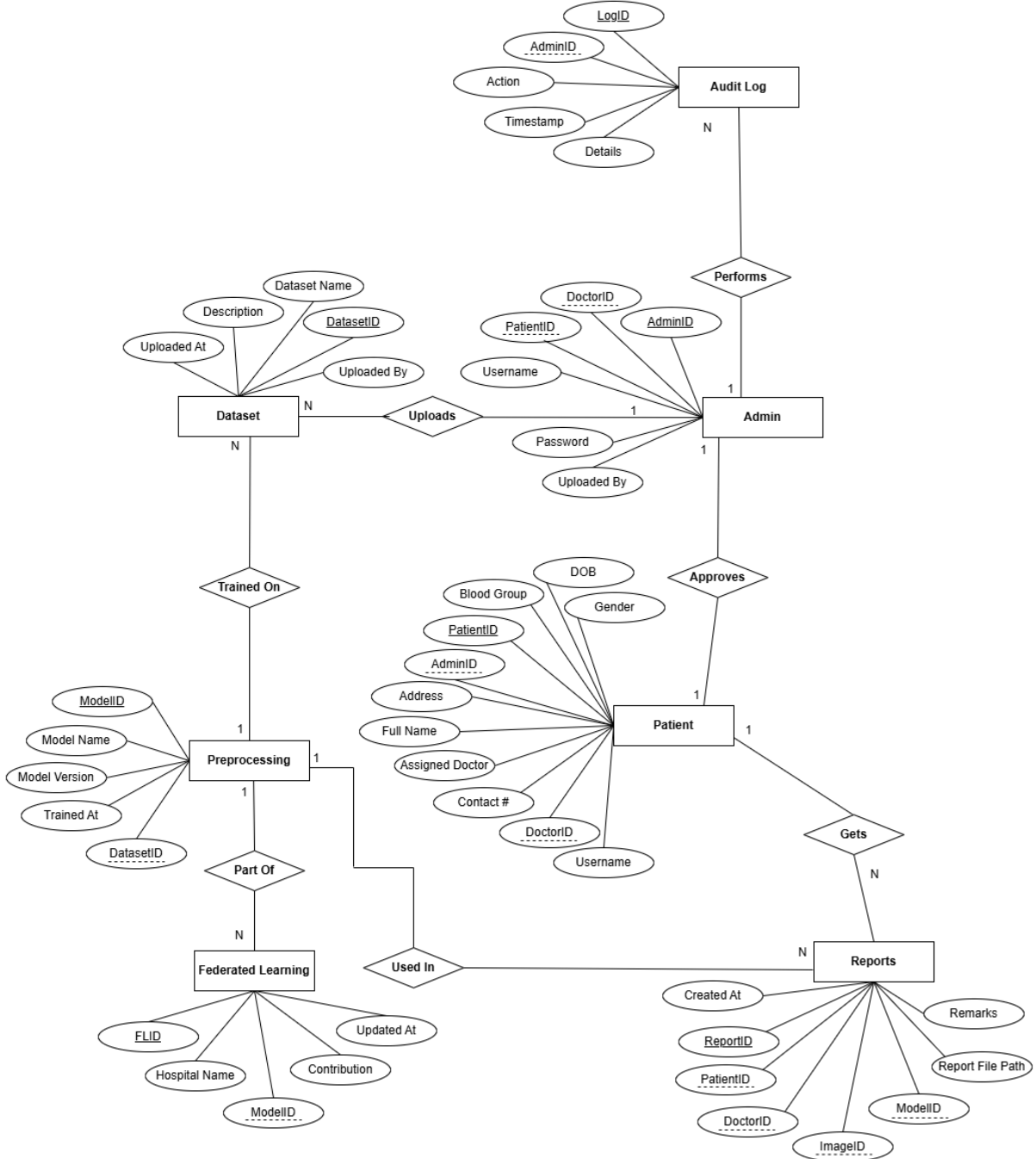


Figure 4.2: Entity Relationship Diagram 1

The ERD below deals with the administrative operations, data management processes, and artificial intelligence (AI) pipeline of the application. It gives a closer perspective to objects that were partially depicted in Image 1, to make the structure, attributes, and relationships among the objects in the system clearer.

The Admin entity is a crucial component in the governance and supervision of the system. It has attributes like Admin ID, Doctor ID, Patient ID, Username, Password and Uploaded By. These features enable the administrator to have control over the doctors and patients as well as ensuring safe authentication and data ownership monitoring. Each activity that the admin takes is logged in the Audit Log which forms a powerful system monitoring mechanism.

The relationships in the admin entity include Approves with Patient (one-to-one), which means that only a verified patient can access system services, and Uploads with Dataset (one-to-many), which means that one admin can upload various datasets to be used in training the model.

Audit Log entity plays a vital role in making sure the transparency, traceability and accountability of the system. It contains the following attributes: Log ID, Admin ID, Action, Timestamp and Details. Every record is documenting a particular event or operation by the administrator like account creation, record update, dataset upload or system settings changes. The timestamp is used to track in time order, whereas the details field is used to give a description of the action carried out.

Patient entity is the individuals who are provided with medical services and has detailed attributes, such as Patient ID, Admin ID, Doctor ID, Full Name, Username, Contact number, Address, Assigned doctor, Blood Group, Gender and Date of Birth (DOB). The qualities make sure that all personal and medical-related data is systematized.

Admin ID and Doctor ID have been included to create distinct relations between the patients, administrators, and healthcare providers so that they can be managed in a coordinated manner. Patients are linked to a doctor who does the diagnosis and treatment. Moreover, Patient entity is connected to Reports (one-to-many) which means that one patient may have a number of diagnostic reports.

The Dataset object is the set of medical data or images that are used to train and enhance the AI model. It has features like Dataset ID, Dataset Name, Description, Uploaded At, and Uploaded By. These properties aid in the identification, organization, and management of datasets effectively.

The Preprocessing entity is linked to the dataset via the Trained On relationship (many-to-one), which implies that several datasets can be used in the training of one preprocessing model.

The Federated Learning entity builds on the AI pipeline by making possible the training of multiple hospitals, without raw data sharing. It has the following attributes FL ID, Model ID, Hospital Name, Contribution and Updated At. This organization catches the contribution of various hospitals to the enhancement of the global paradigm without disclosing the data. The relationship between Preprocessing and Federated Learning (one-to-many) implies that the same preprocessing model may be used in more than one federated learning update.

The Reports entity is the last product of the whole diagnostic pipeline. It contains the following attributes, Report id, Patient id, Doctor id, Image id, Model id, created at, path to report file and remarks. All reports include data about various groups, such as

patient data, physician assessment, medical images added by patients, and predictions made by AI.

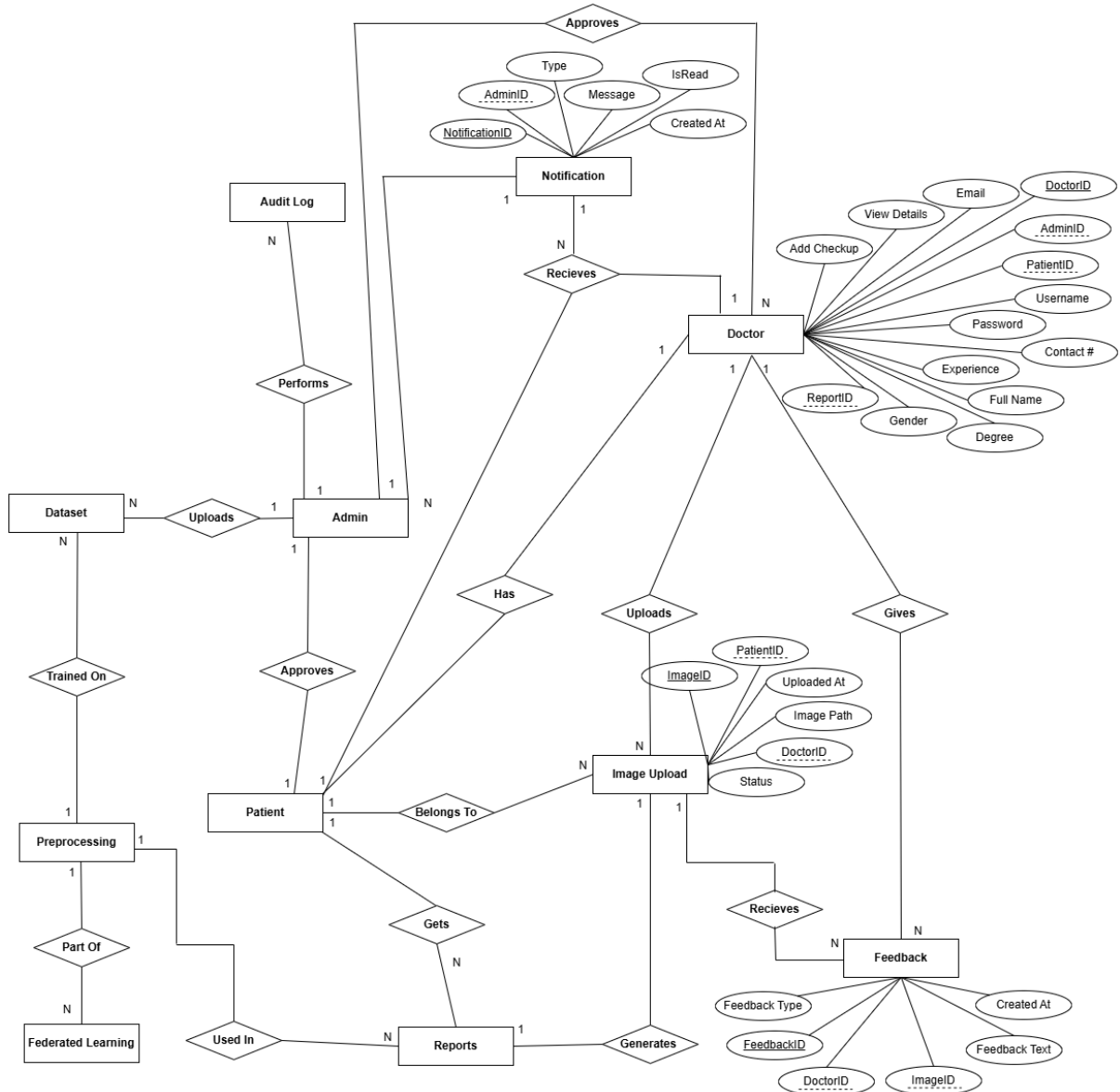


Figure 4.3: Entity Relationship Diagram 2

This portion of the ERD is the functional and interaction-oriented section of the application which is an extension of the initial diagram. Although the above image demonstrated the basic and backend architecture of the system, this figure shows the working aspect, especially the functions of the Doctor, Notification, Image Upload and Feedback in the real time application of the system.

The Doctor entity is the most active and core part of this section, as it is the most essential user of the system. It contains the following attributes: DoctorID, AdminID, PatientID, Username, Password, Email, Contact number, Experience, Full Name, Gender, Degree, ReportID. These characteristics characterize the identity and professional qualifications. The main system processes are View Details and Add Checkup through which the doctor can access the records of the patients and update medical observations.

The Doctor entity is linked to various components, and thus it is the central force of the diagnostic workflow.

The Image Upload entity is directly related to the Doctor via the Uploads relationship. The Image Upload table has the following attributes: ImageID, PatientID, DoctorID, Image Path, Uploaded At, and Status. Each image uploaded corresponds to a unique patient and is the main input to be diagnosed. This dynamic status attribute serves to precisely monitor the processing of each high resolution image (e.g. pending or analyzed) to enable the management of the workflow. Functionally this metadata tracking layer act as a critical abstraction interface between crude medical information and further processing.

Feedback entity is closely connected with Doctor and Image Upload that create a loop of continuous improvement. It has the following attributes: FeedbackID, DoctorID, ImageID, Feedback Type, Feedback Text and Created At. Feedback on particular uploaded images is given by doctors, which is stored and connected by the Receives relationship. Through this mechanism, expert input is enabled to increase the precision of the diagnosis and enhance system activities in the long-term.

The Notification entity helps in communications within the system, and makes sure that doctors are aware of major changes. It has properties like Notificationid, Adminid, Type, Message, Isread, and created at. Doctors receive notifications via a one to many relationship that facilitates effective delivery of system alert, updates, or critical messages. IsRead attribute assists in monitoring the acknowledgment of the notification.

Additional entities like Patient, Admin, Reports, Dataset, Preprocessing and Federated Learning belong to the rest of the system (as indicated in the first diagram) and are only briefly related here to preserve continuity, but their structure and functionality is largely defined in the preceding section.

4.2.2 Relational Schema

Figure 4.4 shows the relational schema [33] of the system. It describes relational tables and associated items that are related to one another. It describes the relationship among the entities and how the data is stored in database.

The following is an overview of the relational database schema for the back-end part of the HistoScanAI cancer detection system. The data organization is performed based on the primary system users: the admin, doctors, patients, and the AI system. The database includes three basic users of this system that are classified according to similar activities. The admin, doctor, and patient tables include personal data (profile, credentials, degree, blood type, etc.) for each user. The associations indicate which administrator authorized the particular user and which doctor is responsible for treating each patient.

The database also contains information on medical reports, uploaded images, and AI prediction results. Each patient entry is accompanied by one or several diagnostic reports to save the patient's medical history. Doctors may study these reports and give comments or confirm the results produced by AI software. Information on AI prediction is stored in the AI system table that includes prediction results, probability of the disease, timestamps, and other necessary data. Other important data for the system include

notification and audit logs.

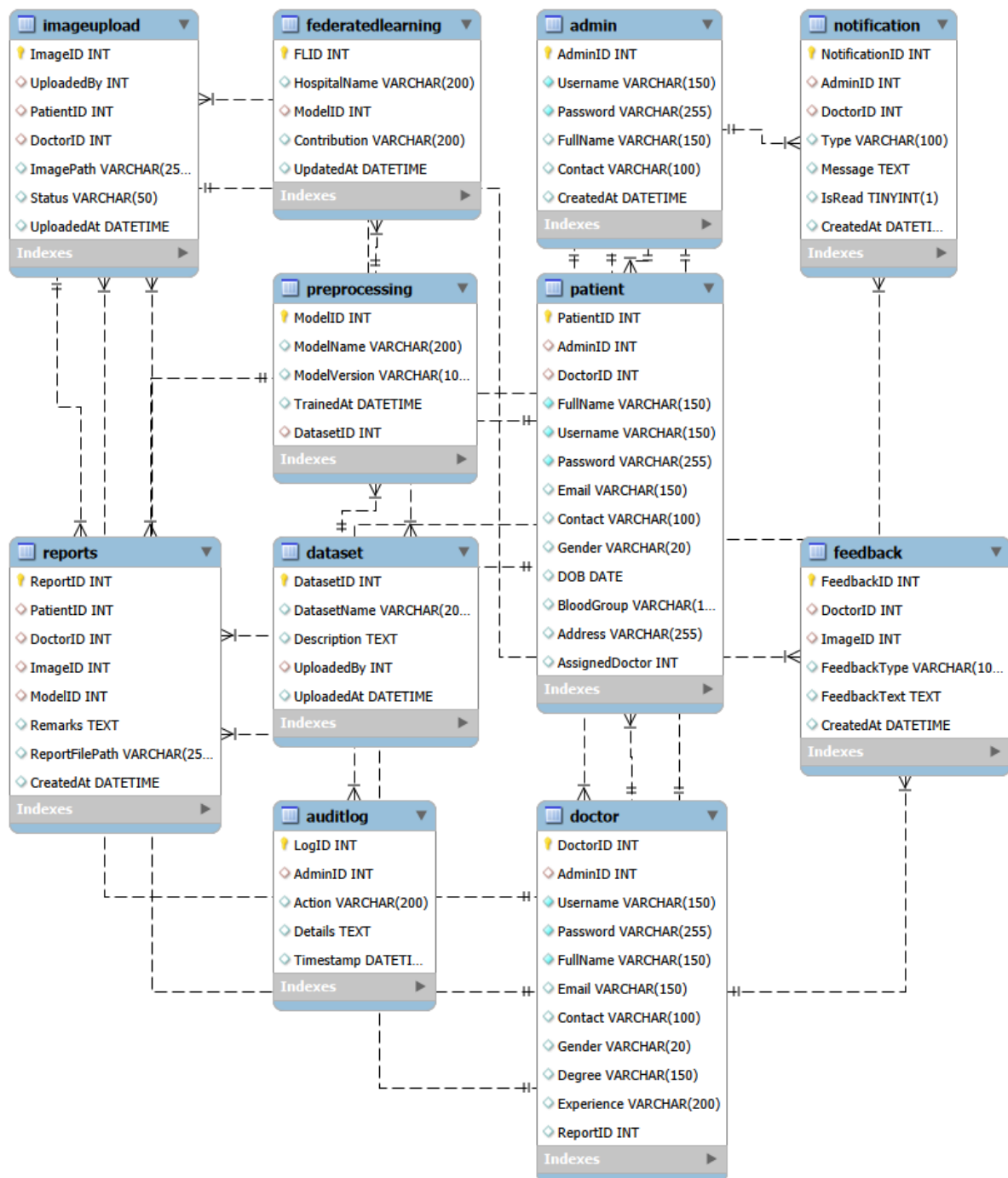


Figure 4.4: Relational Schema Of Entire Project

Our diagnostic workflow is stored in three tables: imageupload, reports and feedback. Starting with imageupload, this table logs the file path to the uploaded histopathology image, along with the relevant patient and attending physician. The file then is sent through the company's AI engine, for which the system logs the pre processing steps taken on the raw image, such as resizing and normalization. Following the AI's prediction, the file is logged in the reports table along with the subsequent observations of the attending physician and any additional comments made by the treating doctor. The feedback table

is where the attending doctor can subsequently correct any incorrect AI predictions and have that correction feed back into the system. Patients are also able to view their diagnostic reports in their online patient portal.

We also have two tables that support our machine learning infrastructure: the dataset and federated learning tables. The dataset table contains information about training data for our AI model, including which image collections were used for training and when they were uploaded by the user. federated learning allows us to train our model globally without transferring patient data to a central location, instead aggregating updates from individual hospitals. The federated learning table stores information about each hospital's participation in global model training and timing of local updates being aggregated into our central models. The pre processing table connects datasets to the models that were trained on them.

The schema is well balanced to support a robust medical AI platform. Data integrity is enforced with foreign keys. Patient privacy is ensured with federated learning architecture. Auditable record is maintained for compliance. The design is scalable to support data from multiple hospitals while maintaining confidentiality of images and reports with patient and doctor information linked correctly.

4.2.3 Data Flow Diagram

A data-flow diagram [34] is a visual representation of how data moves through a system or a process. The DFD additionally gives details about each entity's inputs and outputs as well as the process itself.

4.2.3.1 DFD Level 0

Figure 4.5 shows the DFD LEVEL 0 of the system. It is a baseline summary of the system or process being analyzed or modeled. This fundamental blue-print is carefully designed to be an at-a-glance view, showing the system as a single high-level process, with its relationship to external entities that interact or influence the core operation.

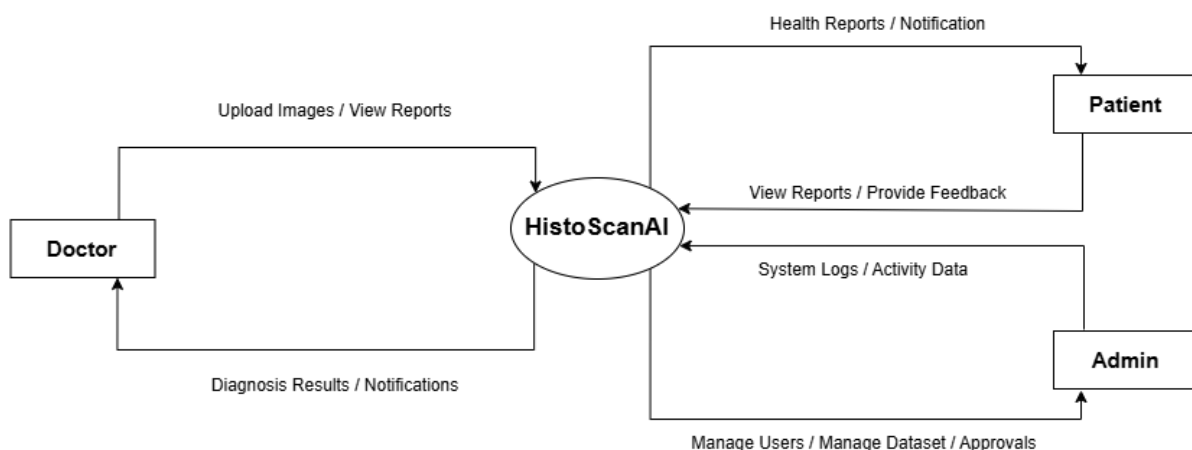


Figure 4.5: DFD Level 0

As shown in this figure, the center piece is HistoScanAI, which is connected to three main parties. Doctors will be able to upload images of tissues and subsequently receive

the diagnoses that have been performed by the AI algorithm. The patients will receive information about their health condition and updates via the application, and will also be able to give feedback about their experience using the application. Finally, the administrator will manage all users' accounts and monitor the application's activities using audit logs.

4.2.3.2 DFD Level 1

Figure 4.6 shows the DFD LEVEL 1 of the system. Each of the sub-processes that are primary are together making up the complete system is noted by a level 1 DFD.

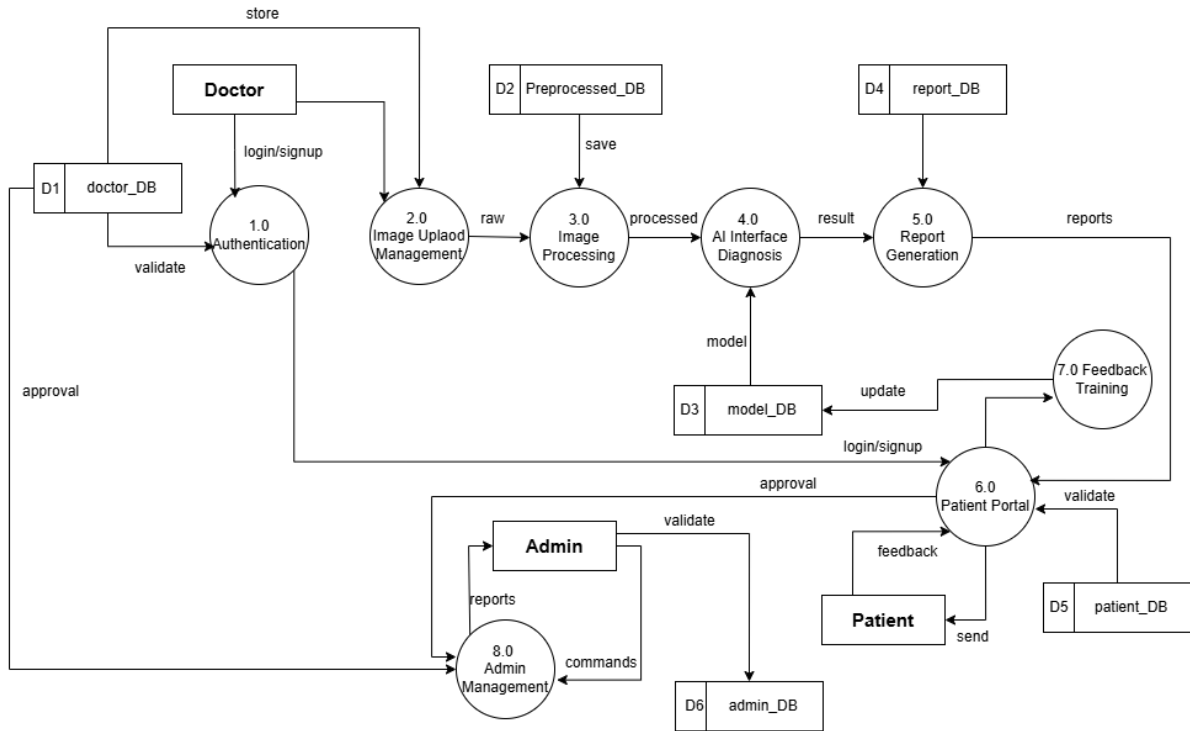


Figure 4.6: DFD Level 1

The figure shows how HistoScanAI diagnoses diseases based on the histopathology data from uploading the image to providing diagnosis. The doctors sign into the system through authentication, uploading the images that undergo preprocessing before being stored. The images are processed by the AI model that provides predictions about cancer based on the training of the model data in the model database.

The patients receive their report from the portal after signing in, from where they can download their data. Their feedback is collected and used to enhance the system through training the models.

The administrators run the system through user approval and activity monitoring to provide security to all participants.

4.2.3.3 DFD Level 2

Figures 4.7, 4.8, and 4.9 shows the DFD LEVEL 2 of system. Parts of level-1 DFD are further explored in level-2 DFD. It can be used to plan or keep track of the accurate or

vital information about how the system is working.

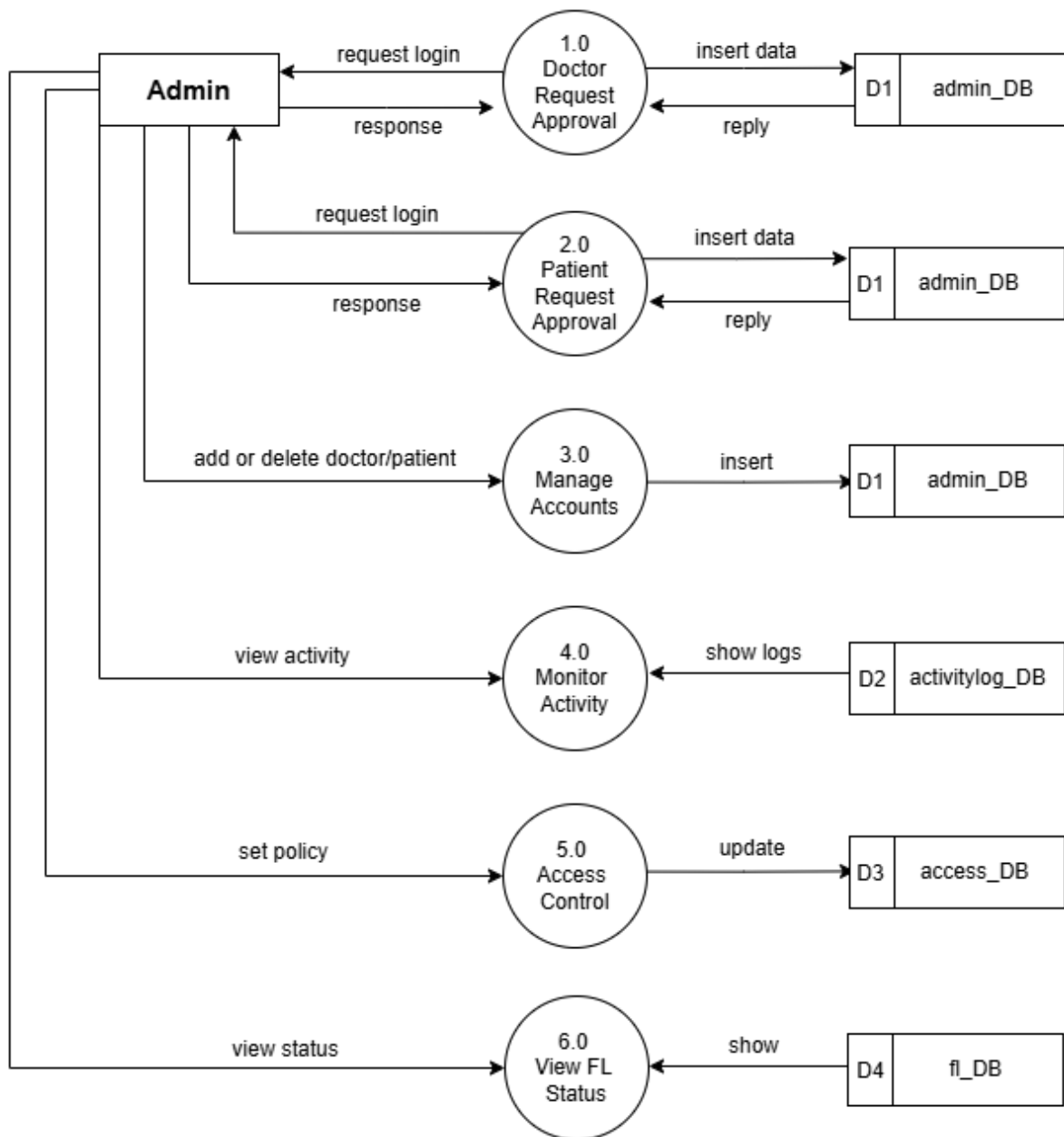


Figure 4.7: DFD Level 2 for Admin

The following figure illustrates the detailed workflow of the administration in His-toScanAI. In case a doctor or patient requests permission, the admin will evaluate and approve the requested accounts through process 1.0 and 2.0, and record their account details in the admin database.

Admins have the ability to either add or delete the account in the process 3.0, and retrieve the activity log information from the activity log database in the process 4.0. Using process 5.0, admins determine the access control policy to modify the access control database to authorize who will have access to the medical records.

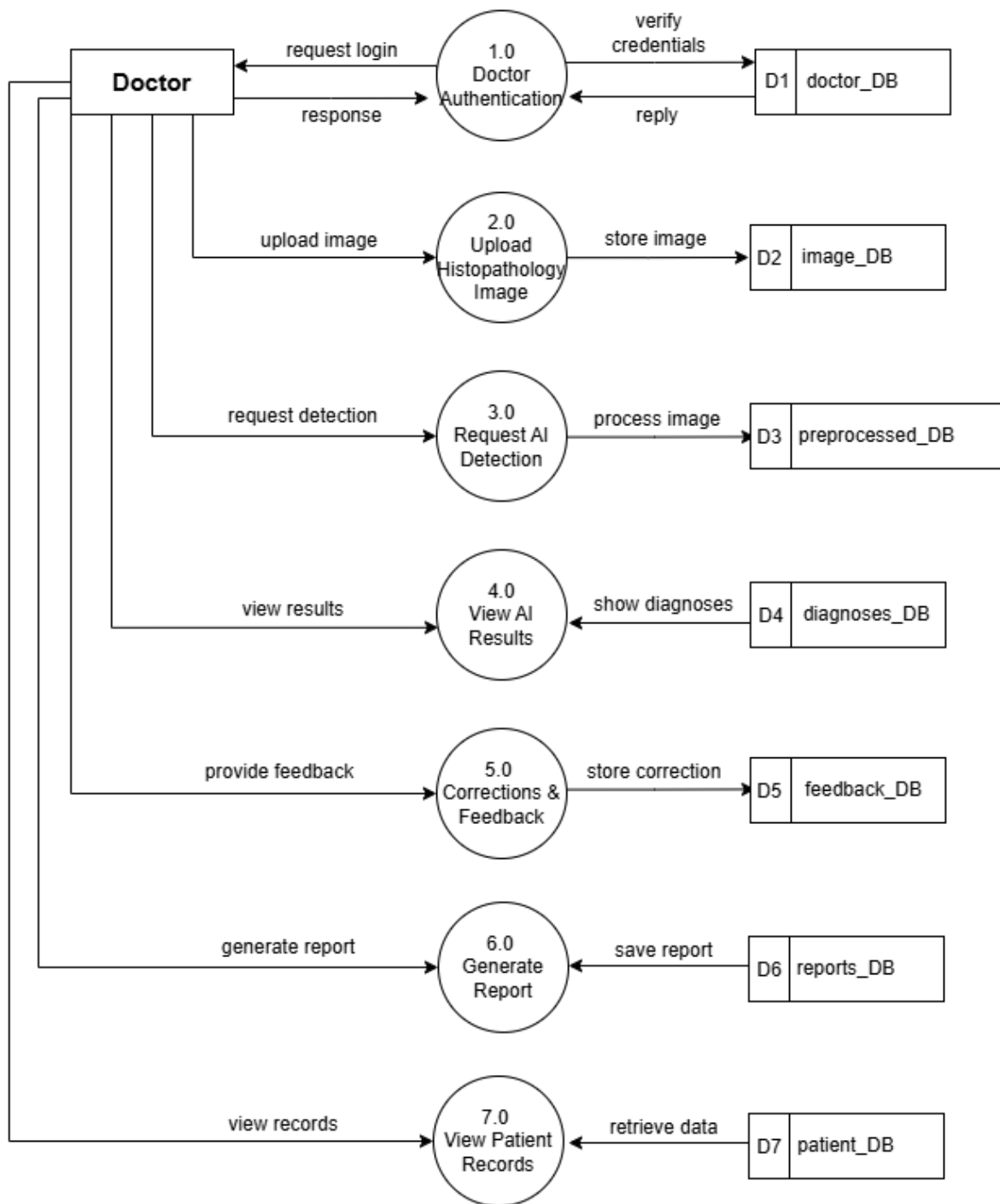


Figure 4.8: DFD Level 2 for Doctor

The workflow begins when doctors log in using process 1.0, after their credentials are checked to match those found in the doctor database. Upon successful authentication, the doctors use process 2.0 to upload the histopathology images. When they are ready for analysis, the doctors invoke process 3.0 in order to ask the AI to make detection predictions, whereby the images are processed and saved in the preprocessed database. The analysis having been done, the doctors use process 4.0 in order to view the results of the AI-based prediction, which have been obtained from the diagnoses database. If the AI-based prediction requires modifications, doctors may provide their feedback by

invoking process 5.0. Process 6.0 is responsible for the creation of reports. Lastly, doctors can view the entire history of patients by invoking process 7.0 in the patient database.

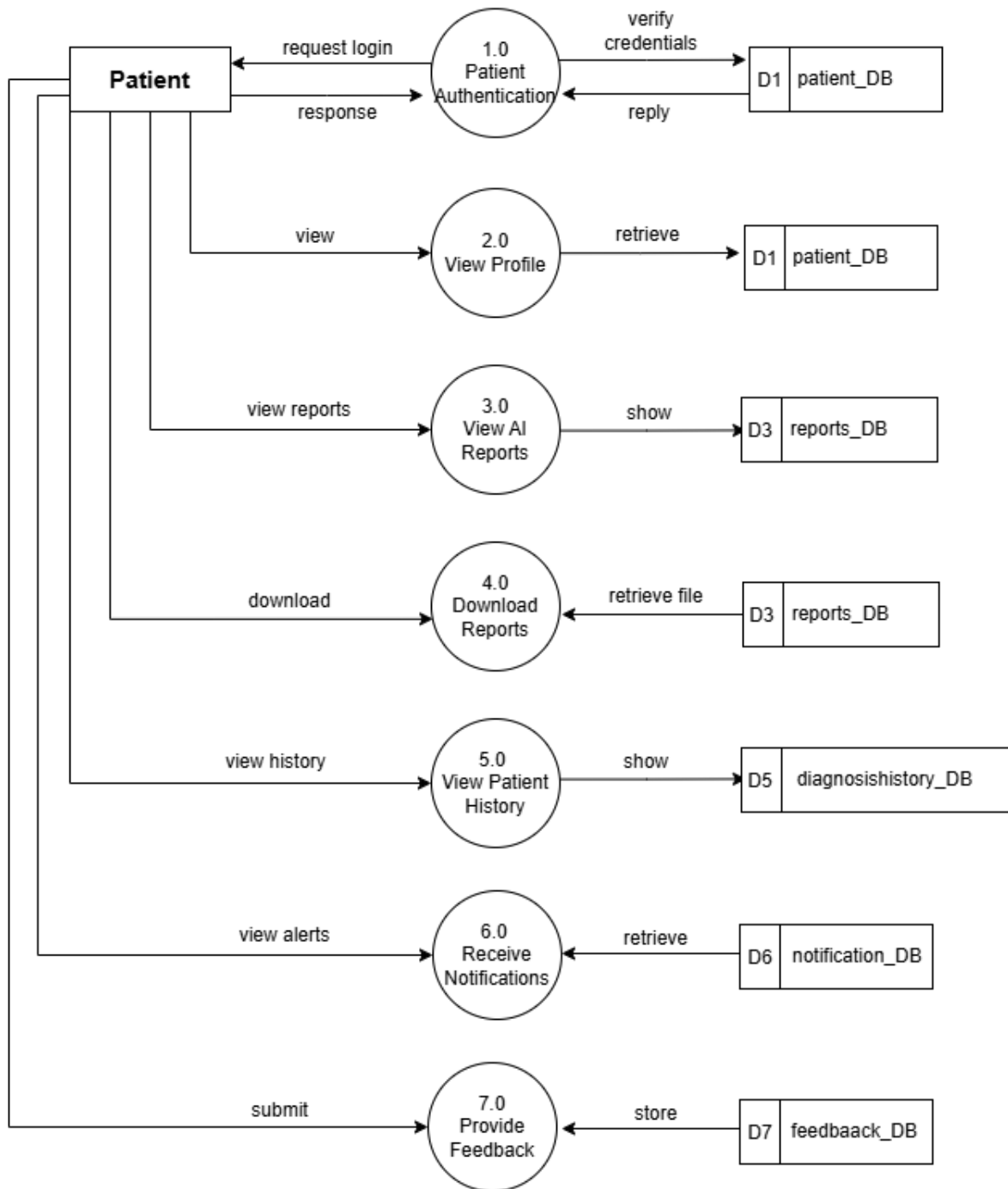


Figure 4.9: DFD Level 2 for Patient

The patients will sign in to check their own dashboard that displays their medical information and AI diagnoses report. The patients can even download these reports to view them later offline and look at all their previous diagnoses history from time to time. These alerts will be sent to them when there are any changes in their results or diagnoses via this platform. Moreover, patients can even provide feedback or raise concerns. Their activities will be saved in their own records only.

Chapter 5

SYSTEM DESIGN

System design marks a crucial phase where the conceptual vision begins to take a well-structured and implementable form. This stage focuses on majorly on outlining the complete systems architecture, including its key elements, modules, and how they communicate with each another. It not only defines the technical blueprint of the platform but also makes sure that each part of the system is inline with user needs, business objectives, and functional requirements. By translating intangible requirements into a tangible, organized framework, this phase serves as the baseline for a scalable, maintainable, and fast solution. Ultimately, system design fills the gap between planning and development, guiding the project from idea to reality.

5.1 Structure Diagrams

Structure Diagram is a conceptual modeling tool that is used to describe the different components that go into a system, like a database [30]. A structure diagram acts as a blueprint that exposes how different modules within a system are organized and linked with each other. It provides a visual layout of the system's architecture, making it easy to understand how each part plays a role to the overall functionality. Whether it's a database, or a website, the structure diagram shows the hierarchy, arrangements, and connections among the modules, ensuring that they work together seamlessly much like a well-fitted puzzle [35].

Structure diagrams serve as essential tools for illustrating the architecture of the platform. They offer a high-level overview of how important system components such as user modules, services, databases, and controllers interact to send core functions. These diagrams are absolutely instrumental in identifying complex relationships and hidden dependencies between components, streamlining overall system design, and ensuring highly coherent development. By offering a crystal clear detailed depiction of the system's backbone, structure diagrams support effective communication among developers, designers, and stakeholders, and guide the implementation of a stable, well-integrated solution.

5.1.1 Class Diagram

Figure 5.1 shows the class diagram of the system. It shows the structure of our system by displaying its classes, properties, operations, and connection between objects. The system includes 12 classes that stand for the fundamental entities of the AI-based histopathology diagnostics application. Admin acts as the base of the hierarchy and is designed for managing doctor approval, patient registration, notifications, and audit logging. The Doctor class contains the largest number of methods because it is utilized for image uploads, requests for AI processing, creation of diagnosis reports, and providing feedback. As far as the Patient class is concerned, it allows users to register within the system, examine their diagnosis, receive notifications, and download reports.

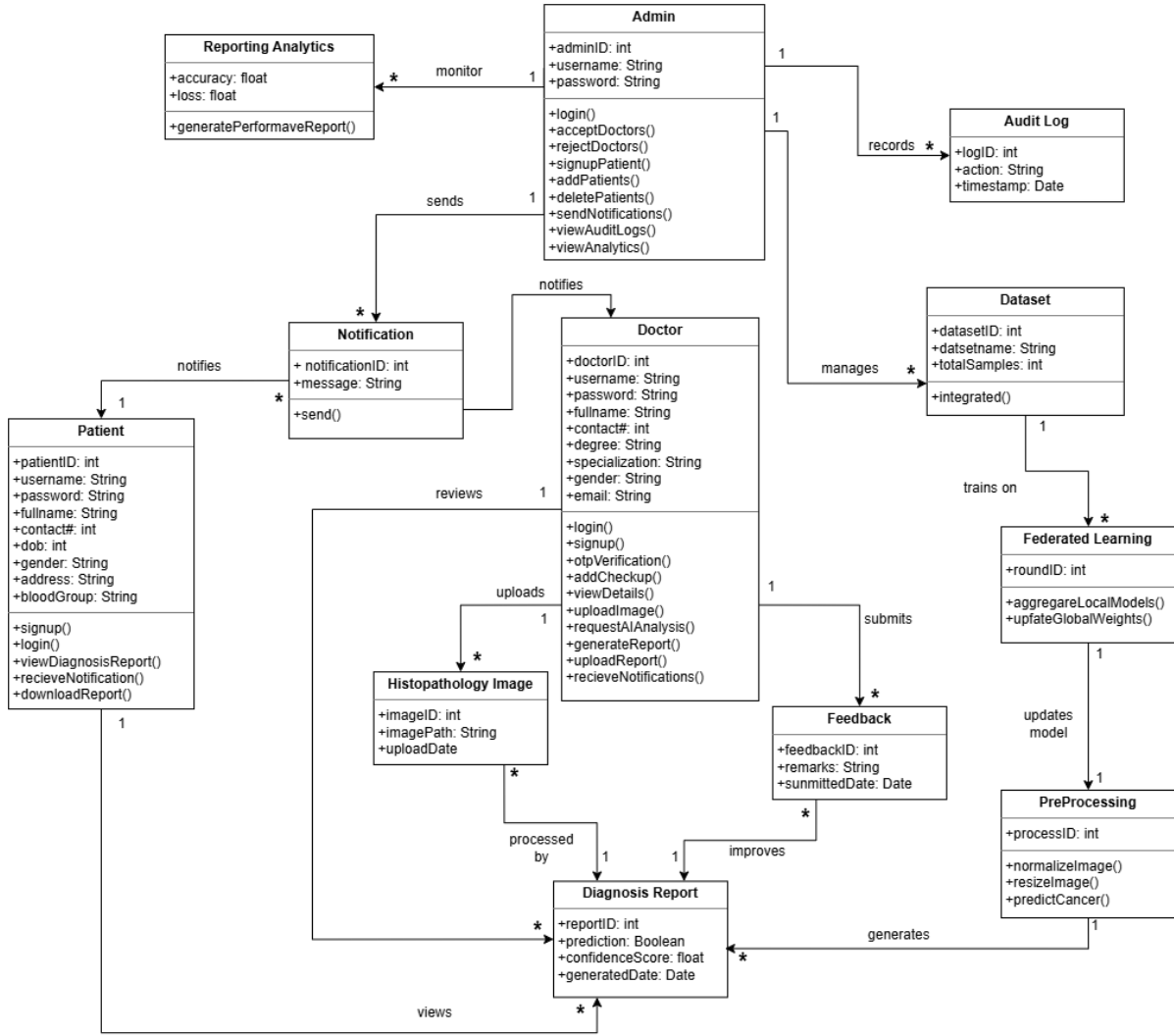


Figure 5.1: Class Diagram Of Complete System

Next, the Histopathology Image class is used to save information about images uploaded by users that are then pre-processed via normalization, resizing, and cancer prediction. Later on, the information obtained from prediction will be saved using the Diagnosis Report class. Additionally, doctors may offer feedback on the results of AI analysis through the Feedback class since such comments are useful for refining the model.

Finally, as for the infrastructure of AI, there are such classes as Dataset, Federated Learning, and Reporting Analytics utilized for processing the training dataset, training models, and conducting accuracy and loss analytics.

5.1.2 Deployment Diagrams

Figure 5.2 shows the deployment diagram of our system. It shows our system's execution architecture, showing nodes like hardware or software environments and the middleware that links them. It is visual representation of physical hardware and software of our system.

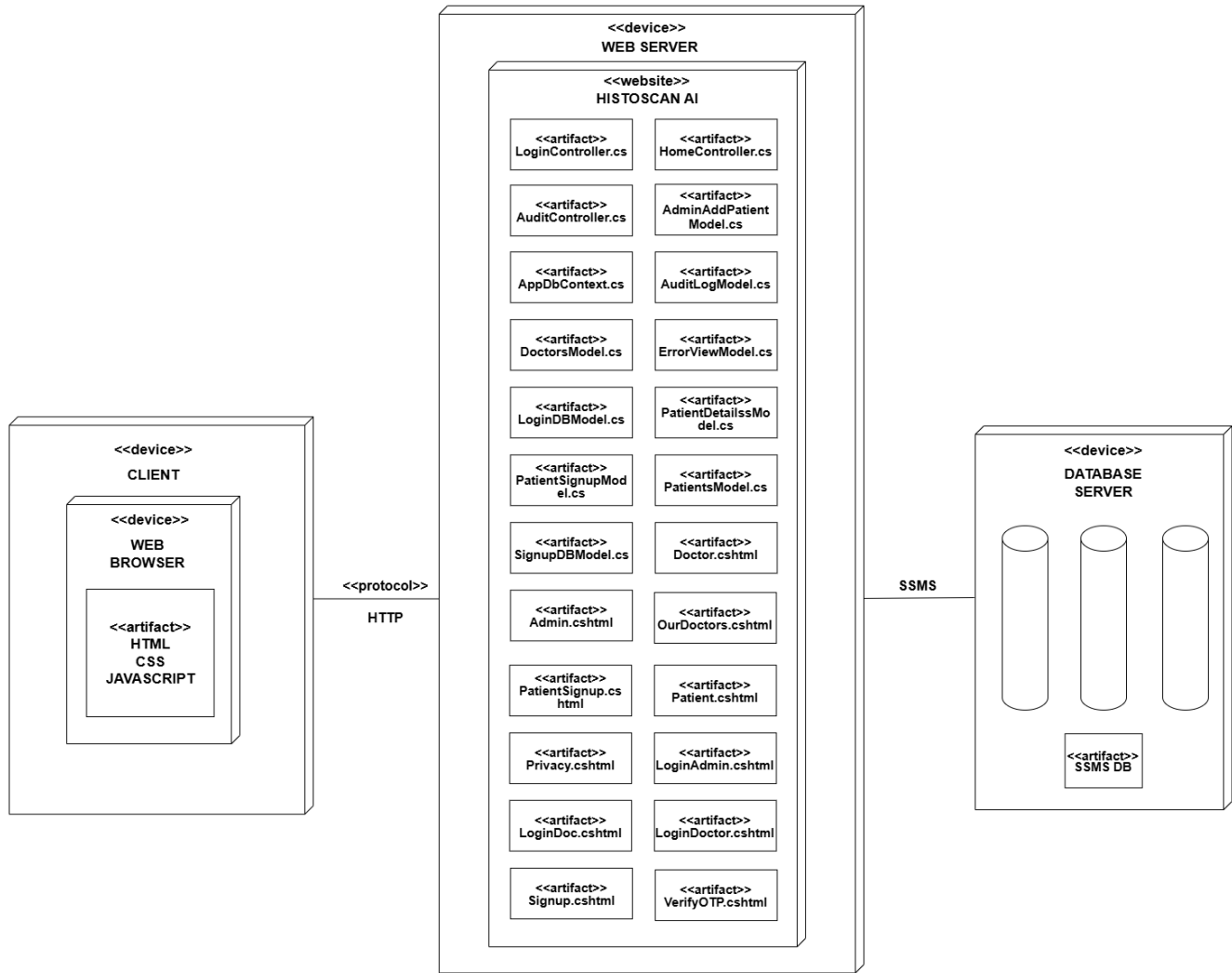


Figure 5.2: Deployment Diagram

This diagram represents the 3 layered architecture of the HistoScanAI platform. On the client side the user opens the platform with a standard web browser. This browser renders the HTML/CSS/JS webpage and processes user interactions locally on the client side. It sends the requests to the central web server where the HistoScanAI web application is deployed using the HTTP protocol. The web server processes the information and displays to the user dashboards, upload forms, results of analyses and diagnostic reports.

The core application is running on the Web Server where the HistoScanAI website is hosted. The website is built using ASP.NET/C# web application technology and includes the corresponding controller files (LoginController, HomeController, AuditController) as well as the model classes (DoctorsModel, PatientsModel, AuditLogModel) and the views (.cshtml files for the web-based Admin, Doctor and Patient interfaces). The business logic is realized on this server and it also contains the authentication mechanism as well as the integration of the AI models. The server communicates with the Database Server hosting the SQL Server Management Studio (SSMS) where the patients' data, images, audit logs and results are stored.

5.2 Behavioral Diagrams

A behavior diagram is used to alluminate things like internal procedures, corporate processes, or how different systems interact. The diagrams chosen will determine which aspect is shown. Objects with the ability to modify their states through behavior are modeled in the Unified Modeling Language (UML).

5.2.1 Activity Diagrams

Activity diagrams show the complete workflow of our system. They visually display a sequence of operations and the control flow, similar to a flowchart or data flow diagram. These diagrams help in understanding how different components of the system interact with each other during execution. They also illustrate decision points, parallel processes, and the overall sequence of activities performed by users and the system. Activity diagrams are highly valuable in the design of business process models and the user interface design in a well-defined manner.

5.2.1.1 Admin's Activity

The workflow diagram illustrates the activities in the processes involved in system administration that include user management, datasets management, and system monitoring. These diagrams illustrate how admin processes guarantee secure access and proper system functioning.

Admin Login Activity Diagram:

Figure 5.3 depicts the procedure of admin authentication in the system. First, the admin logs into the system using their credentials; if they are correct, they proceed further to approve/reject a doctor, add/delete a patient, or view audit logs, but if the credentials are incorrect, access is denied.

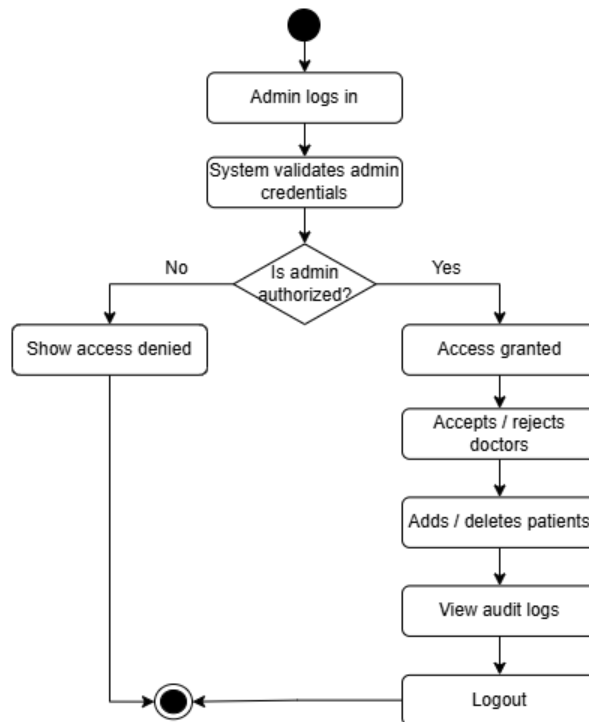


Figure 5.3: Admin Login Activity Diagram

When an admin logs in, our system first checks for valid credentials. If the credentials are incorrect or do not have permission to log into the management portal, the system displays an "Access Denied" message and logs the admin out. When valid credentials are provided, the logged in admin has full access to the management portal where he/she can review a list of pending or approved doctor registrations, approve/ reject or add patient accounts as required, and review the System's Audit logs. The admin logs out after completing the management tasks.

User and Role Management Activity Diagram:

Figure 5.4 indicates the flow involved in handling users and their roles. The process involves user listing, doctor approval or rejection, role assignment, and user status update.

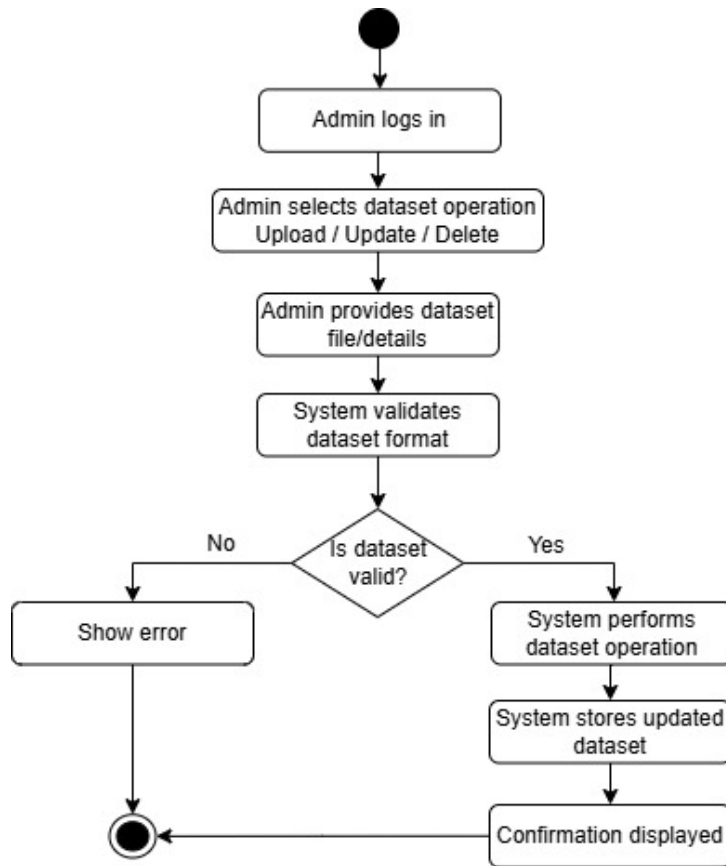


Figure 5.4: Admin User and Role Management Activity Diagram

This figure presents the processes of managing the training datasets of the AI Model by an admin. The admin selects what should be done to the datasets (new dataset upload, updating the current dataset, or deleting a dataset), performs this action via uploading a file or inserting the required information. The system checks whether the file format used is appropriate. If not, the admin gets the corresponding message, and the process stops here. If the format is good, the system executes the required action and gets the confirmation message.

Dataset Management Activity Diagram:

Figure 5.5 illustrates the upload, update, and management of medical datasets by the admin. It showcases the validation process, storage of datasets, and their availability for training.

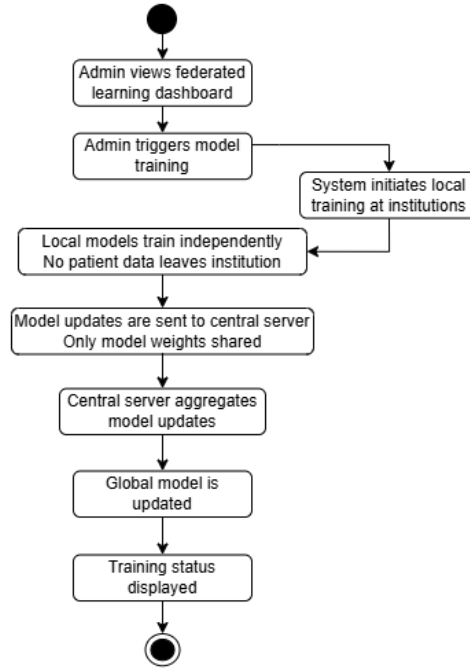


Figure 5.5: Admin Dataset Management Activity Diagram

Admin logs into the dashboard and starts a FL round. Task is shared with participating hospitals, where local models are trained on their data. Only updated model weights are sent back to the central server for aggregation. The admin updates the global model using combined knowledge and monitors training progress through dashboard.

Audit Activity Diagram:

Figure 5.6 explains how the admin monitors system logs. It covers auditing system actions to ensure transparency and reliability.

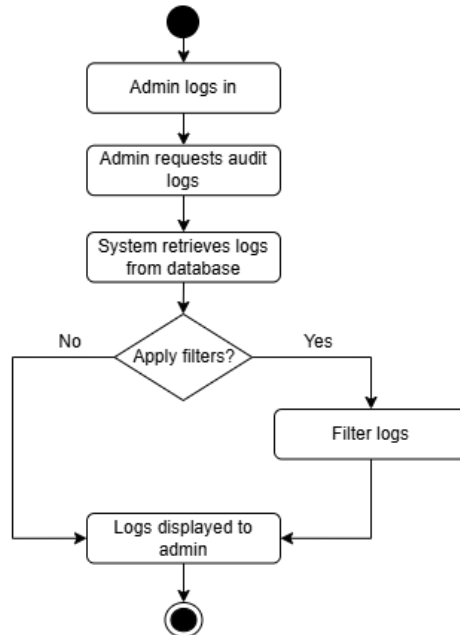


Figure 5.6: Admin Model Monitoring and Audit Activity Diagram

This diagram shows the workflow for an admin reviewing system audit logs. The action begins with an admin user logging into the site, then they want to view the audit logs, at which point the site fetches all of the audit logs from the database. The admin then has the option to apply filters to the history, such as searching for specific activity within a time period, activities performed by a specific user, or a specific type of action. The system can then return the filtered history before it is displayed to the admin, or the full history if no filters have been applied, and then the admin reviews the logs.

Federated Learning Activity Diagram:

Figure 5.7 explains how the admin oversees the federated learning process. It covers the complete round life cycle from model distribution to global model deployment.

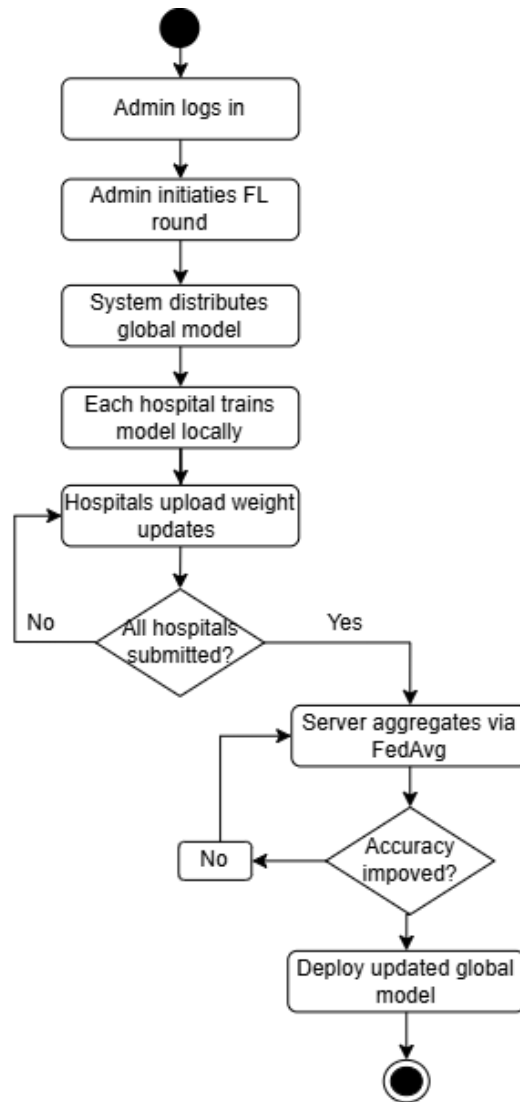


Figure 5.7: Admin Federated Learning Activity Diagram

This diagram shows the workflow for an admin managing a federated learning round. The process begins with the admin logging in and initiating a new FL round, at which point the system distributes the current global model to all participating hospitals. Each hospital then trains the model locally on its own patient data and uploads only the encrypted weight updates back to the central server. The system checks whether all

hospitals have submitted their updates; if not, it waits. Once all updates are received, the server applies FedAvg aggregation to produce a new global model. The system then checks whether accuracy of model has improved: if it has, the updated global model is launched in all hospitals; if not, the updates are rejected and the existing model is retained.

Dashboard and Analytics Activity Diagram:

Figure 5.8 explains how the admin oversees system-wide statistics and AI performance. It covers the all analytics viewing workflow from login through data review.

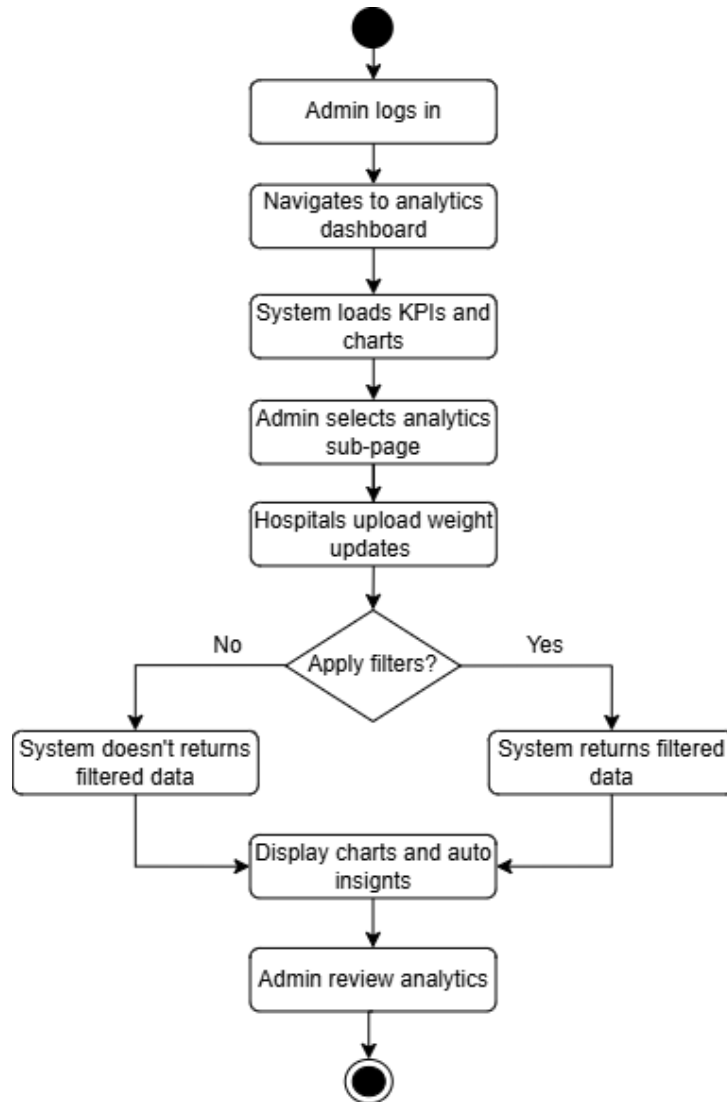


Figure 5.8: Admin Dashboard and Analytics Activity Diagram

This diagram illustrates the workflow for an admin reviewing system analytics. The action initially begins with the admin logging in and navigating to the main analytics dashboard, at which point the system automatically loads all nine core KPIs and their associated charts. The admin selects a specific analytics sub-page, such as AI performance, patients, doctors, hospitals, usage, dataset, or user feedback. The admin then has the direct option to apply various filters, such as narrowing data by date range, doctor, hospital, or cancer type. The system returns the promptly returns filtered dataset and

renders the relevant charts and auto-generated insights, which the admin then reviews for decision-making.

Feedback Activity Diagram:

Figure 5.9 explains how users submit feedback form and then how admin reviews and analyzes the collected responses. It caters the entire feedback life cycle across all three user roles.

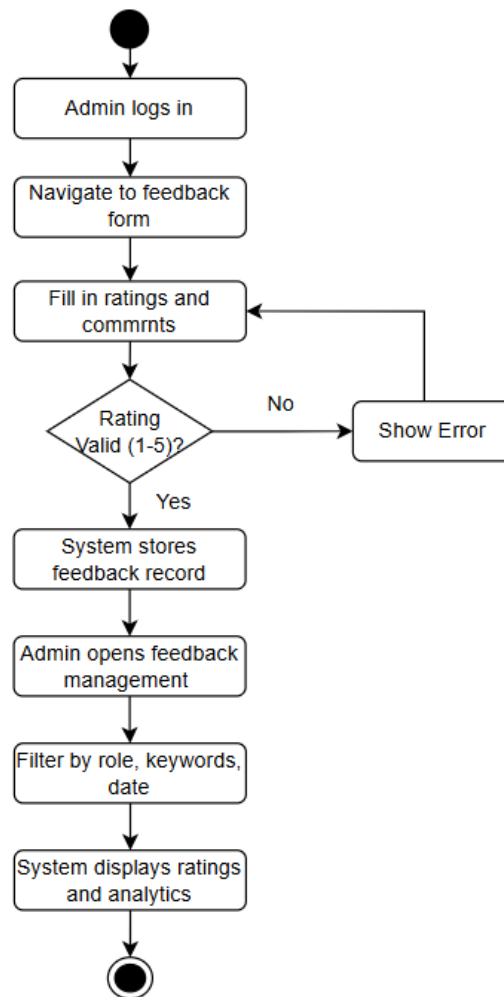


Figure 5.9: Admin Feedback Management Activity Diagram

This diagram shows the workflow of collecting and maintaining user feedbacks. Whoever logs in whether admin, doctor, or patient, navigates to the feedback form and fills in the multi-question rating fields along with an optional comments field. The system actively validates that all submitted ratings fall within the 1–5 range; if any rating is invalid, an error message is shown and the user is prompted to correct it.

Once valid, the system securely stores the feedback record with the user’s role, user-name, and exact timestamp attached. Subsequently, the admin opens the specific feedback management page, where they can filter and search submissions by role, keyword, or date range. The system then displays the filtered feedback list alongside various com-

puted analytics including per-role satisfaction scores, per-question averages, and overall rating trends.

Admin Notification Activity Diagram:

Figure 5.10 explains how the admin is automatically notified when a new user registers. It covers the notification flow for both doctor and patient registrations.

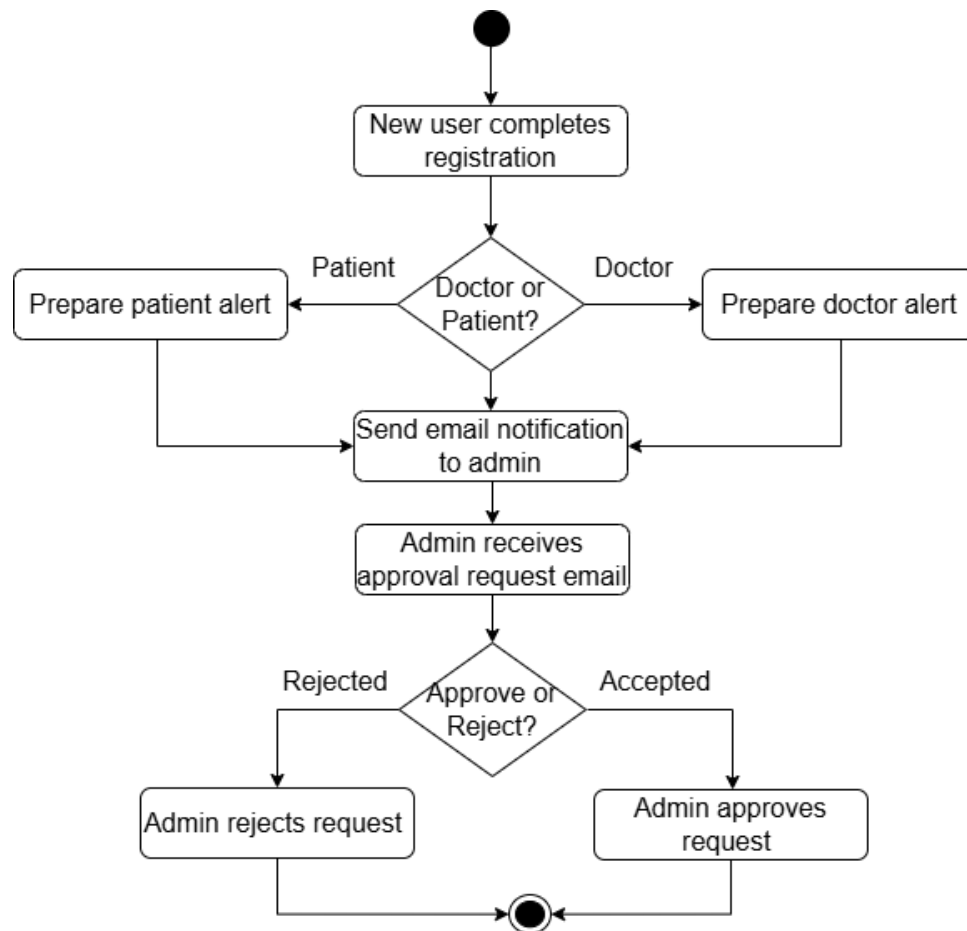


Figure 5.10: Admin Notification Activity Diagram

This diagram shows the workflow for notifying the admin when a new user registers on the system. The process is triggered automatically as soon as a new user completes registration. The system checks whether the registrant is a doctor or a patient and prepares the appropriate email message accordingly. In both cases, the notification email is sent to the admins registered email address informing them that a new approval request is pending. The admin receives the email and then logs into the system to either approve or reject the new account.

5.2.1.2 Doctor's Activity

It involves the series of processes that are done by the doctors like authentication, upload of the images, analysis using AI, and creation of a report. Such flowcharts show the process of diagnosis using the system and how doctors interface with various modules of the system. The flowcharts also show the decision points for doctors to analyze and validate the result generated by the AI.

Doctor Sign Up Activity Diagram:

Figure 5.11 shows the process for registering doctors. It requires providing personal and professional information. After admin approval, doctors can log into the system.

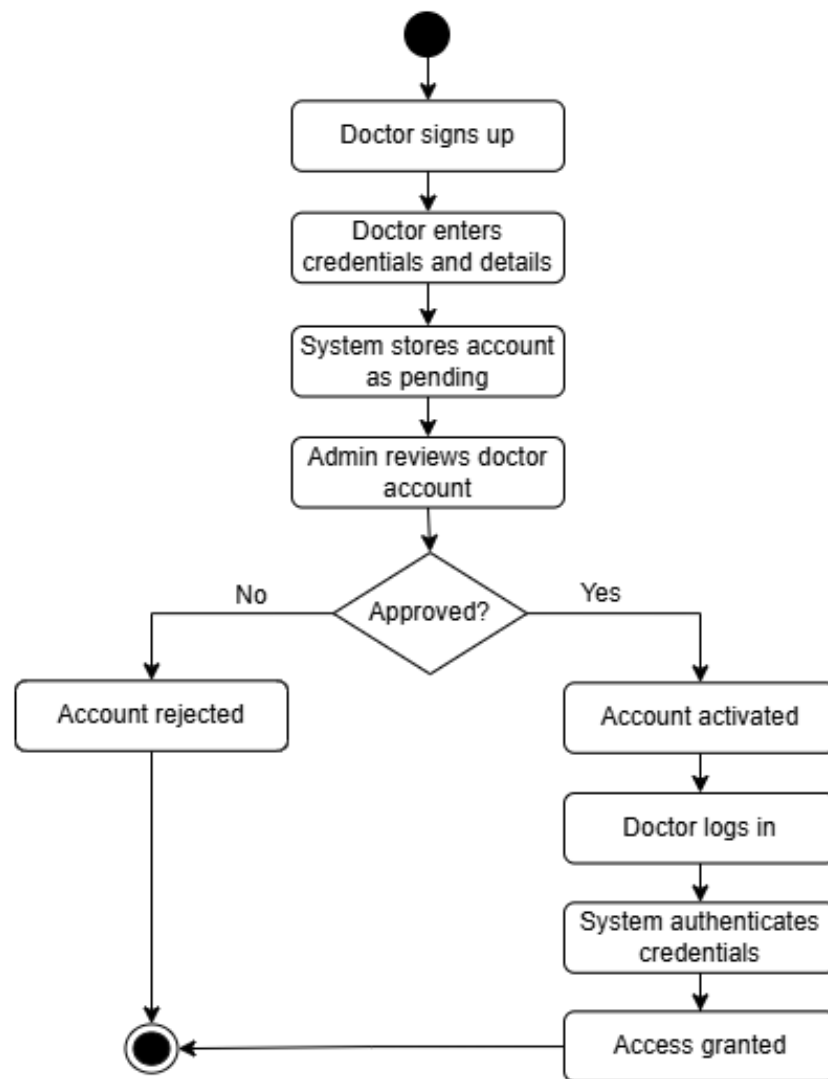


Figure 5.11: Doctor Sign Up Activity Diagram

The doctors would first register themselves on the system by providing required details like username, password, full name, contact number, degree, specialized years, gender and email address. The system would then render the account as “pending” until the admin approves it. If admin rejects the account, the process ends there but if account gets approved, doctor would then login to the system and the system would ask for credentials for authentication and entering otp is must for authorized doctors and then doctor would be able to utilize the whole system.

Doctor can add checkup details view patients history all the diagnosis then upload the histopathology images, afterwards apply artificial intelligence for the results, and at last the report is generated that consists of diagnosis that whether it is cancerous or non-cancerous, confidence score along with xai explanation that anyone can understand easily.

Doctor Login Activity Diagram:

Figure 5.12 illustrates the authentication procedure. It validates credentials and grants controlled access to functions specific to doctors.

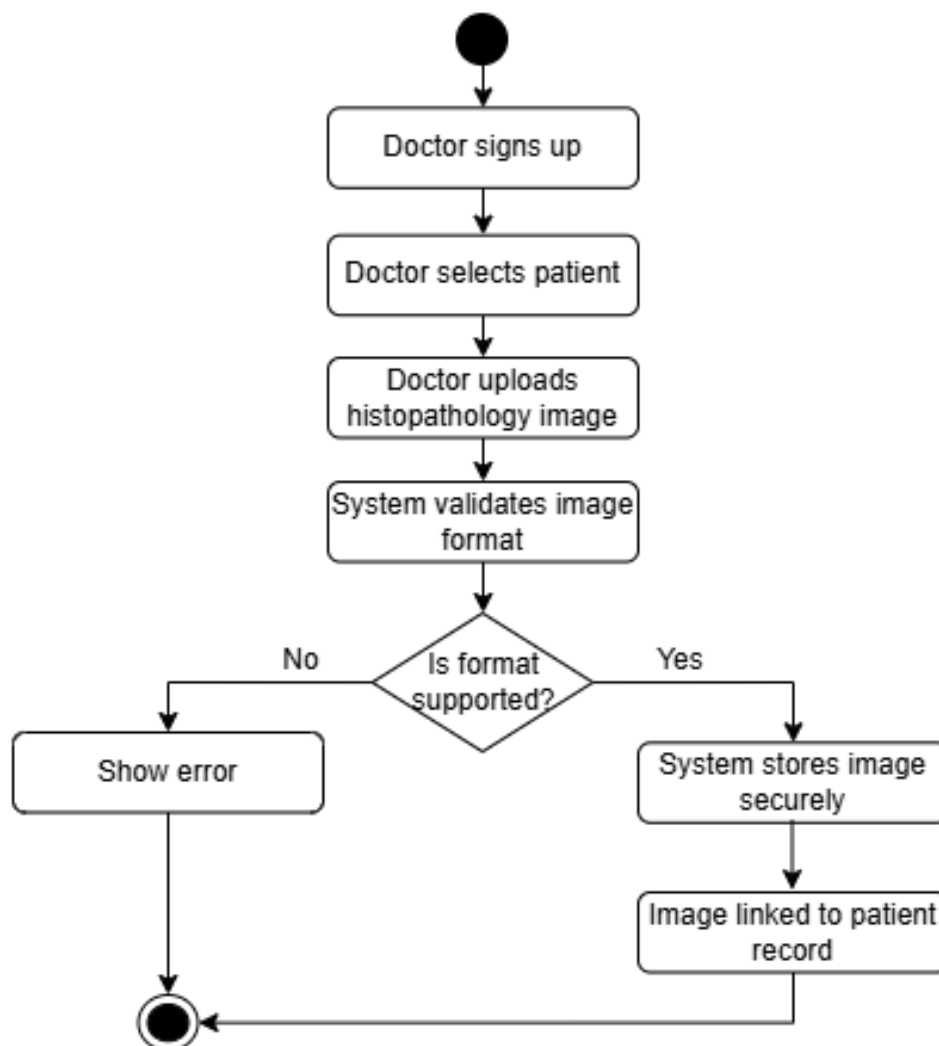


Figure 5.12: Doctor Login Activity Diagram

The activity diagram models the workflow of a doctor uploading a histopathology image into a medical system, beginning with authentication where the system validates the doctor's credentials before granting access to the dashboard. Once logged in, the doctor searches the patient database by name or ID, and upon finding the correct patient, opens their medical record.

The doctor then uploads a tissue sample image via the interface, after which the system automatically validates the file for correct format, size, and integrity rejecting unsupported files with an error message.

Doctor Uploads Patient Case Activity Diagram:

Figure 5.13 Illustrates the process for uploading histopathological images by a physician. Such as selecting patient information, uploading images, verification of images, and transferring data for AI analysis.

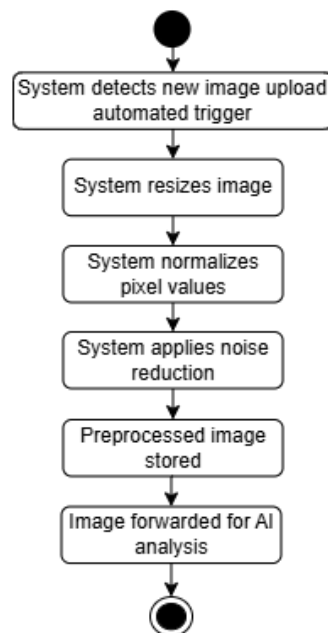


Figure 5.13: Doctor Uploads Patient Case Activity Diagram

Once a new image is uploaded to the system, it is preprocessed in system where the image is resized, normalized and smoothened to reduce noise to produce image of consistent quality.

AI Prediction Review Activity Diagram:

Figure 5.14 shows the perspective through which physicians can view and evaluate the predictions generated using AI.

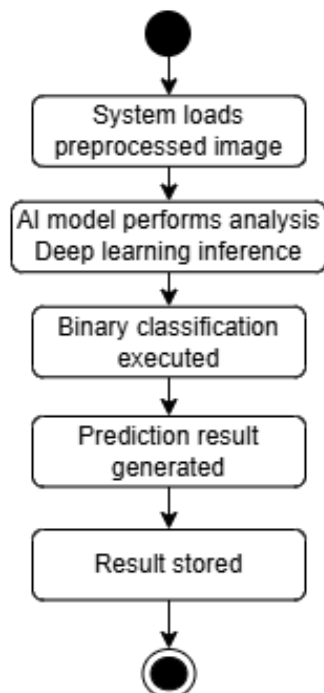


Figure 5.14: Doctor AI Prediction Review Activity Diagram

The system puts the preprocessed tissue image through the deep learning AI model for analysis. It performs a binary classification in order to determine if the sample contains cancerous or benign cells. A prediction result along with the AI's confidence scores is then produced by the system and logged into the database for the doctor to review as needed and for the patient to access.

Report Generation Activity Diagram:

Figure 5.15 diagnostic reports production procedure. This procedure entails gathering AI reports, doctor comments, and producing or distributing the final report safely.



Figure 5.15: Doctor Report Generation Activity Diagram

After the histopathology image is processed, the system automatically generates a comprehensive diagnostic report based on the model analysis results. This report is then linked directly to the corresponding patient's record, ensuring all diagnostic data remains organized and traceable within internal medical system. The doctor can then view the generated report through the secure interface to review the findings.

Additionally, the doctor has access to the patient's full diagnostic history, allowing upclear comparison of current results with previous reports to track active disease progression or treatment response. Finally, all reports are stored persistently in the system, maintaining a secure and complete medical archive for future reference, audits, or further clinical decision-making.

Doctor FL Activity Diagram.:

Figure 5.16 explains how the decentralized federated learning pipeline operates. It covers local model inference, privacy-preserving gradient sharing, and final global model aggregation.

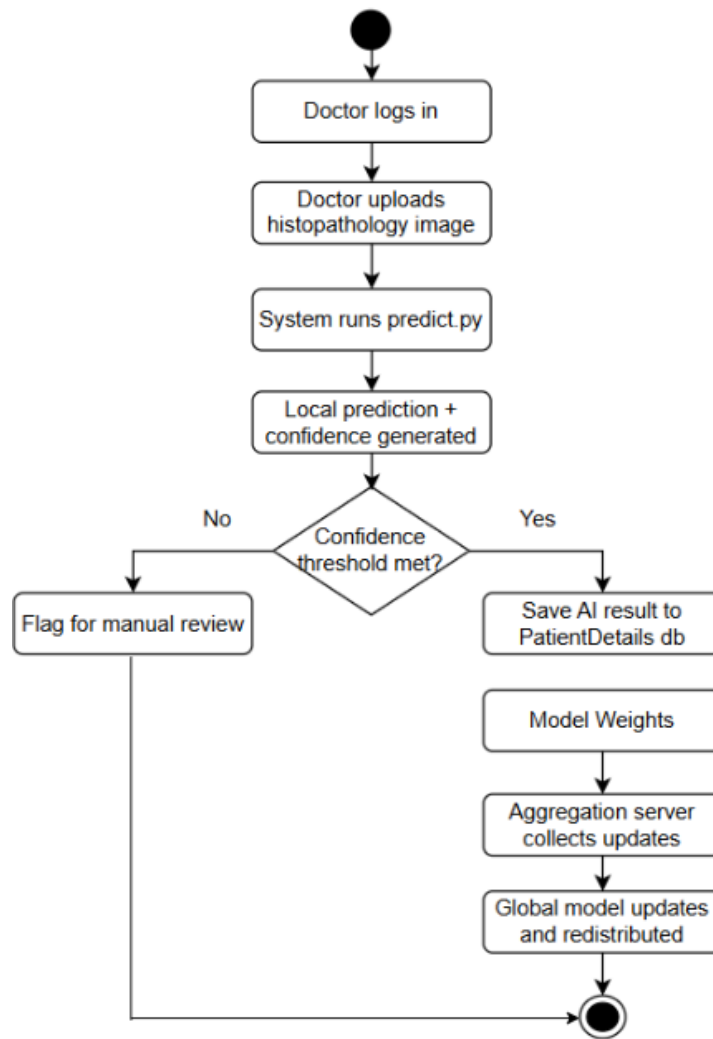


Figure 5.16: Doctor Federated Learning Activity Diagram

Once the doctor logs in and uploads the histopathology images, the underline system executes a local prediction script (`predict.py`) which runs the diagnostic AI model on the uploaded image and generates a local prediction along with a calculated confidence score. The system then evaluates whether this confidence score meets a predefined performance threshold if it does not, the entire case is flagged for manual review by a medical expert to ensure maximum patient safety. If the confidence threshold is met, the AI prediction result is automatically saved to the PatientDetails database, confirming the final diagnosis is reliable enough for permanent storage.

Following this, the locally trained model weights updated from processing this new image are successfully extracted and sent to an aggregation server, which is part of a federated learning architecture. The aggregation server collects these weight updates from different hospitals and combines them to produce an improved global model.

Doctor Feedback Activity Diagram.:

Figure 5.17 explains how a doctor submits system feedback. It covers the approval check, five-category rating form, validation, and audit logging upon successful submission.

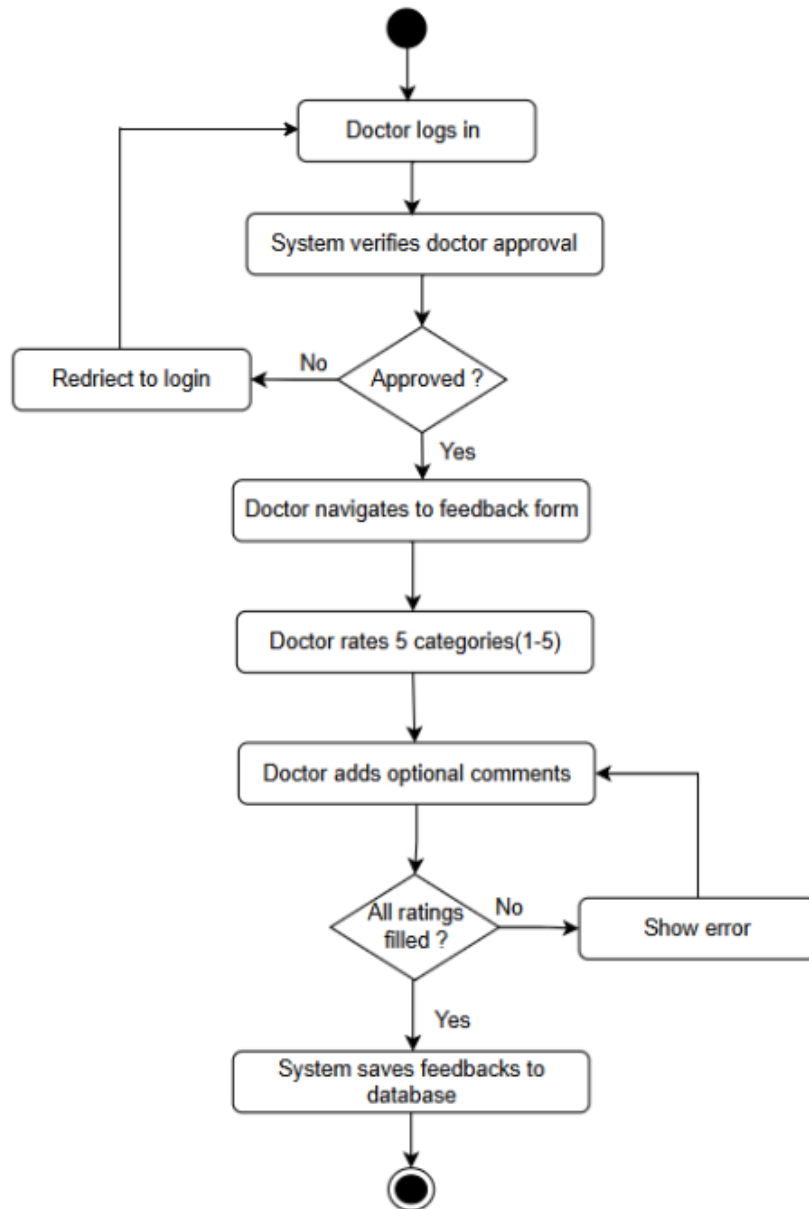


Figure 5.17: Doctor Feedback Activity Diagram

It shows that the flow begins with the doctor logging in, after which the system verifies their approval status. Unapproved doctors are redirected immediately. Approved doctors navigate to the feedback form (SubmitDoctor), rate five categories (admin support, system performance, AI quality, report upload, dashboard ease), and optionally add technical issues and suggestions.

If any required rating is missing, the system shows an error and re-presents the form. On valid submission, feedback is persisted to the feedback table and a FEEDBACK_SUBMITTED audit log entry is created.

Doctor Feedback Notification Diagram:

Figure 5.18 explains how a patient is notified when a doctor adds an AI analysis report. It covers AI model invocation, result persistence, email validation, SMTP dispatch, and patient report download.

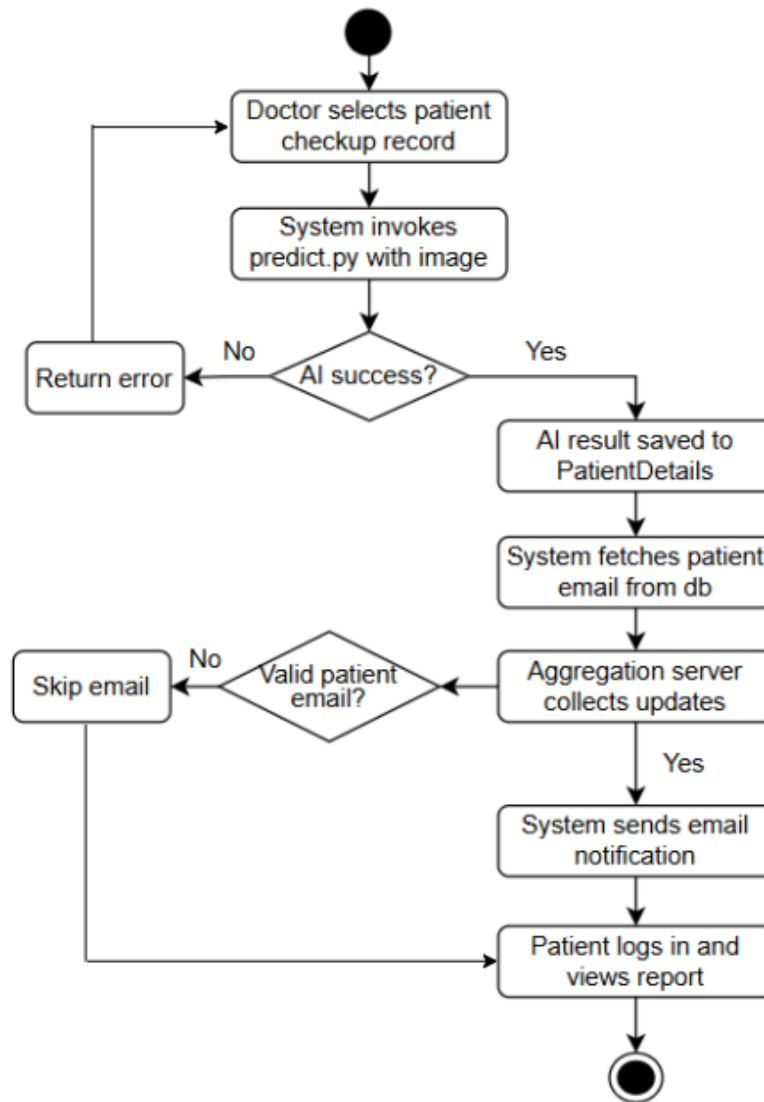


Figure 5.18: Doctor Notification Activity Diagram

It covers the notification flow triggered when a doctor runs AI analysis. After the doctor selects a patient checkup and requests analysis, the system calls `predict.py` with the histopathology image. On failure, an error is returned to the doctor UI. On success, the AI result and timestamp are written to `PatientDetails`, and the system retrieves the patient's email.

If a valid email exists, an SMTP notification is dispatched via Gmail ("Your doctor has added your report"); otherwise the email step is skipped and logged. The patient can then log in to view and download their HTML report from the patient portal.

5.2.1.3 Patient's Activity

The patient activity diagrams show how patients interact with the system to access diagnostic results. They emphasize secure log-in, viewing diagnostic reports, then downloading them and managing access to health information. Its users choice if he wants to only view reports or download too. And can also give review by giving ratings.

Patient Sign Up and Login Activity Diagram:

Figure 5.19 describes the process of patient account creation and authentication. It includes entering personal details, credential verification, and accessing patient-specific features upon successful login.

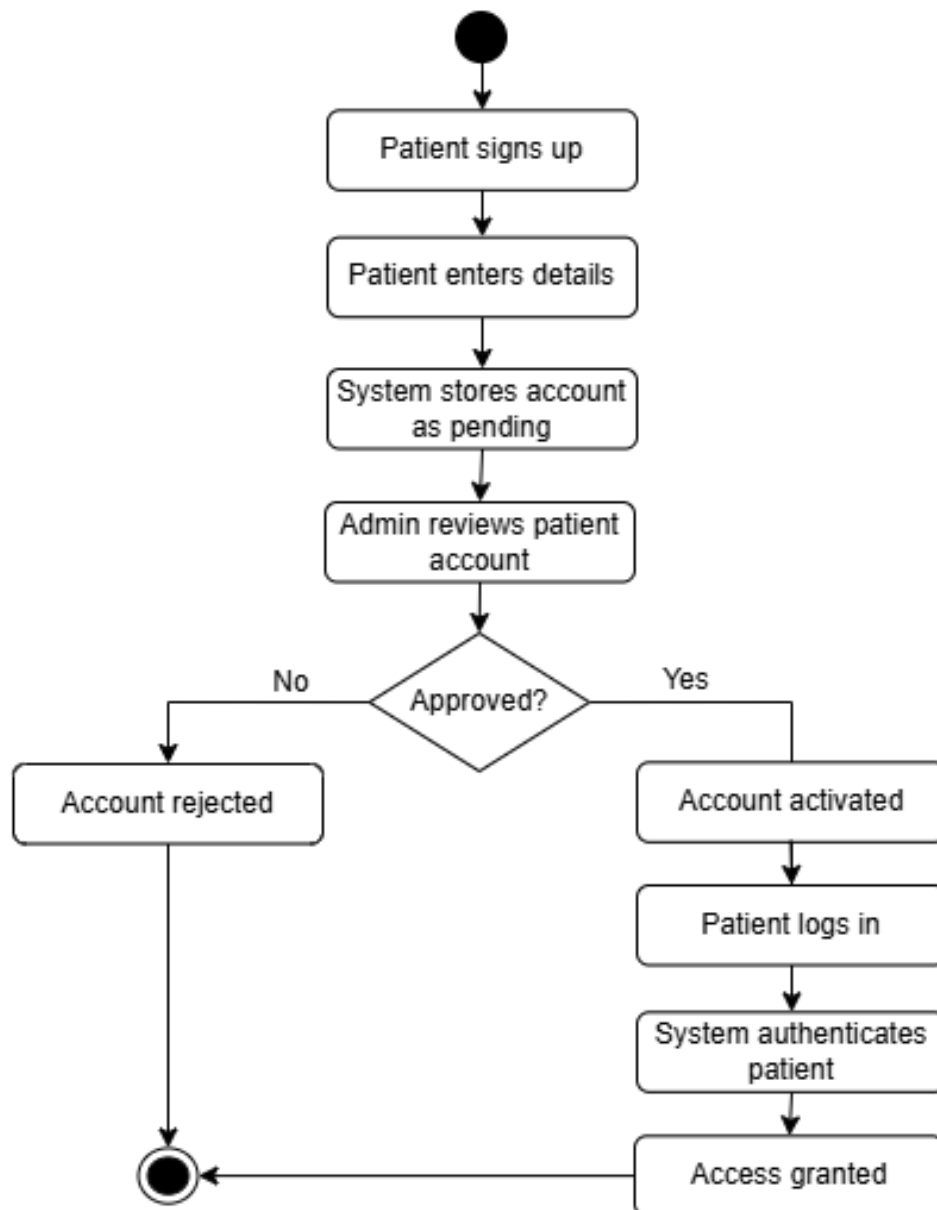


Figure 5.19: Patient Sign Up and Login Activity Diagram

You register as a patient on the site by entering in your and your family member's information, which is held in a "pending" status until an admin reviews/ verifies the account. If admin rejects registration process is over. If approved then patient can log in on the site with his/her username and password after validating his/her credentials. He/ She can access the patient's medical/health dashboard.

View Diagnosis Report Activity Diagram:

Figure 5.20 illustrates how patients will be able to get to their reports. This is through checking the AI-generated reports, doctor notes, and medical results.

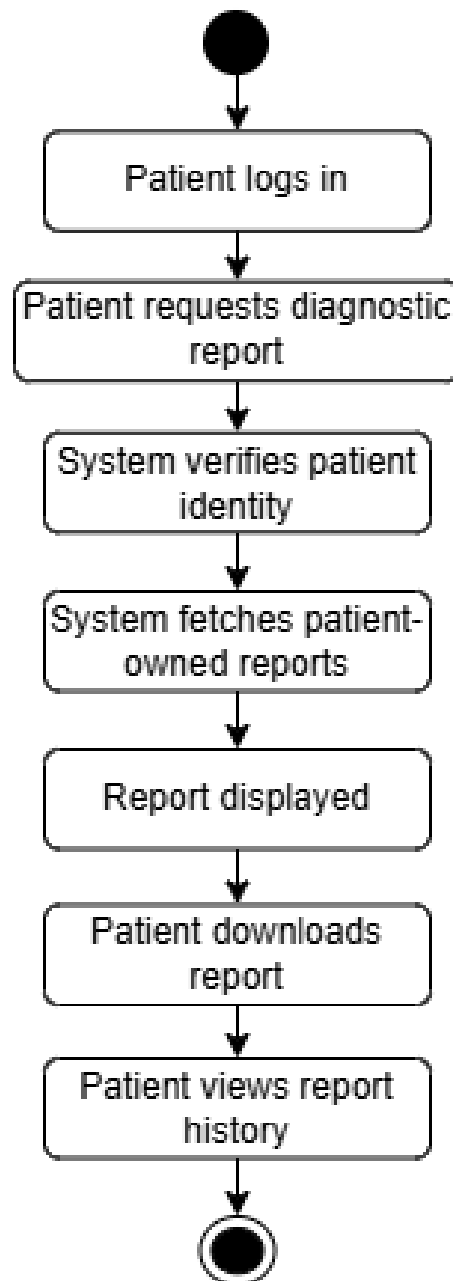


Figure 5.20: Patient View Diagnosis Report Activity Diagram

The patient starts with logging into the system using their credentials, after which they can submit request to access their diagnostic report. Before any report is retrieved, the system verifies the patient's credentials to make sure that only the real user can access his medical data. Once identity is proved, the system shows all reports that are exclusively linked to that patient's record, maintaining strict data privacy.

The retrieved report is then displayed on the interface for the patient to review. If the patient wishes to retain a copy, they have the option to download the report directly. Additionally, the patient can view their full report history, allowing them to track their diagnostic journey over time, monitor changes in their condition, and share past results with other healthcare providers if needed.

Patient Federated Learning Activity Diagram:

Figure 5.21 explains how the federated learning pipeline processes a patient's histopathology image. It covers local model inference, privacy-preserving weight sharing, and global model aggregation ensuring raw patient data never leaves the hospital.

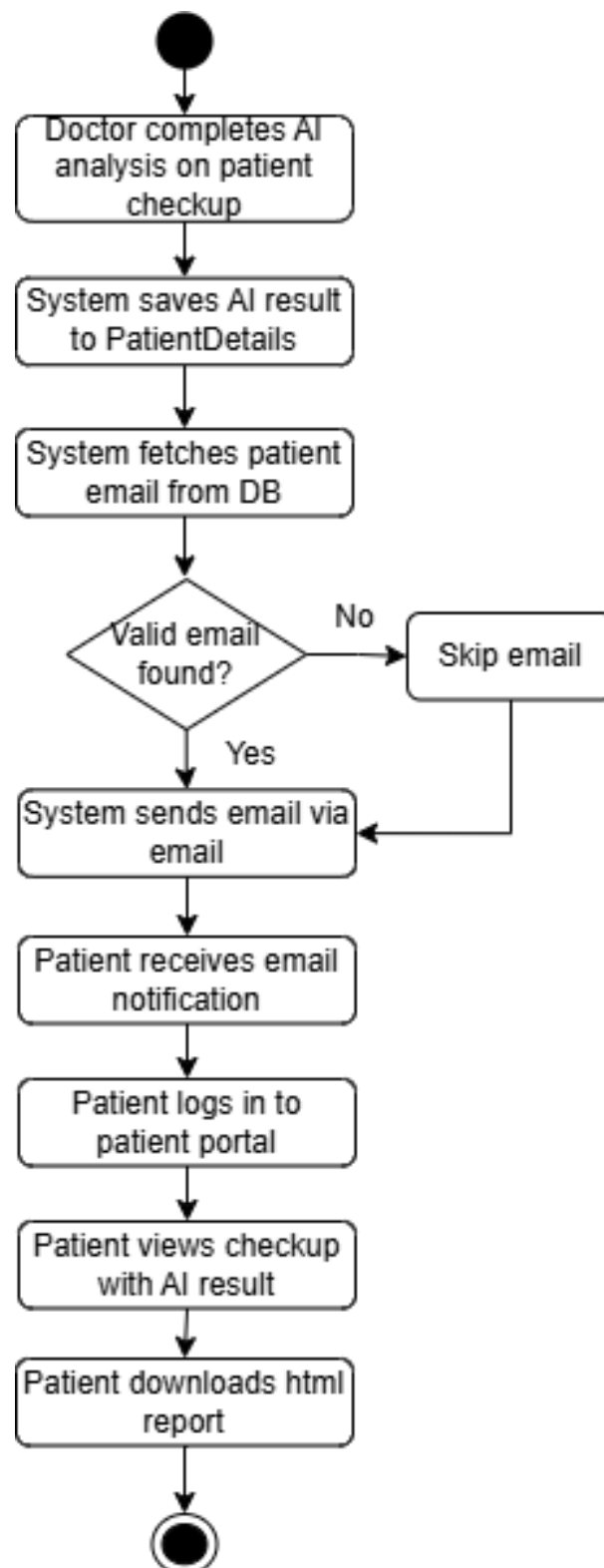


Figure 5.21: Patient Federated Learning Activity Diagram

It explains how the federated learning pipeline processes a patient’s histopathology data. The patient logs in and the system runs the local AI model (`predict.py`) on the uploaded image. If confidence is sufficient, the result is stored in `PatientDetails`; otherwise it is flagged for manual review. Critically, only model weights — never the raw image — are sent to the aggregation server, which applies FedAvg to update the global model that is then redistributed to all participating hospitals.

Patient Feedback Activity Diagram:

Figure 5.22 explains how a patient submits system feedback. It covers session verification, the five-category star rating form, validation, and audit logging upon successful submission.

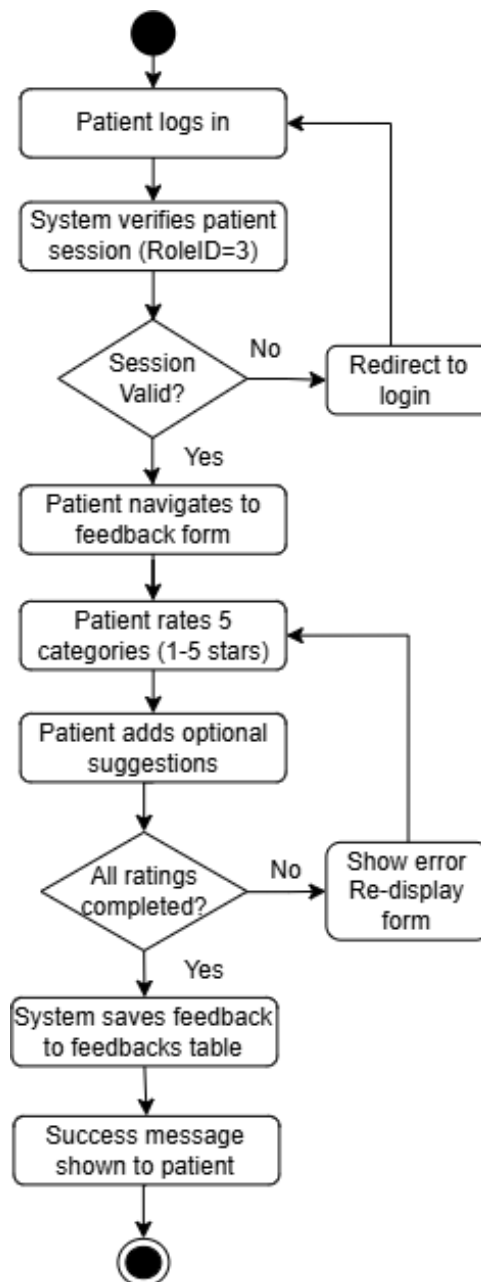


Figure 5.22: Patient Feedback Activity Diagram

It explains how a patient submits feedback. After logging in, the system verifies the

session role (RoleID = 3); an invalid session redirects to the login page. The patient then accesses the SubmitPatient feedback form and rates five categories — doctor behavior, report quality, usability, download ease, and overall satisfaction — with an optional suggestions field. If any required rating is missing, the system shows a validation error and re-presents the form. On valid submission, the feedback is persisted to the Feedbacks table, a FEEDBACK.SUBMITTED audit entry is written, and the patient sees a confirmation message.

Patient Notification Activity Diagram:

Figure 5.23 explains how a patient is notified when their AI report becomes available. It covers AI result persistence, email validation, SMTP dispatch, patient login, and report download.

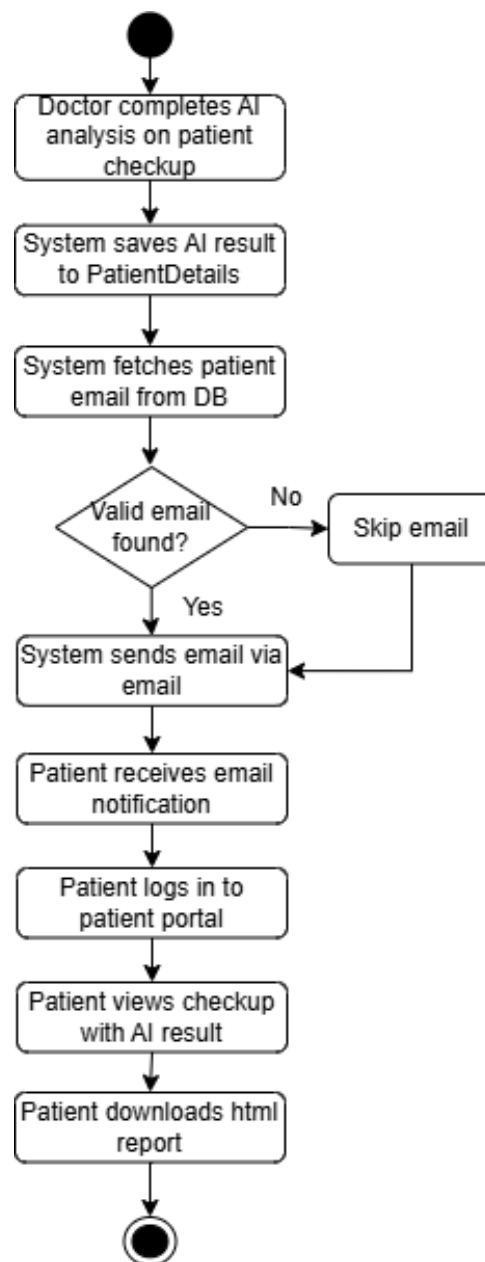


Figure 5.23: Patient Notification Activity Diagram

Once the doctor completes the AI-assisted analysis on a patient's checkup, the system automatically saves the AI result to the PatientDetails database, ensuring the findings are securely recorded. The system then attempts to fetch the patient's registered email address from the database to send a notification. If no valid email is found, the email step is skipped entirely and the flow continues without notification.

If a valid email exists, the system sends an automated email alerting the patient that their results are ready. Upon receiving the notification, the patient logs into the patient portal using their credentials and navigates to their checkup section, where they can view the AI-generated diagnostic result in full detail. Finally, the patient has the option to download the report in HTML format, providing them with a portable and readable copy of their diagnostic findings for personal records or further medical consultation.

5.2.2 Communication Diagrams

A communication diagram represents the interactions of objects or components as a series of sequentially numbered messages, providing a clear and structured view of how different parts of a system collaborate to achieve specific goals. Communication diagrams combine data from Class, Sequence, and Use Case Diagrams to describe the system's static structure as well as its dynamic behavior, offering a comprehensive perspective that bridges both architectural and operational aspects of the system. They focus on how different components are linked and how messages are passed between them to perform specific operations, making the flow of control and data exchange across the system immediately visible to developers and analysts alike.

Communication diagrams also help in understanding the relationships and dependencies among system elements, making it easier to analyze system functionality and identify potential bottlenecks or points of failure in the communication chain. The numbered sequencing of messages allows developers to trace the exact order in which operations are triggered, which is particularly valuable during debugging, system testing, and design review phases. Furthermore, they are useful for visualizing real-time communication and ensuring that all interactions are properly coordinated within the system, reducing the risk of missed or misrouted messages during implementation.

Communication diagrams serve as an effective collaboration tool between technical and non-technical stakeholders, as their object-link structure is intuitive and easier to read than some other UML diagram types. They are especially beneficial in object-oriented system design, where multiple objects must interact across layers such as the user interface, business logic, and data access layers. By mapping out these interactions explicitly, communication diagrams support better system planning, promote modular design, and ensure that each component's responsibilities and communication pathways are well-defined before development begins.

5.2.2.1 Admin Communication Diagram

Figure 5.24 shows the communication diagram of the admin. Showing the working with its components of authentication, managing doctors and patients, checking on the audit and activity logging along with the notifications admin receive from doctors and patients.

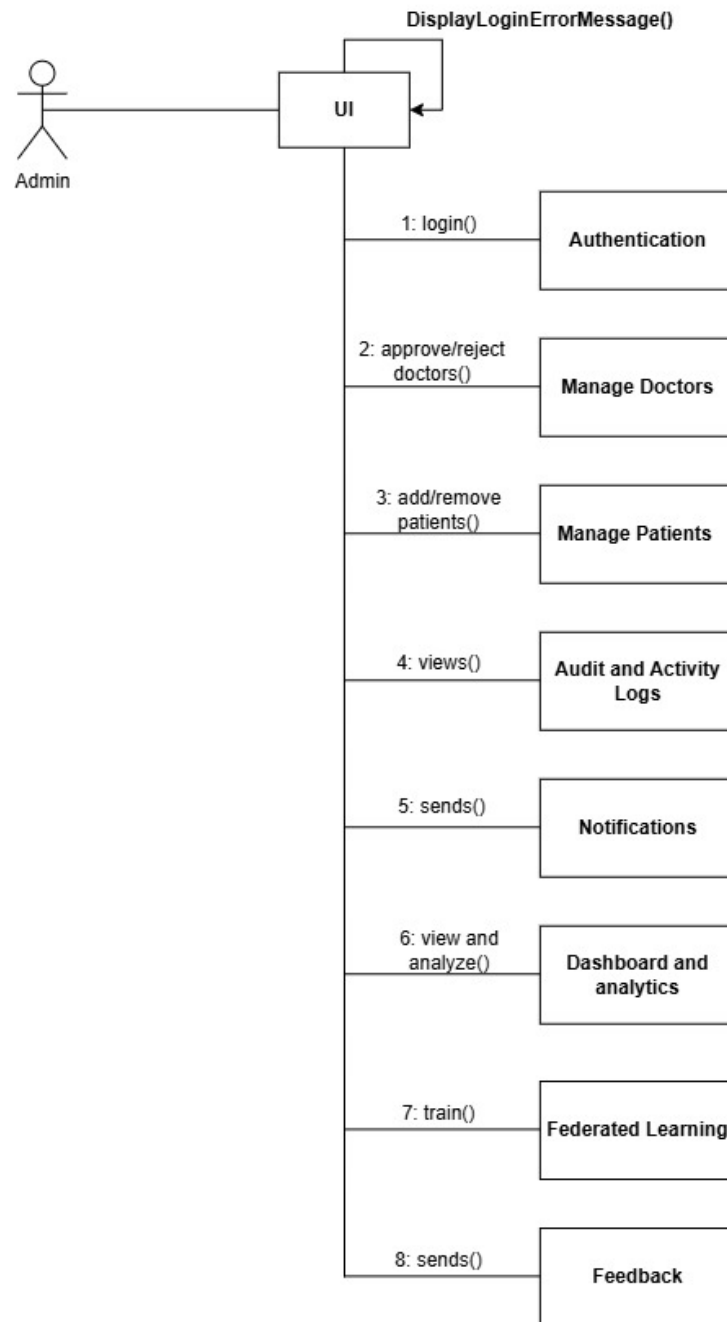


Figure 5.24: Communication Diagram of Admin

It illustrates how the admin interacts with the system to perform various operations and how messages are exchanged between the admin and different system components. By first logging in, then approving or rejecting doctors, adding or removing patients, managing activity and audit logs, and most importantly, the notifications.

5.2.2.2 Doctor Communication Diagram

Figure 5.25 shows the communication diagram of the doctor. Showing the working with its components of authentication, treating the patient afterwards viewing and then generating its report. Doctors sends notification to patient about intimation of report generation and providing with feedback.

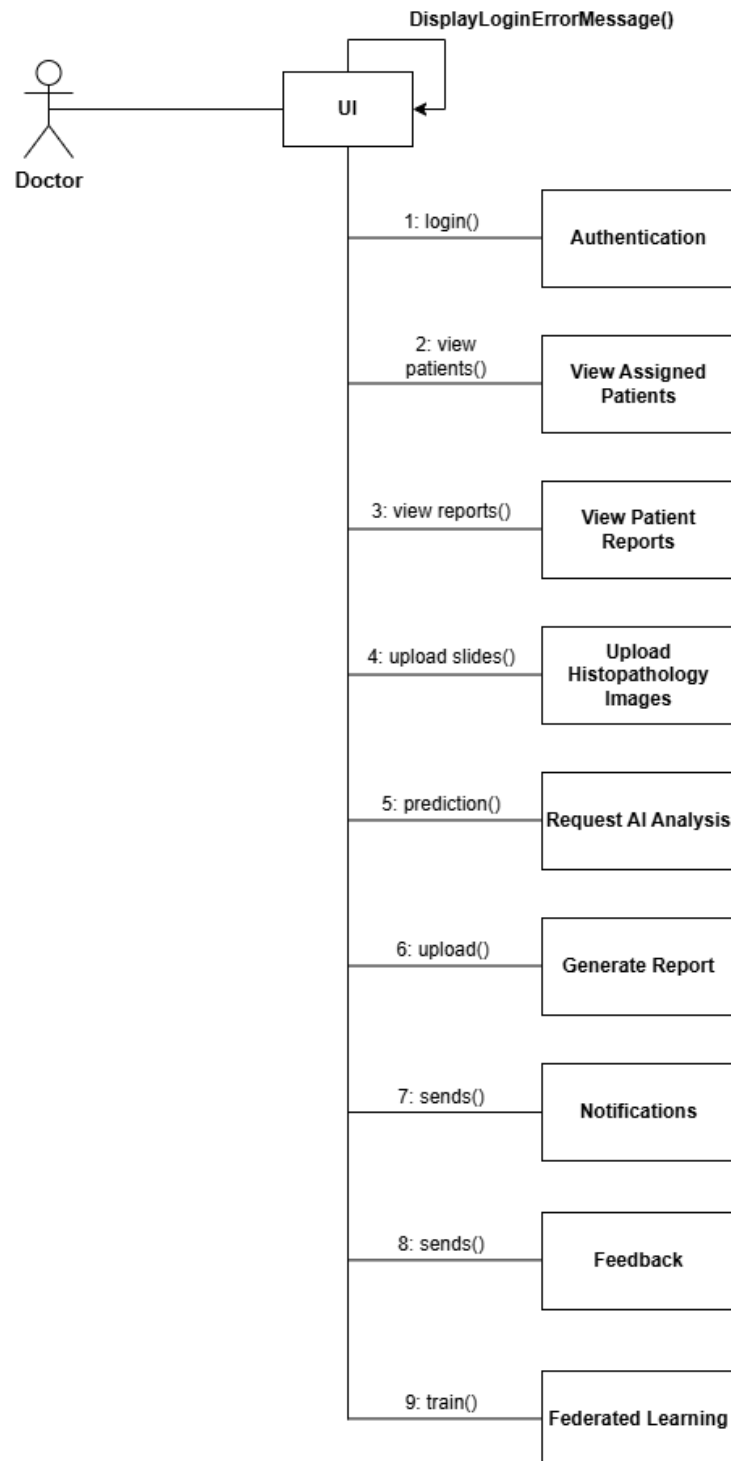


Figure 5.25: Communication Diagram of Doctor

It illustrates how the doctor interacts with the system to perform various operations. By first authenticating and verifying by otp, then viewing the assigned patients, their reports, the doctor then uploads histopathology images and requests for AI analysis, afterwards in report generation it shows the detection with explainable AI giving the feedback.

5.2.2.3 Patient Communication Diagram

Figure 5.26 shows the communication diagram of the patient. Showing the working with its components of authentication, viewing profile and reports, also downloading the reports, notifications, feedback and federated learning are also done in the patient portal.

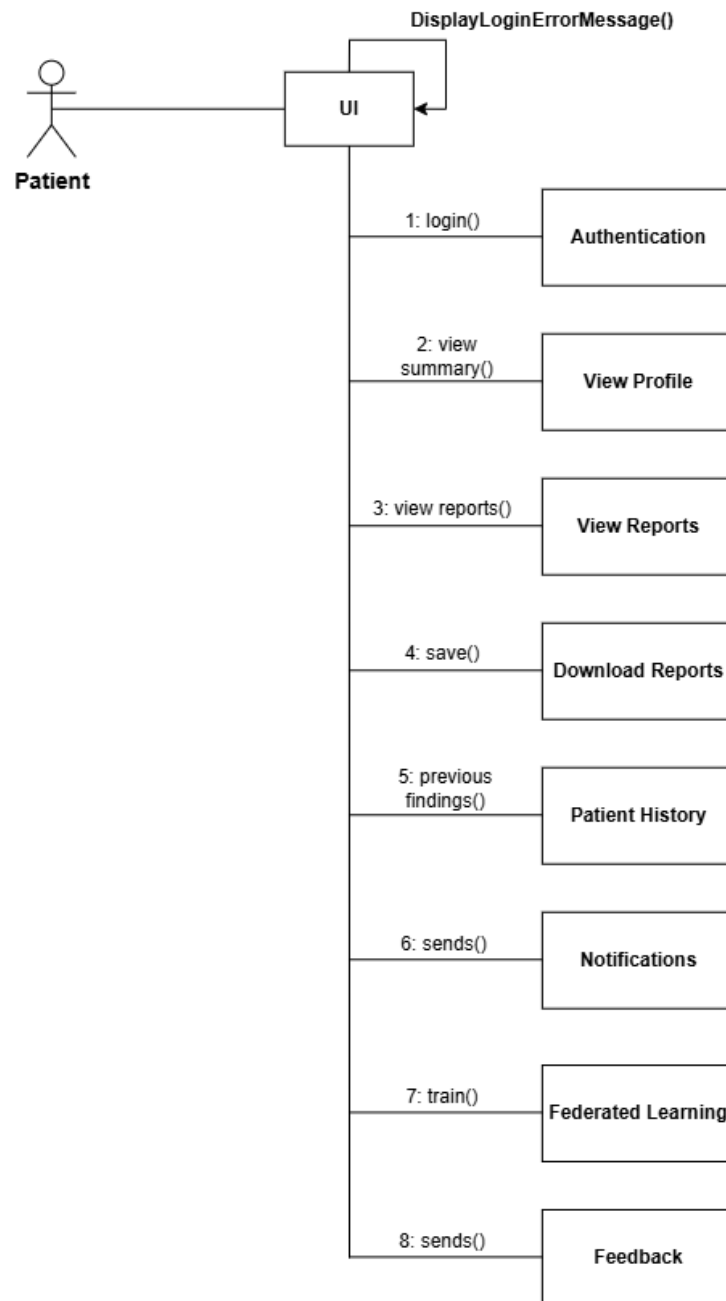


Figure 5.26: Communication Diagram of Patient

It illustrates how the patient interacts with the system to perform various operations like viewing and downloading reports and having a check on history. Patient can view all the previous reports and download as many times they want. If a new patient signs in, the notification will be given to admin that new patient is there to approval.

5.2.3 Sequence Diagrams

Sequence Diagrams are interaction diagrams that describe the steps used to complete an operation. They depict how items interact within the framework of a cooperation. By using the vertical axis of the diagram to represent time and the messages that are transmitted and when, sequence diagrams, which have a time focus, can visually depict the order of an interaction.

Here, now sequence diagrams of admin, doctor and patients are seen.

5.2.3.1 Admin's Sequence Diagrams

The admin is responsible for managing system configuration, users, and overall platform integrity. Admin functionalities ensure secure access control, proper dataset handling, privacy-preserving AI training, and continuous system monitoring for transparency and accountability.

User and Role Management:

Figure 5.27 demonstrates that how admin manages system users by creating, accepting, or rejecting accounts. The system validates admin privileges and securely updates user records in the database to ensure controlled access.

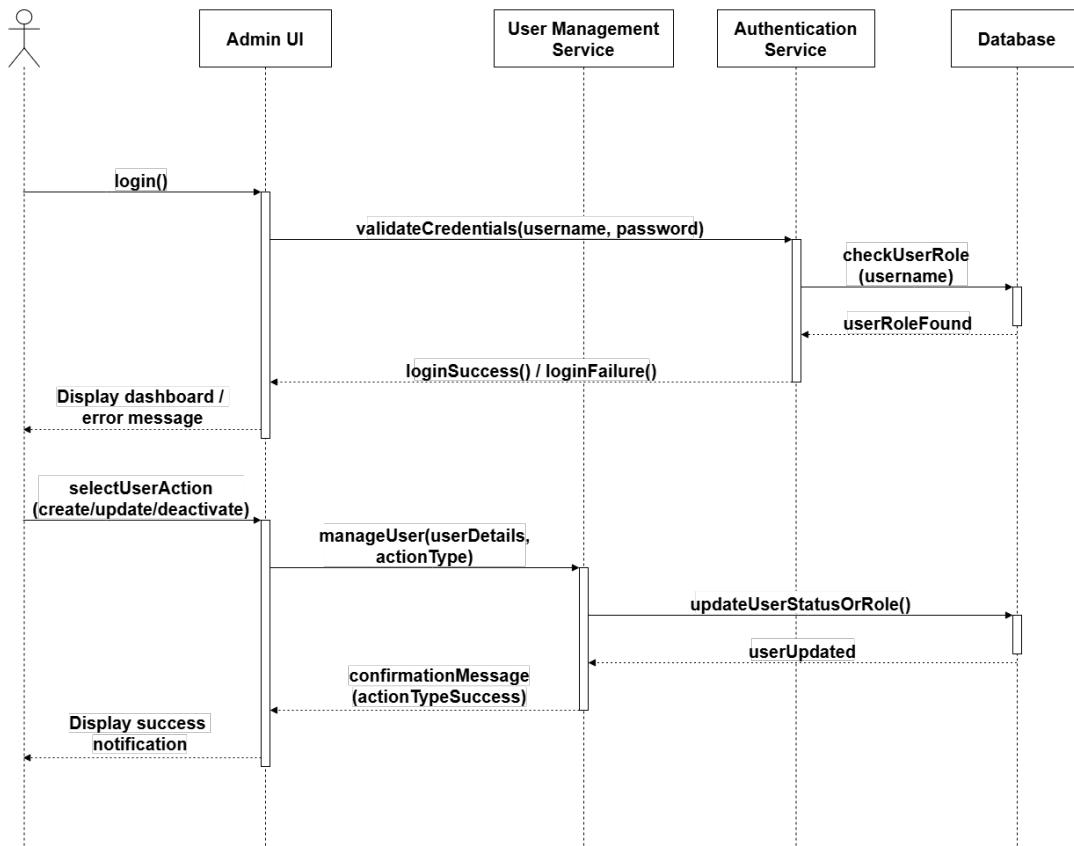


Figure 5.27: Admin User and Role Management

Login starts when the admin provides the username and password using the Admin

UI. After that, the username and password are transferred to the Authentication Service and checked in the database based on the user's username, password, and role. In case of failed validation, an error message is returned to the Admin UI.

In case of correct authentication, the admin will have access to the system and can work with dashboards. Moreover, it is possible for the admin to manage operations related to creating and updating users. For these actions, the Management Service communicates with the Authentication Service to modify the database.

Admin Dataset Management:

Figure 5.28 describes the working of histopathology datasets created by the admin used for AI training and validation. The system restricts data set access to admins only and preserves dataset integrity during all operations.

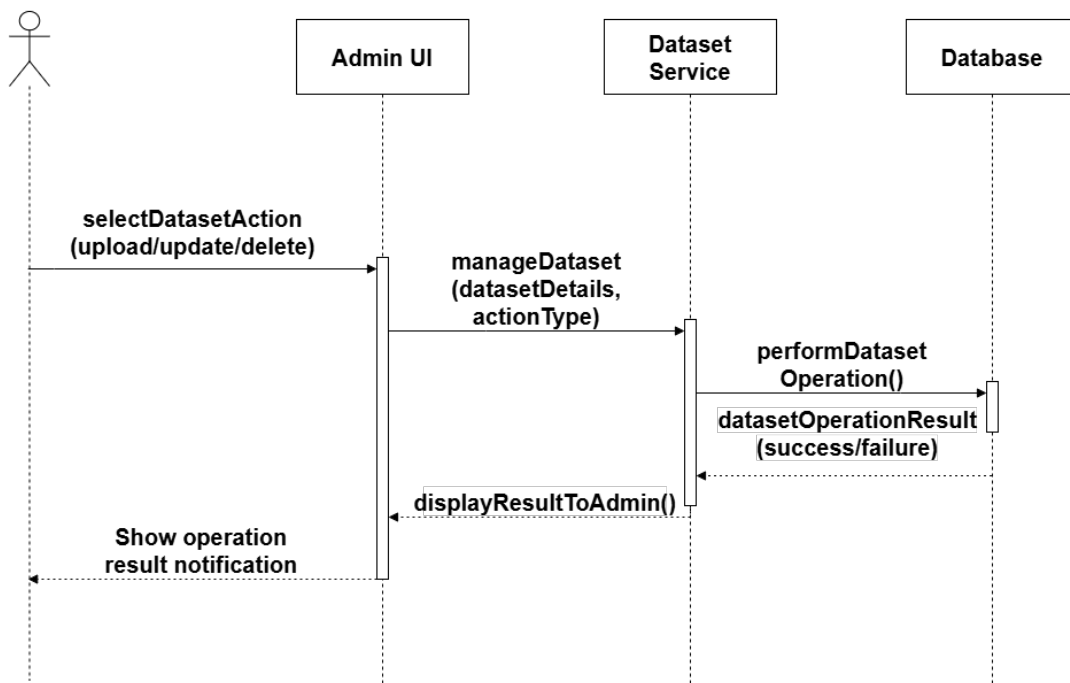


Figure 5.28: Admin Dataset Management

This sequence diagram represents the Dataset Management workflow among the Admin UI, Dataset Service, and Database. The admin makes a request from the Admin UI; it is then processed via the Dataset Service and implemented at the Database. The output (whether successful or not) is received and presented to the admin.

Admin Federated Learning Management:

In Figure 5.29, the admin monitors federated learning across decentralized institutions, where models are trained locally and only parameters are shared with the central server to ensure data privacy.

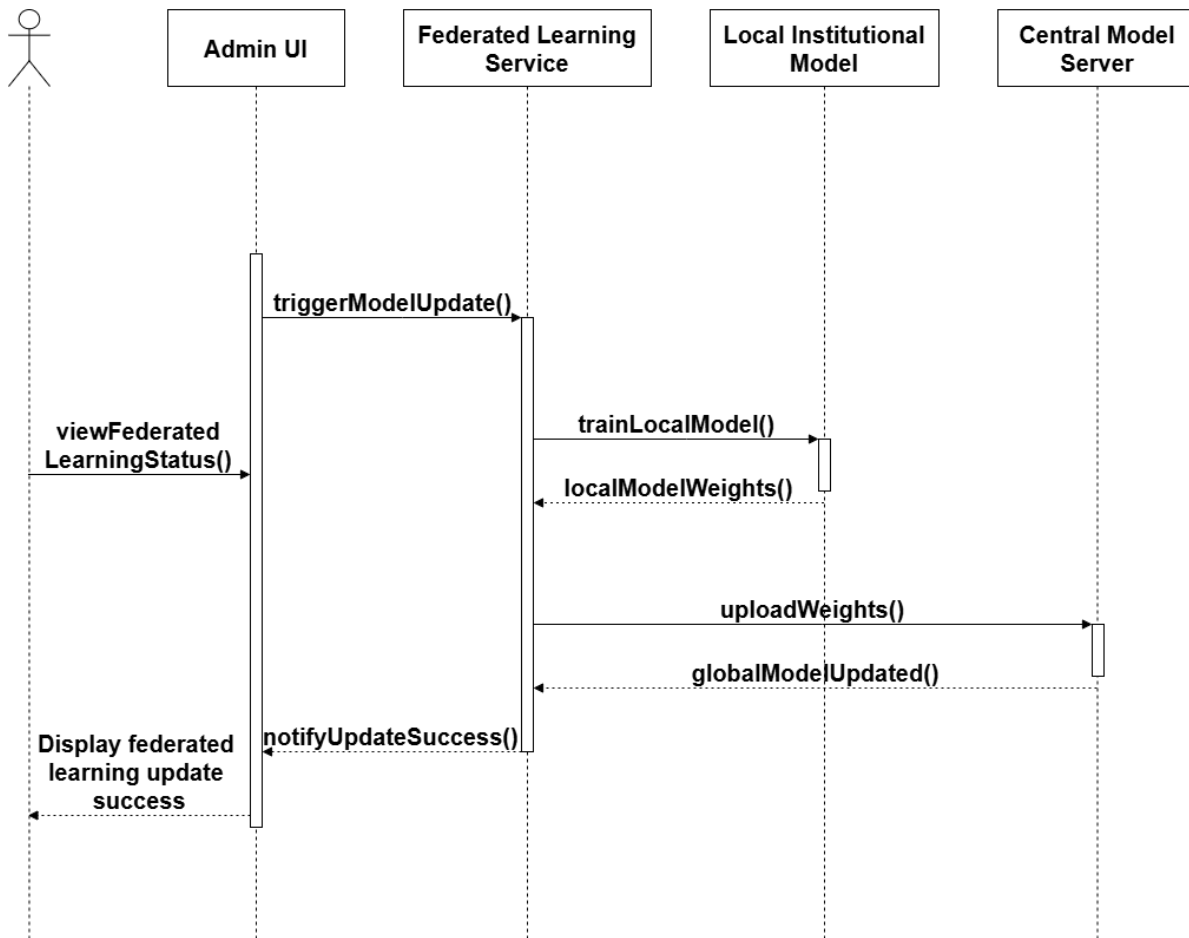


Figure 5.29: Admin Federated Learning Management

This procedure commences when the admin requests for an update in the status of federated learning together with triggering a model update using the Admin UI. This request is passed to the Federated Learning Service, which triggers local model training on-site without transferring raw data. The weights obtained from training the local model are passed to the central server, and they are used to update the global model. Finally, once done, the system notifies the Admin UI successfully updating the model.

System Monitoring and Audit Logs:

Figure 5.30 depicts the admin accessing the system activity logs for transparency and accountability purposes, whereby all important events are documented along with their time stamps and user information, but are visible only to authorized admins.

Activity logging plays a vital role in any software application by ensuring the maintenance of a consistent trail of events within the software system. It entails collecting relevant information regarding what actions were carried out, who carried out the activities, what was the time during which the operations took place, and where from the activity was performed.

On the other hand, audit logs go further than just keeping a record of all activities. They provide a chronologically ordered trail that can later on be referred to in case of

an audit process or security analysis. While activity logs serve the function of keeping track of all activities performed in a software application, audit logs provide more specific information concerning each action or transaction such as authentication attempts, information about successful and failed actions, and data concerning the access of information, among others. Audit logs also provide additional attributes such as severity levels and timestamps associated with actions.

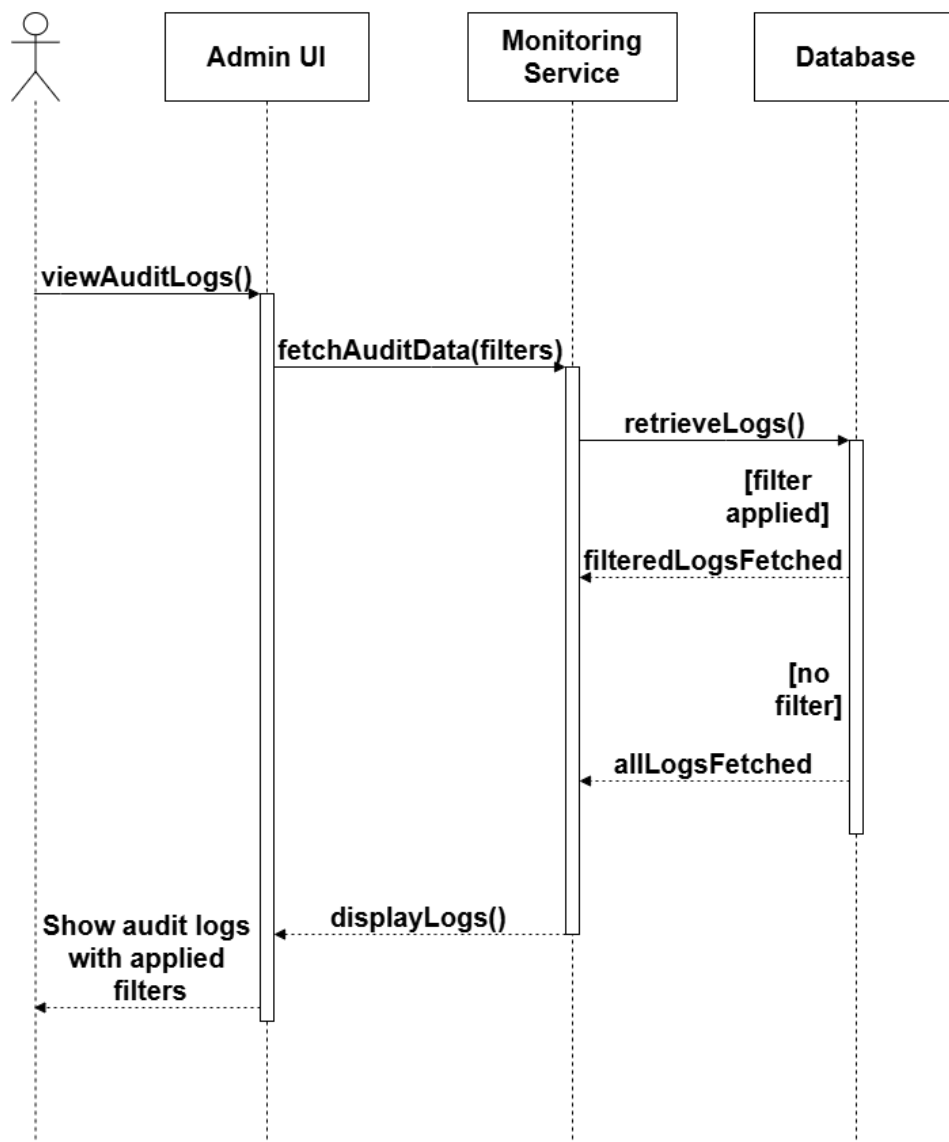


Figure 5.30: Admin System Monitoring and Audit Logs

The entire process commences when the admin requests viewing of audit logs using the Admin UI. The request is passed to the Monitoring Service, which collects the log information from the database based on the request. In case the request requires filtering the logs, filtered log information is collected, and, in the case where no filtering is requested, all logs are collected. The collected information is passed to the Admin UI.

Admin Dashboard and Analytics Management:

In Figure 5.31, the admin views real-time system analytics across patients, doctors, hospitals, and AI performance, with optional filters to narrow the data.

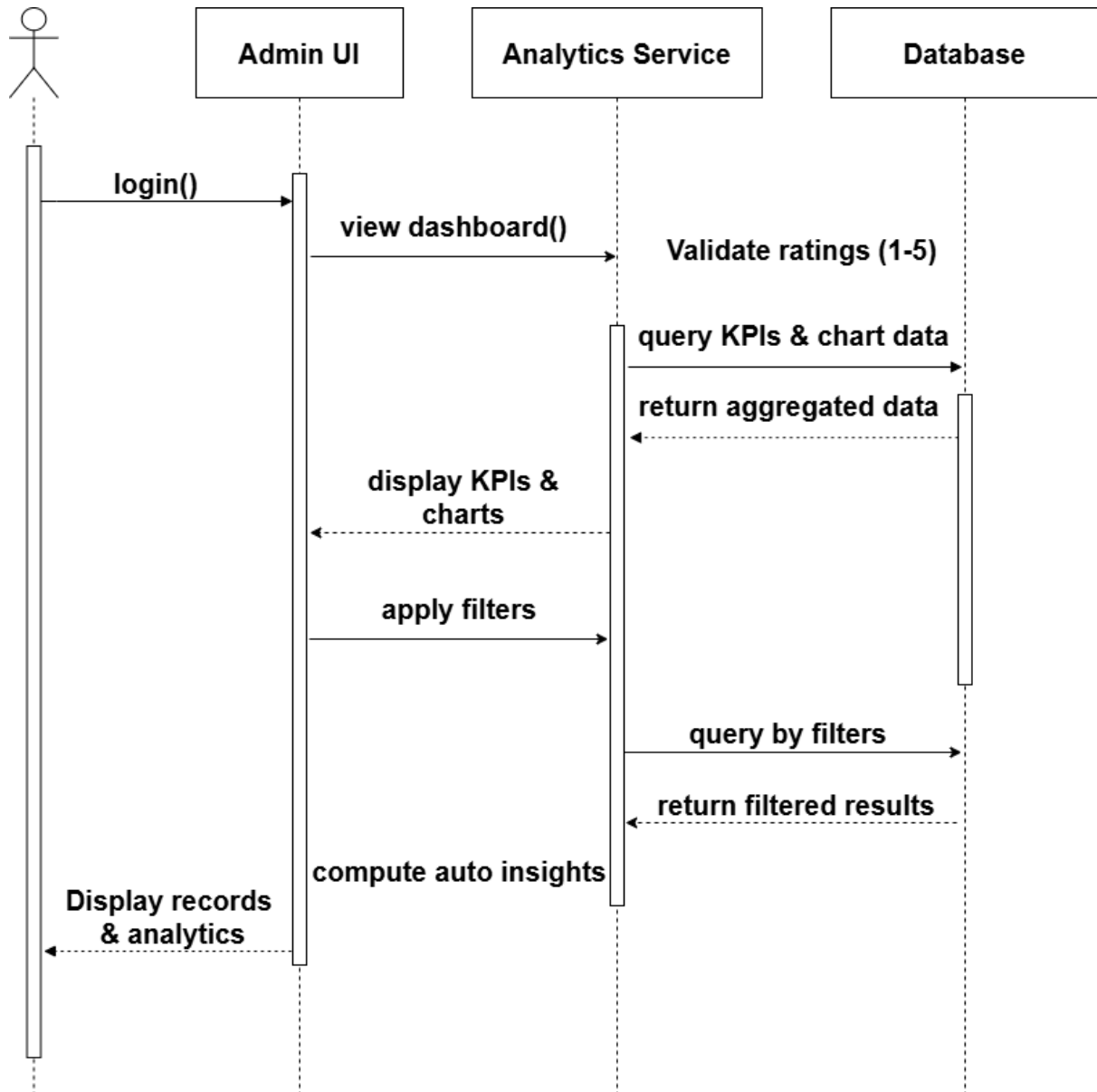


Figure 5.31: Admin Dashboard and Analytics Sequence Diagram

This procedure commences when the admin navigates to the analytics dashboard via the Admin UI. The request is forwarded to the Analytics Service, which queries the database for KPI totals and chart data. The database returns the aggregated results, and the Analytics Service passes the rendered KPIs and charts back to the Admin UI for display. The admin may then apply filters such as date range, doctor, hospital, or cancer type. The Analytics Service re-queries the database with the specified filter parameters, retrieves the filtered results, computes the automated plain-English insights, and returns the final filtered data and insights to the Admin UI for the admin to review.

Admin Feedback Management:

In Figure 5.32, users across all roles submit satisfaction feedback, and the admin reviews the aggregated responses with per-role and per-question analytics.

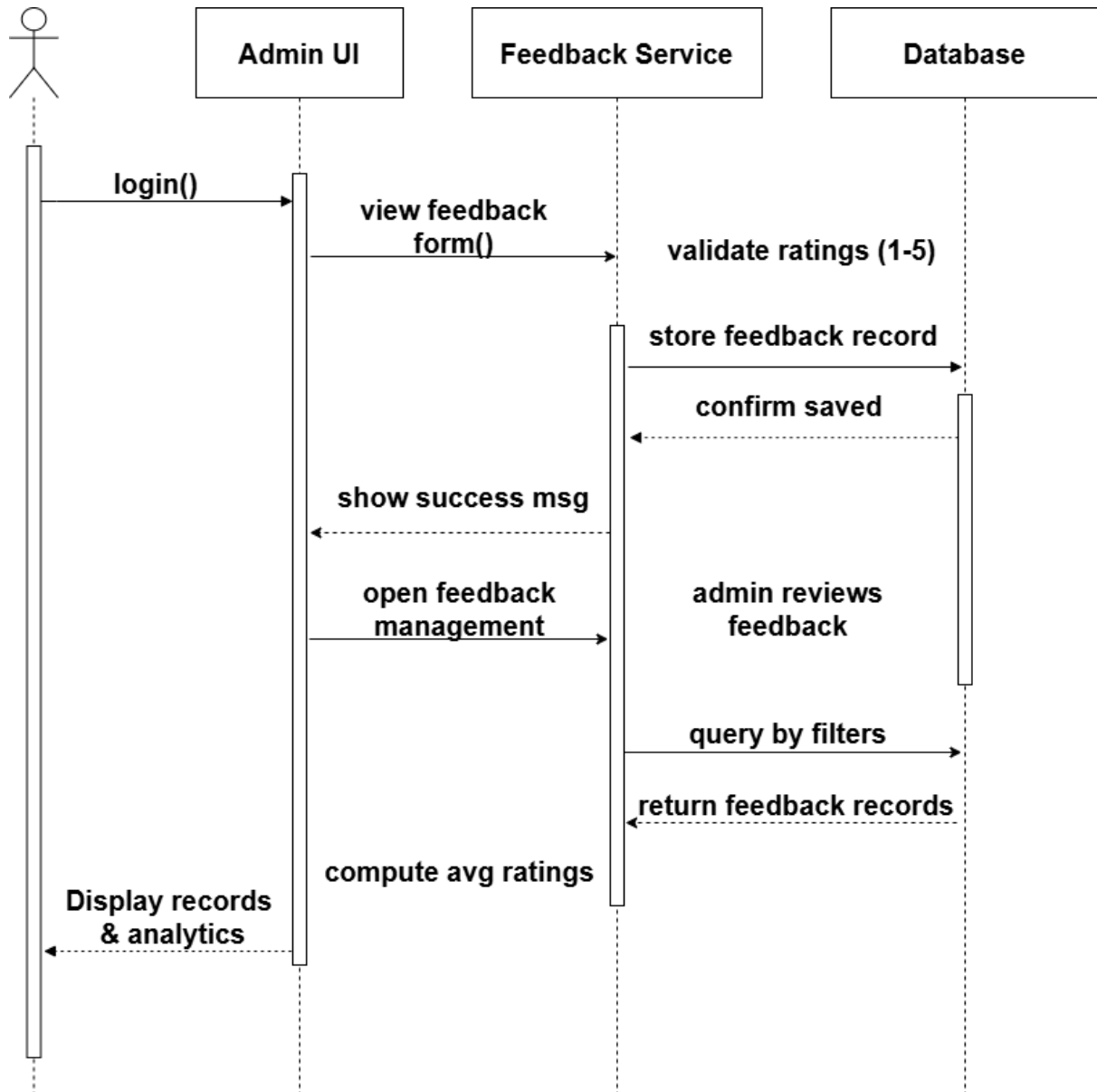


Figure 5.32: Admin Feedback Management Sequence Diagram

This procedure commences when a user submits the feedback form via the User UI. The Feedback Service validates that all submitted ratings fall within the 1–5 range; if any rating is invalid, an error is returned to the user. If validation passes, the service stores the feedback record in the database with the user’s role, username, and timestamp attached, and returns a success confirmation to the user. In a separate flow, the admin opens the feedback management page. The Feedback Service queries the database using the admin’s selected filters such as role, keyword, or date range, retrieves the matching records, computes the average ratings and per-role analytics, and displays the full results

to the admin for review.

Admin Notification:

In Figure 5.33, the system automatically notifies the admin via email whenever a new doctor or patient completes registration, prompting the admin to approve or reject the account.

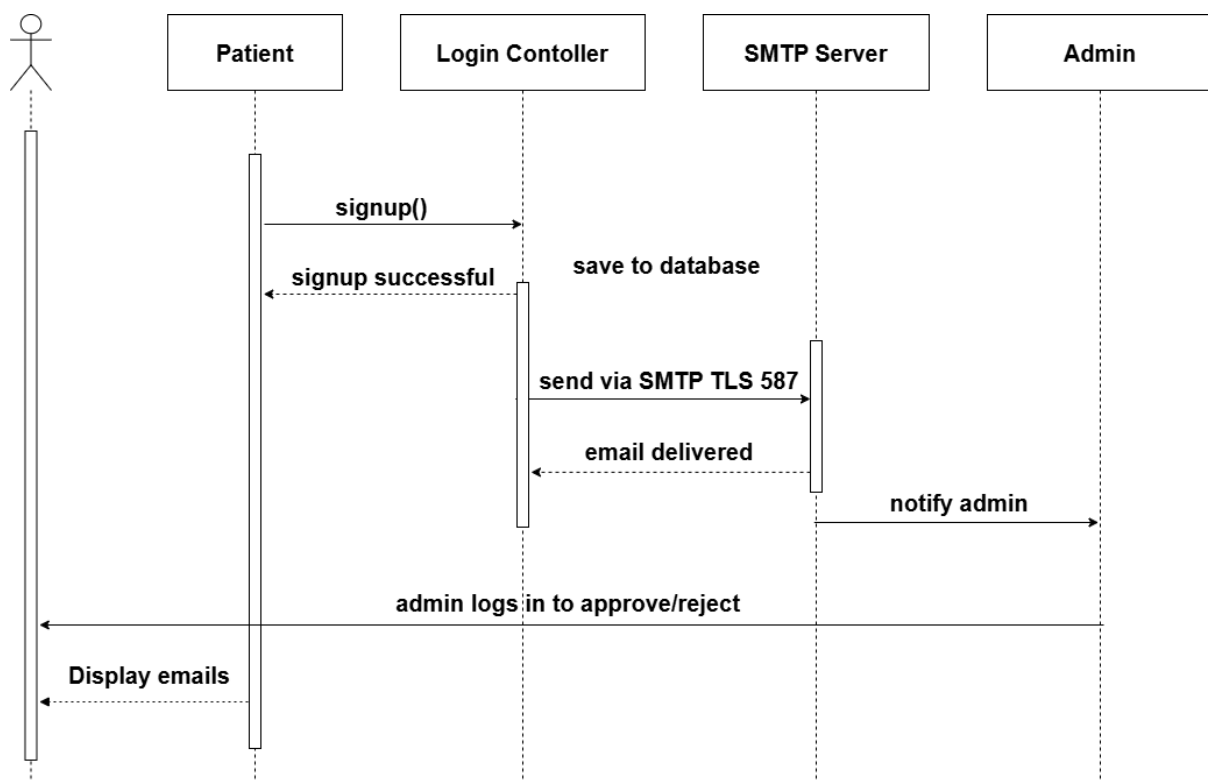


Figure 5.33: Admin Notification Sequence Diagram

This procedure commences when a new doctor or patient completes the registration form via the User UI. The Login Controller saves the new user record to the database and returns a registration success message to the user.

The controller then prepares the appropriate notification email, labeled `docAdded` for a doctor registration or `patientAdded` for a patient registration, and dispatches it to the admin's email address via SMTP using TLS on port 587. The SMTP server delivers the email and confirms delivery to the controller. The admin receives the notification and subsequently logs into the system to either approve or reject the pending account.

5.2.3.2 Doctor's Sequence Diagrams

These doctors have been granted permission by the government to become legitimate practitioners of medicine who will be able to use the system for uploading the histopathology images, analyzing them through AI technology, and producing reports for diagnosis.

Secure Sign Up, Approval, and Authentication:

Figure 5.34 shows the procedure for doctor registration and authentication, in which credentials and information submitted by the doctor stay on hold until being authenticated by the admin, and once their account is authenticated, then they can login to the system using RBAC.

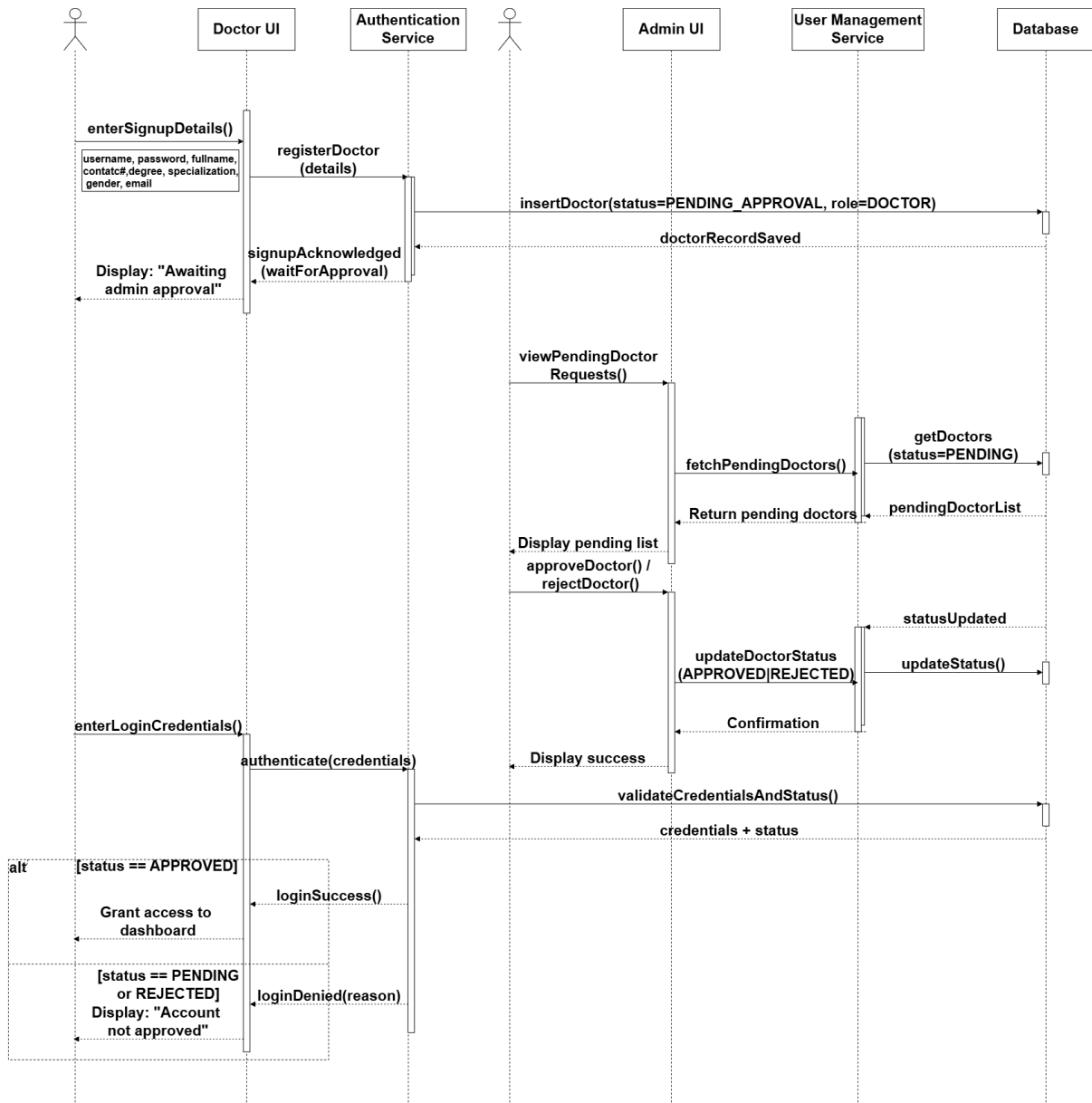


Figure 5.34: Doctor Sign Up

The doctor first fills up his signup information including the username, password, education, and specialization by using the Doctor UI, thus triggering a registerDoctor(details) call to the Authentication Service. The service saves the doctor record in the Database with status PENDING_APPROVAL and returns signupAcknowledged message to the doctor indicating that he needs to wait for admin approval. The admin sees the list of doctors who have applied for signup. The User Management Service checks the Database

to get the list of those pending signup requests. The admin can then choose to either accept or deny the doctor's application.

Histopathology Image Upload:

Figure 5.35 illustrates the physician uploading histopathology pictures of a chosen patient through a secure user interface, where the application verifies the format of the picture file, links the picture to the patient's medical record, and stores it safely for further processing.

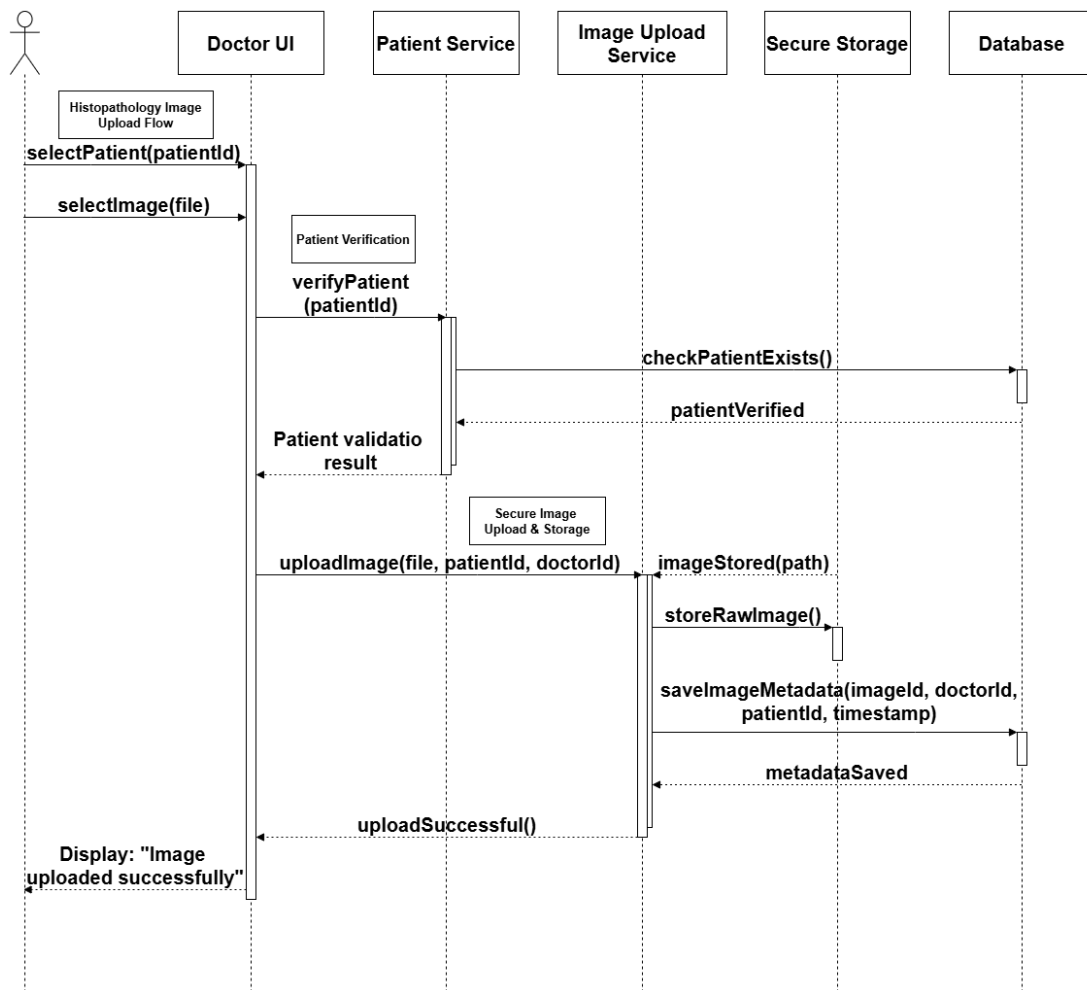


Figure 5.35: Doctor Histopathology Image Upload

Then the doctor uses Doctor portal to pick one patient and one file with an image; after that, Patient Service verifies the existence of the selected patient. Further, the doctor uses uploadImage. Metadata for the uploaded image, meanwhile, will be stored in the Database by Image Upload Service. Once metadataSaved comes from Image Upload Service, the end process will give rise to the following: uploadSuccessful() that means “Image uploaded successfully” in Doctor UI.

Image Pre-Processing:

Figure 5.36 shows the automated process used to preprocess images. This process includes resizing the images, normalizing the pixels' values, and decreasing the amount of noise present in the images to produce high-quality images that can be accurately analyzed using artificial intelligence.

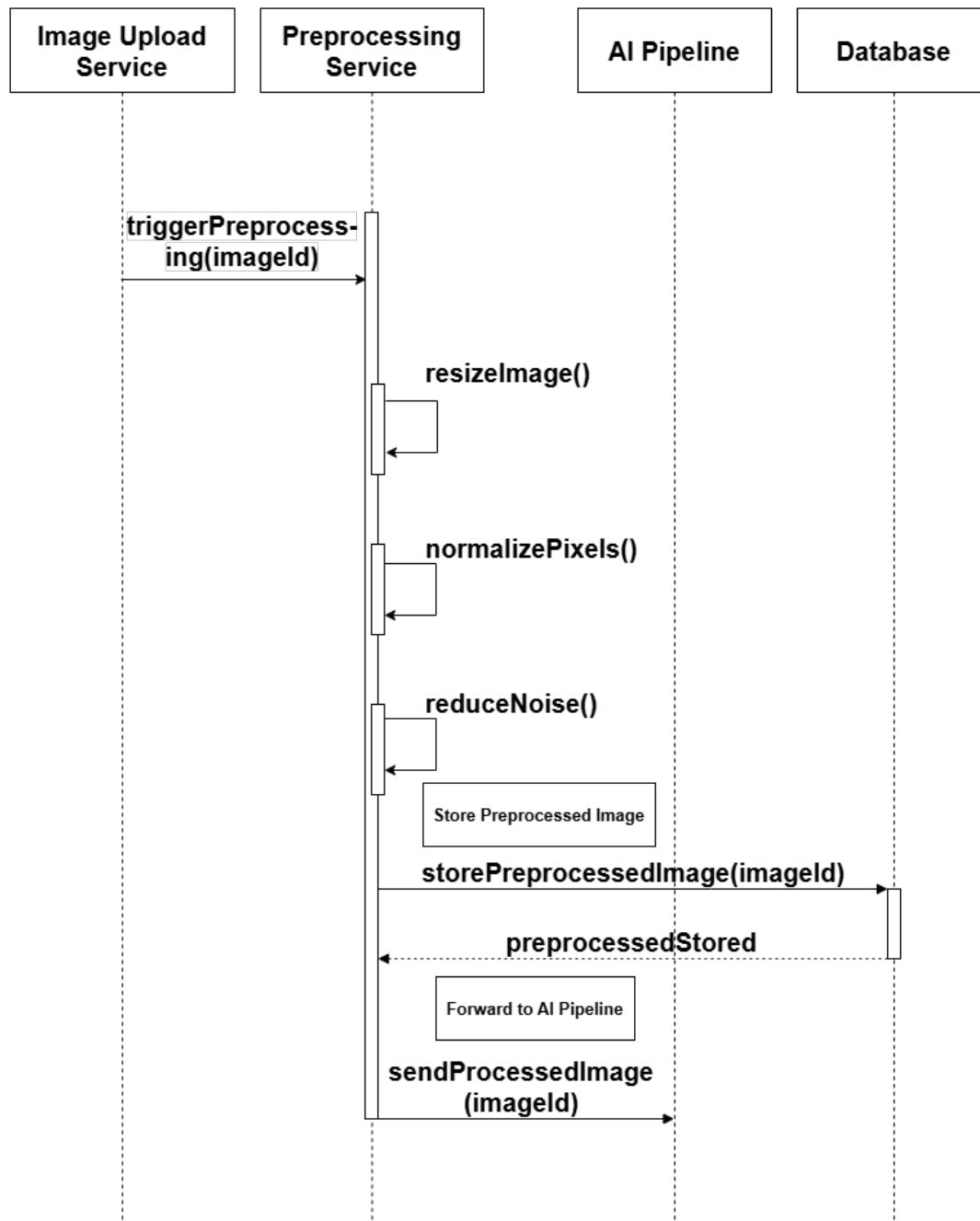


Figure 5.36: Doctor Image Pre-Processing

Additionally, augmentation techniques can be applied to improve model generalization and robustness. Common augmentation methods include rotation, flipping, scaling, and contrast adjustment, which artificially expand the dataset and help the model learn

invariant features. Normalization ensures that pixel intensity values are brought to a common scale, improving convergence during model training. Noise reduction techniques, such as filtering, further enhance image quality by removing irrelevant distortions.

Cancer Detection Using AI Model:

Figure 5.37 explains that after pre-processing the image, the AI model analyzes the preprocessed image to classify it as either malignant or benign through binary classification. Predictions are generated and presented in a reasonable response time before generating the report.

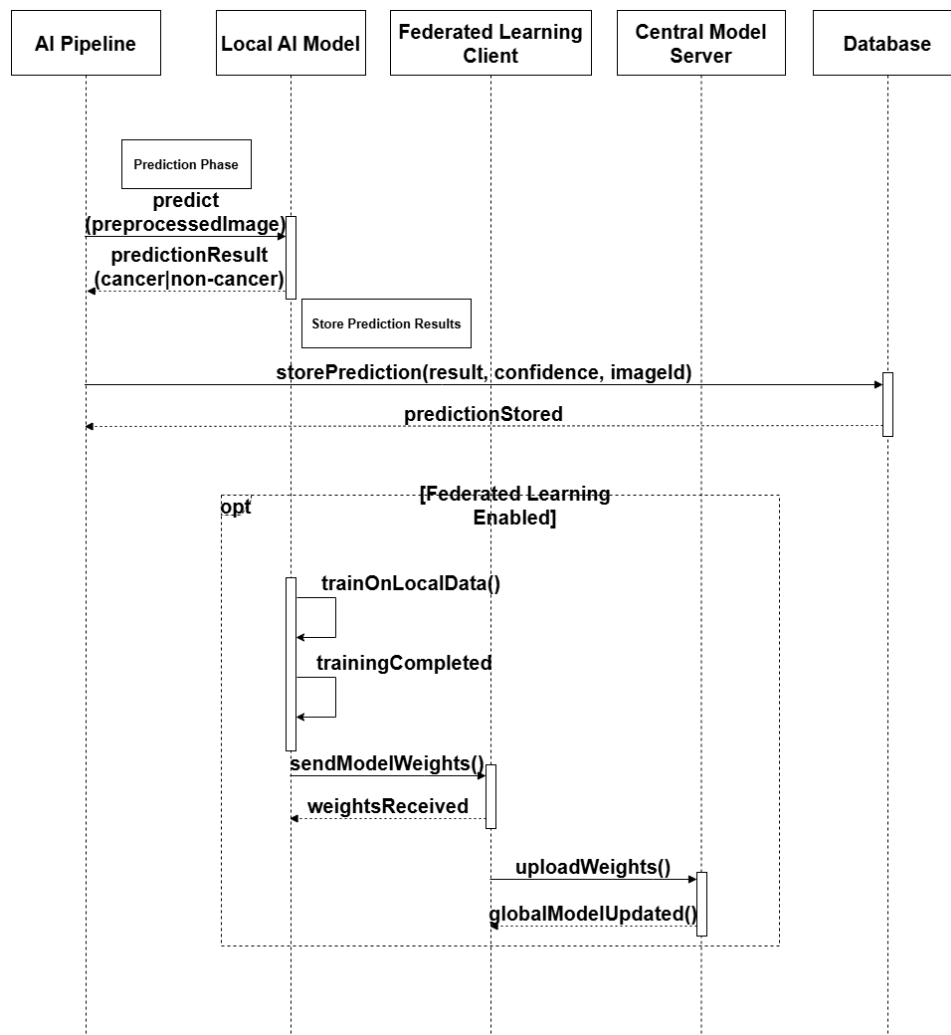


Figure 5.37: Doctor Cancer Detection Using AI Model

The AI Pipeline calls the Local AI Model predict, where it predicts a cancer or non-cancer outcome and stores it in the Database. This returns a predictionStored event. When FL is used, the Local AI Model trains the local data using, and after completing its training process, the model sends the weights to the FL Client. The weights are uploaded to the Central Model Server by the FL Client using the uploadWeights() function, where they are aggregated and return a globalModelUpdated() event.

Diagnostic Report Generation and History Access:

Figure 5.38 explains that the system will produce a report showing the predictions made as well as patient information. It is possible for doctors to check and access the full medical history of their patients from their diagnostics. This will help them monitor how a disease progresses and make better decisions. Furthermore, the reports can be saved for future use. Finally, it ensures that the medical history of the patient will be available when needed.

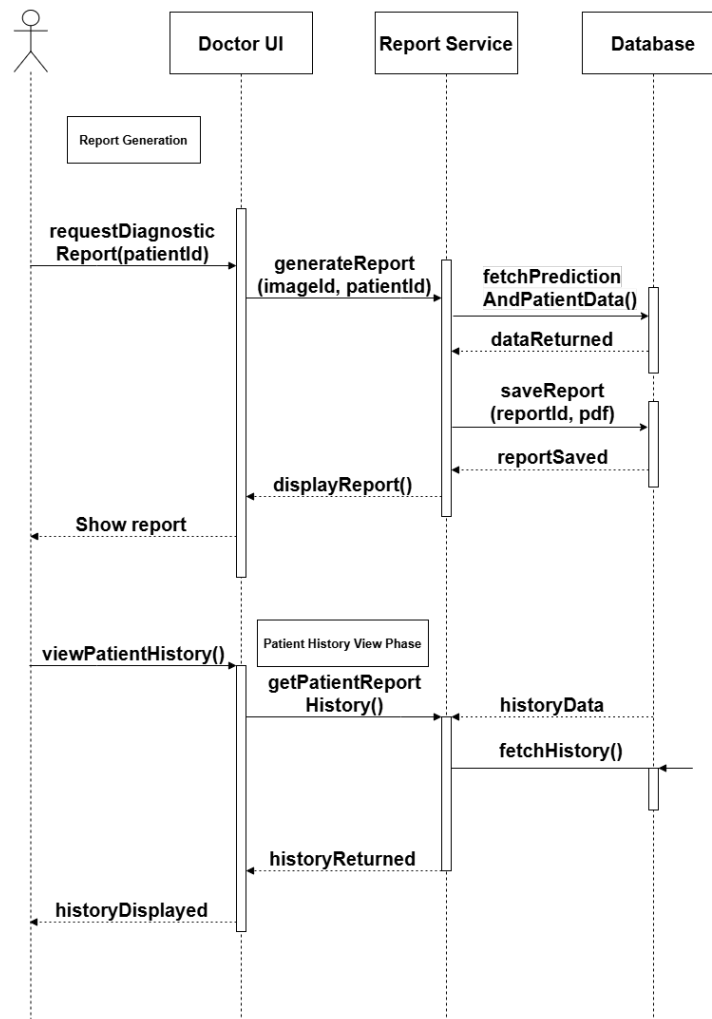


Figure 5.38: Doctor Diagnostic Report Generation and History Access

The request for the diagnosis is made by the doctor via the Doctor UI. It then passes on this request to the Report Service, which gathers all the required information about the patient and his prediction from the database and makes a report in PDF format. In the following step, the doctor gets access to patient history, in which Report Service provides the previous reports of the patient from the database.

Doctor Feedback:

In Figure 5.39, the doctor submits a structured satisfaction questionnaire rating admin

support, system performance, AI quality, report upload, and dashboard ease.

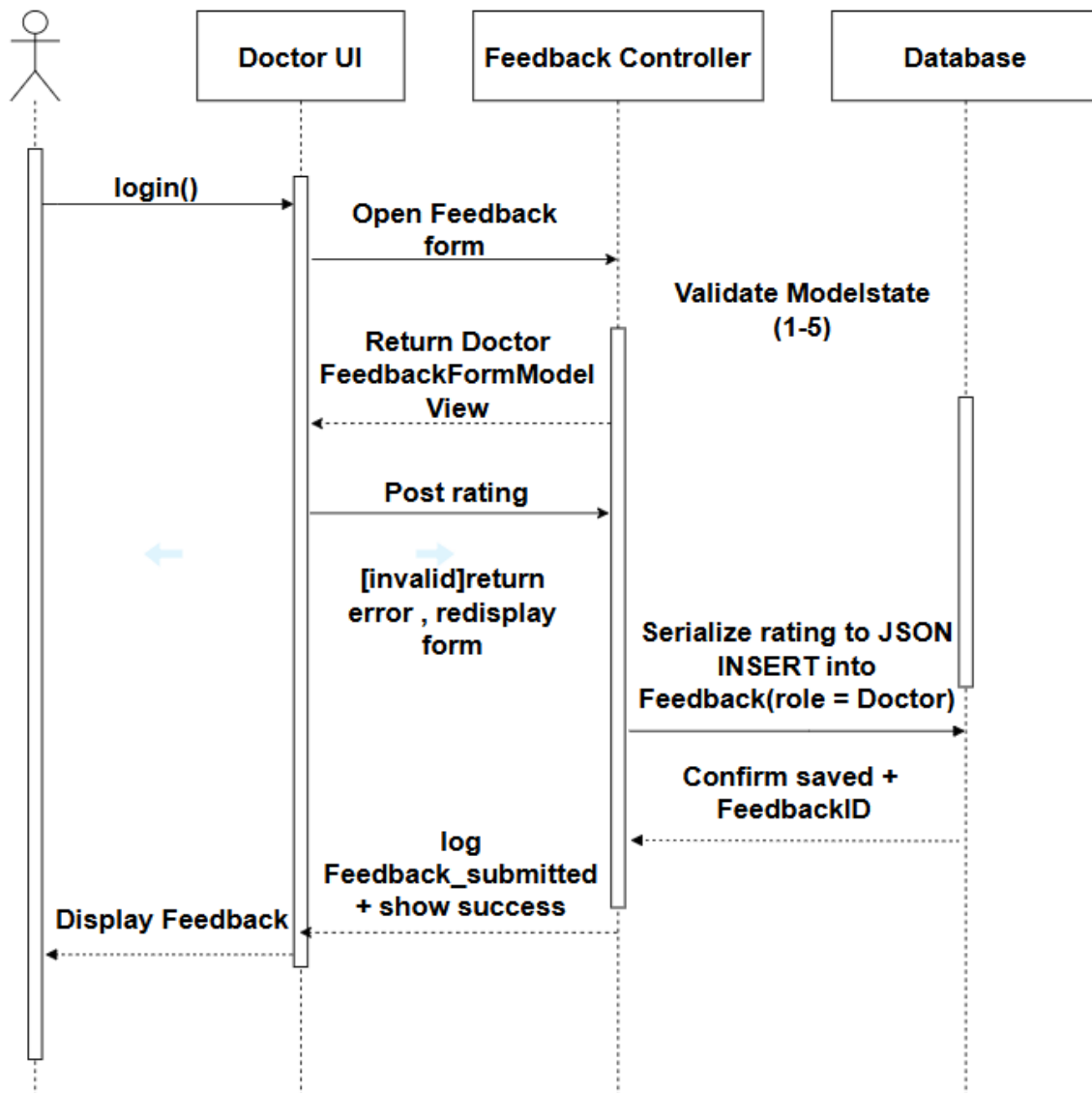


Figure 5.39: Doctor Feedback Sequence Diagram

This procedure commences when the doctor opens the feedback form via GET /Feedback/SubmitDoctor. The Feedback Controller returns the DoctorFeedbackFormModel view. The doctor fills in five rating fields and submits the form via POST. The controller validates the model state, ensuring all ratings fall within the 1–5 range; if validation fails, an error is returned and the form is redisplayed. If valid, the ratings are serialized to JSON and the Feedback Service inserts a new record into the Feedbacks table with role set to Doctor and target type set to admin. The database confirms the save with the new FeedbackID, and the controller logs a FEEDBACK_SUBMITTED audit event and returns a success message to the doctor.

Doctor Notification:

In Figure 5.40, the system sends a 6-digit OTP to the doctor's registered email during login to verify their identity before granting dashboard access.

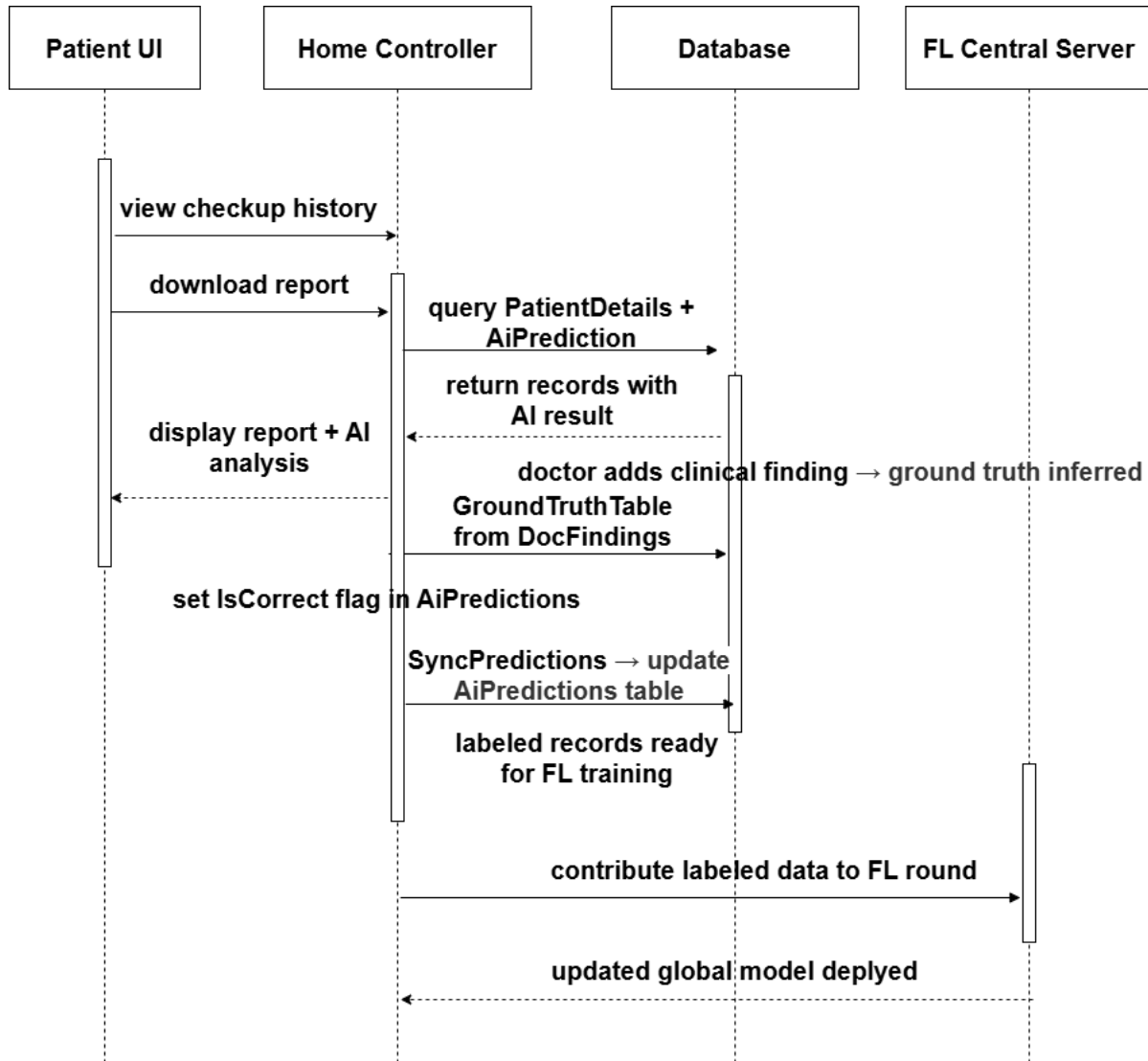


Figure 5.40: Doctor Notification Sequence Diagram

This procedure commences when the doctor submits their login credentials via the Doctor UI. The Login Controller validates the credentials and checks that the account status is approved. Upon successful validation, the controller generates a 6-digit OTP with a 5-minute expiry and dispatches it to the doctor's registered email address via SMTP using TLS on port 587. The SMTP server confirms delivery. The doctor receives the OTP email and enters the code into the verification form. The controller validates the submitted OTP and, if correct, grants the doctor access to the dashboard. If the OTP is incorrect or expired, an error is returned.

Doctor Federated Learning:

In Figure 5.41, the doctor contributes to federated learning indirectly by uploading histopathology images, which are used by the local institutional model during FL training rounds without leaving the hospital.

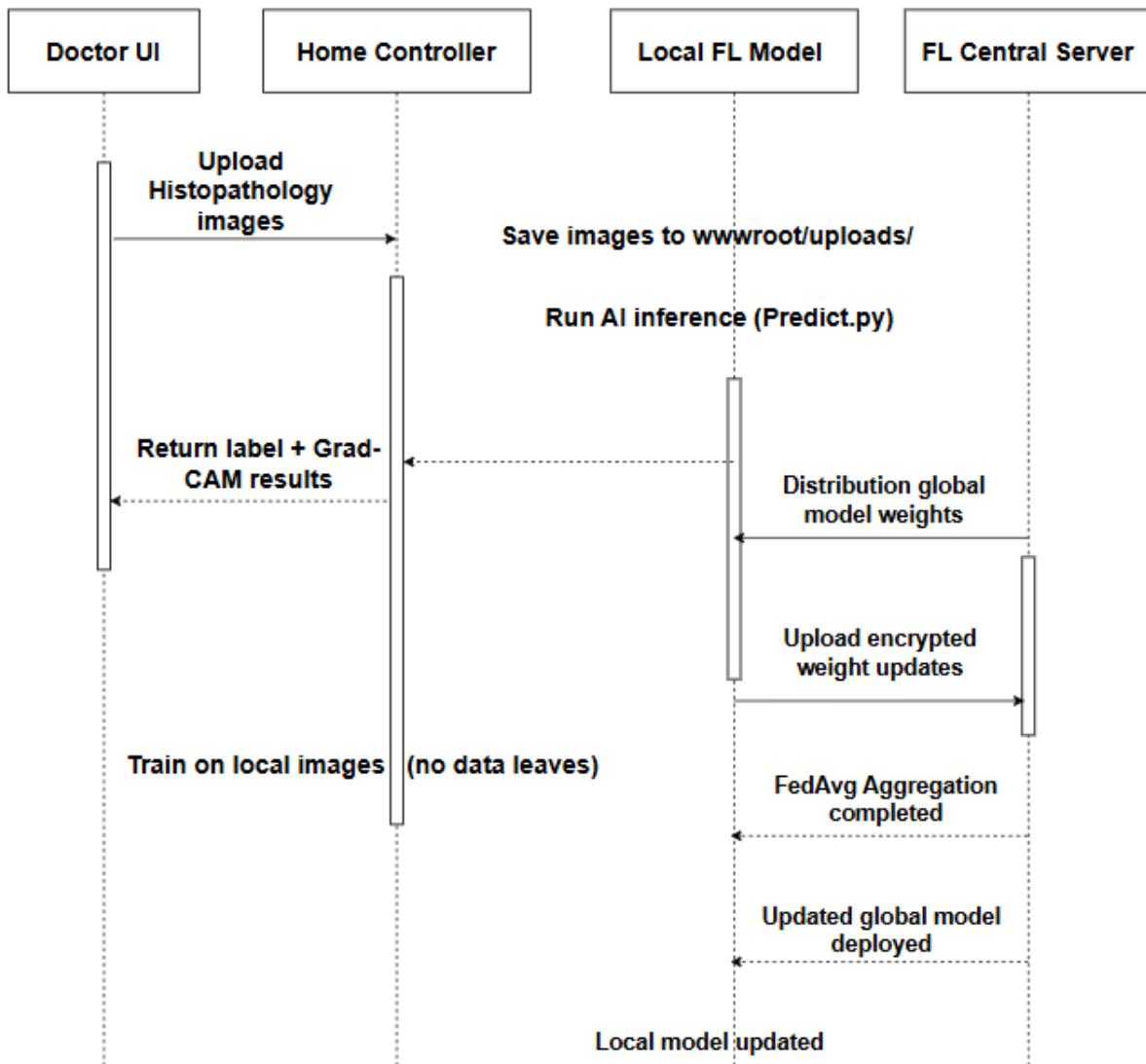


Figure 5.41: Doctor Federated Learning Sequence Diagram

This procedure commences when the doctor uploads a histopathology image via the Doctor UI. The Home Controller saves the image to the server's directory and invokes the Python inference pipeline to produce the AI result, which is returned to the doctor. When the admin initiates an FL round, the FL Central Server distributes the current global EfficientNet-B0 model weights to the Local Institutional Model at each hospital. The local model trains on the doctor-uploaded images. Once local training completes, the local model uploads only the encrypted weight updates to the FL Central Server. The

server applies FedAvg aggregation and, if the updated model improves accuracy, deploys the new global model back to all participating hospitals.

5.2.3.3 Patient's Sequence Diagrams

Patients access the system to view their history, results and reports. Ensures patient data privacy by enforcing secure authentication, admin approval, and restricted access to only the patient's own diagnostic information and history.

Secure Signup, Approval, and Login:

Figure 5.42 describes how the patients sign up and wait for admin approval. Once approved, they can securely log in and access only their own medical data; rejected users are denied access.

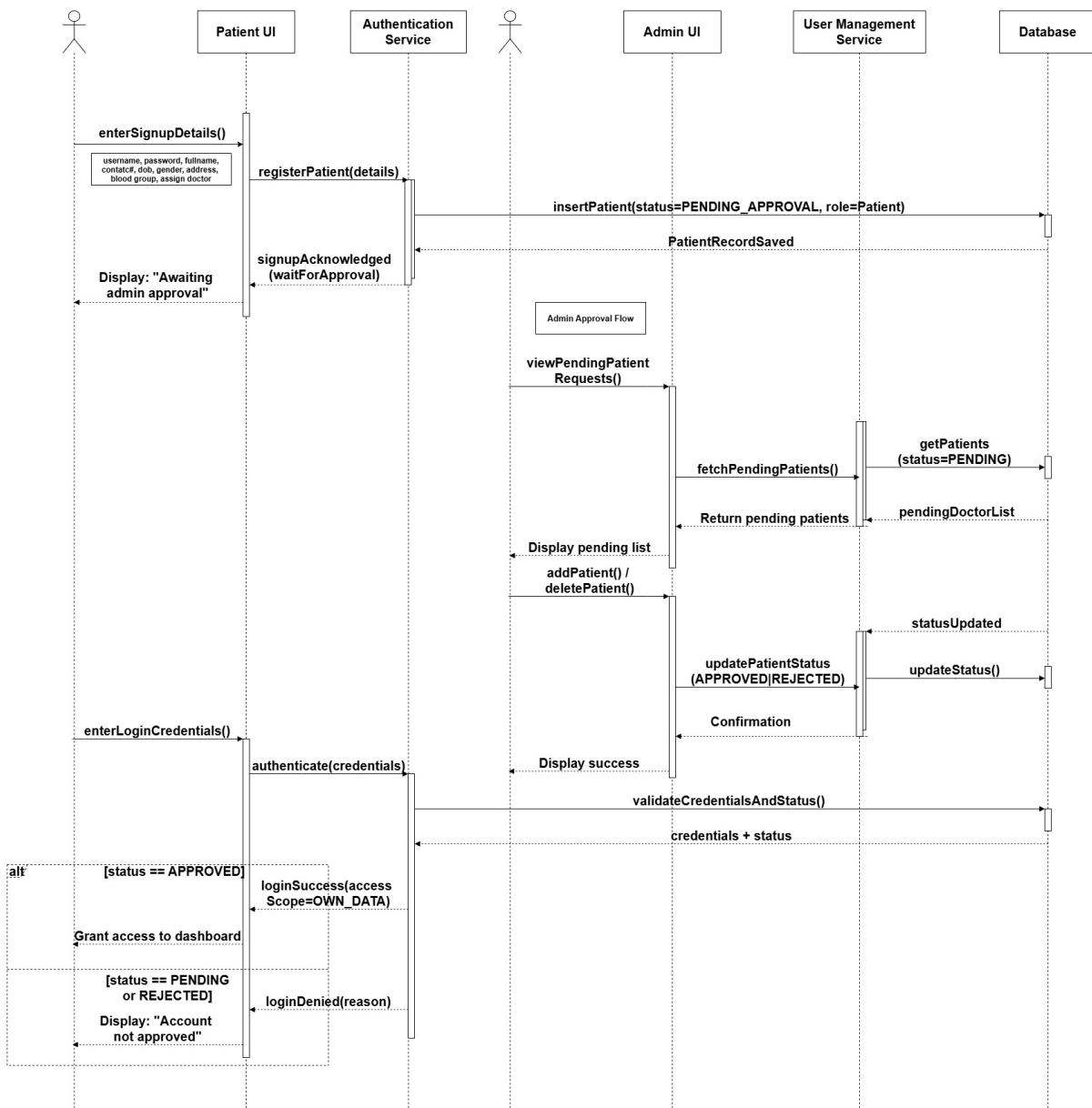


Figure 5.42: Patient Sign Up

Patient inputs his details using the Patient UI and sends them to the Authentication Service. The details are saved in the database under the "pending approval" tag, and then the patient is asked to wait until approval from the admin. An admin checks out these details using the User Management Service and can either approve or reject them depending on the database entry. While logging in, the credentials are checked. In case the patient's credentials have been approved, he will be logged in successfully, and if not, an error message will come stating "account not approved."

View Diagnostic Results and History:

Figure 5.43 explains that the patients request their diagnostic reports. Patients can view, download, and access the complete history of their diagnostic reports while maintaining data privacy.

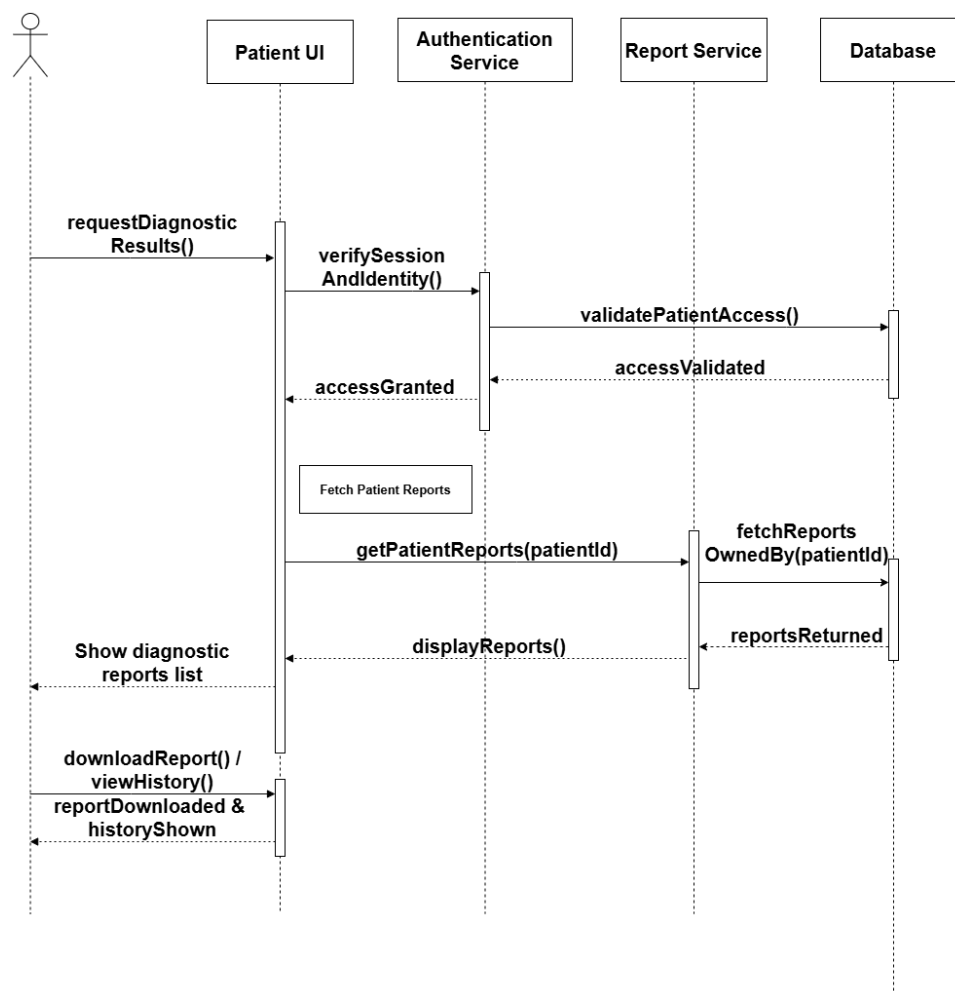


Figure 5.43: Patient View Diagnostic Results and History

The patient will be able to view recent and historic results of the diagnostic test(s). The Patients UI will request diagnostic results for the patient. The authentication service will contact the database to confirm the patients ability to access the results.

Patient Feedback:

In Figure 5.44, the patient submits a satisfaction questionnaire rating doctor behavior, report quality, usability, download ease, and overall satisfaction.

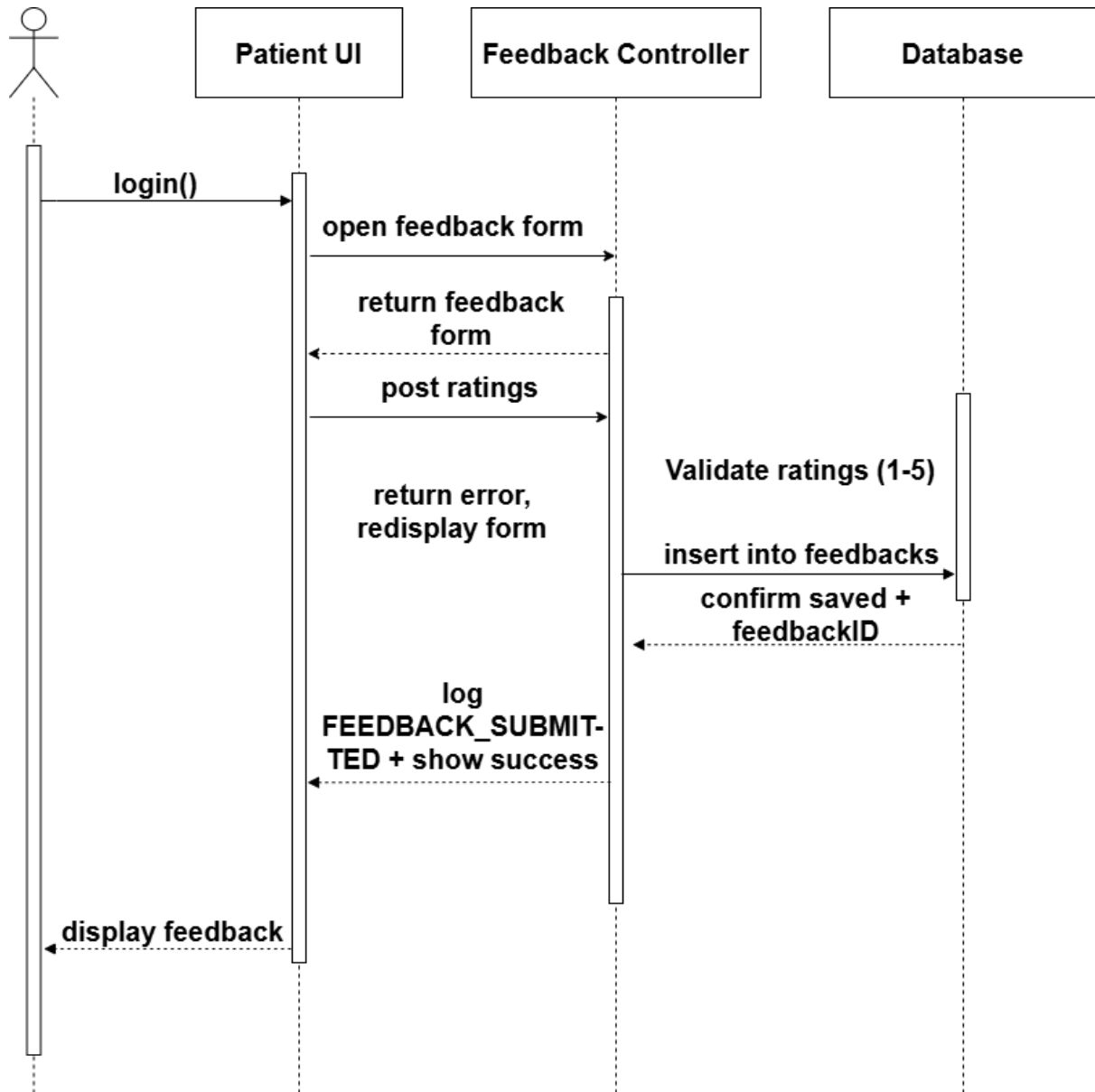


Figure 5.44: Patient Feedback Sequence Diagram

This procedure commences when the patient opens the feedback form via GET /Feedback/SubmitPatient. The Feedback Controller returns the PatientFeedbackFormModel view. The patient fills in five rating fields along with an optional suggestions field and submits via POST.

The controller validates that all ratings fall within the 1–5 range; if any rating is invalid, an error is shown and the form is redisplayed. If valid, the ratings are serialized to JSON and the Feedback Service inserts a new record into the Feedbacks table with role set to Patient and target type set to doctor. The database confirms the save, and

the controller logs a `FEEDBACK_SUBMITTED` audit event before returning a success confirmation to the patient.

Patient Notification:

In Figure 5.45, the system automatically emails the patient when their AI diagnostic report is ready, triggered immediately after the AI analysis completes.

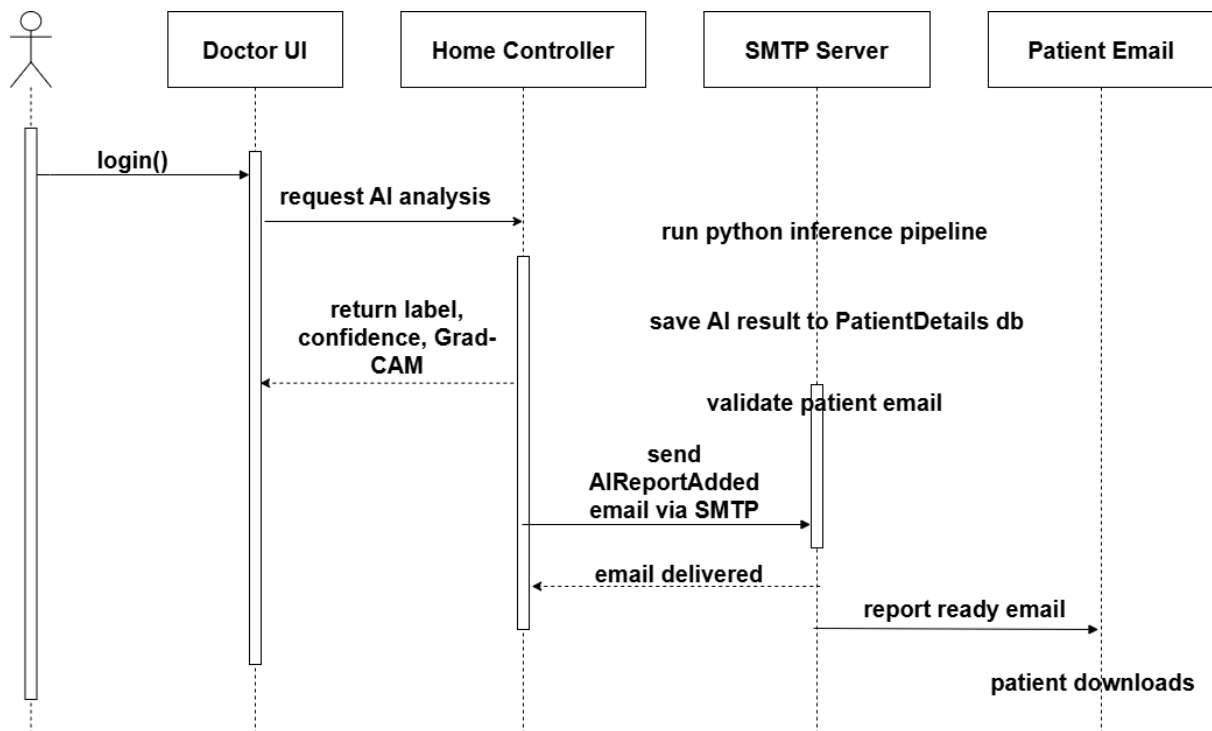


Figure 5.45: Patient Notification Sequence Diagram

The above sequence diagram in the UML notation shows the procedure flow of the AI-enabled medical analysis system with its four significant blocks such as Doctor UI, Home Controller, SMTP Server, and Patient Email. Firstly, the physician logs into the application using the Doctor UI and starts an analysis process requesting an AI service. After receiving a request from the user interface, the Home Controller starts a Python model inference process, stores the obtained AI analysis results such as labels, confidence values, and Grad-CAM visualizations into the PatientDetails database and returns those results to the UI again.

Once the physician obtains the required results, the system continues the workflow and starts sending out notifications about the results availability to the patients automatically. Thus, after checking the patients' email addresses, the Home Controller proceeds with sending out an "AIReportAdded" notification through the SMTP Server. Next, when an email gets sent to a certain address, the server informs the home controller that the email was sent and returns the notification that the "report ready" email is now in patients' mailboxes.

Patient Federated Learning:

In Figure 5.46, a patient's confirmed diagnosis contributes ground-truth labels that passively improve the shared federated learning model without any patient data leaving the hospital.

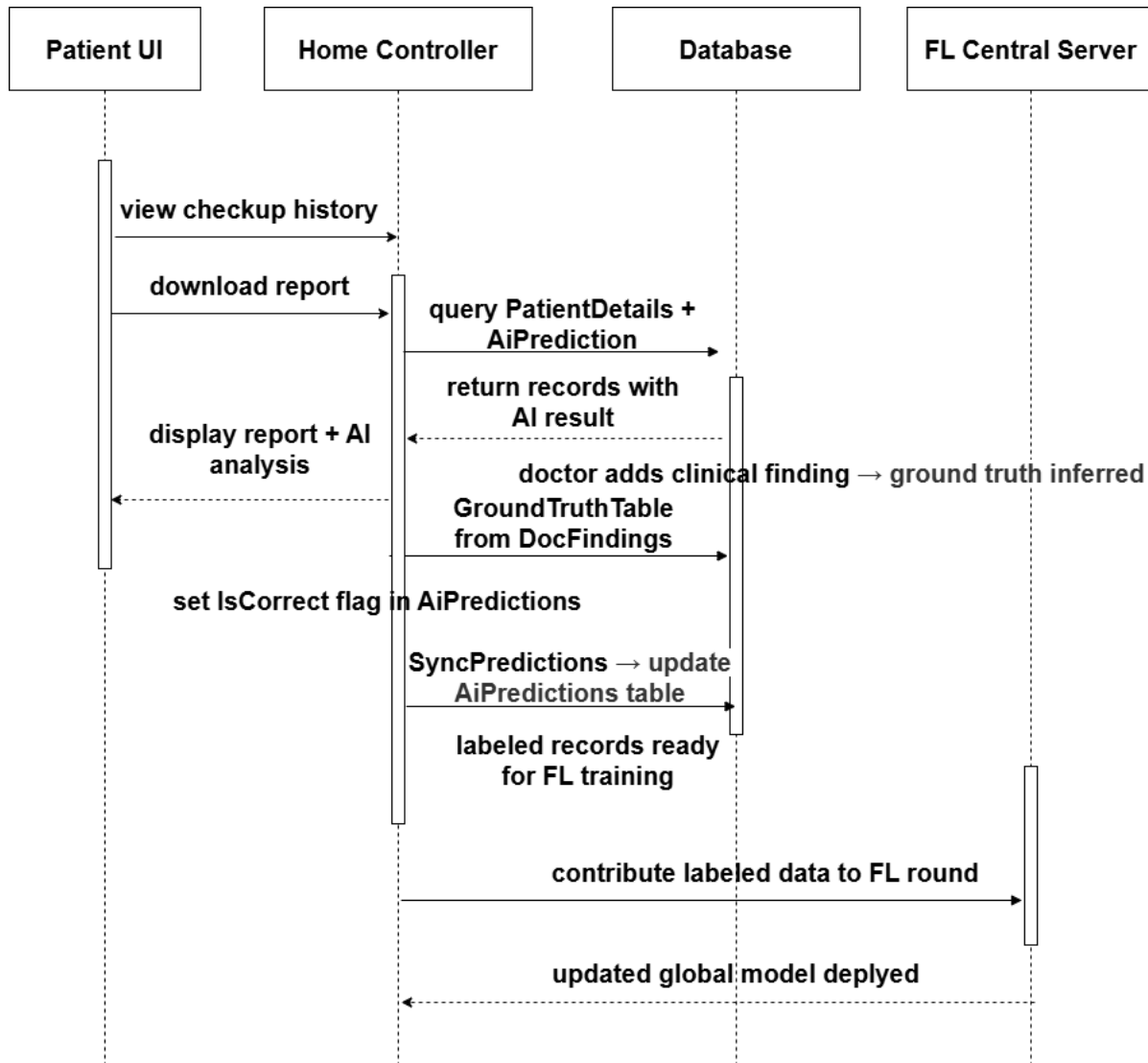


Figure 5.46: Patient Federated Learning Sequence Diagram

This procedure commences when the patient views their checkup history or downloads a report via the Patient UI. The Home Controller queries the PatientDetails and AiPredictions tables and returns the records including the AI result and Grad-CAM explanation. When the attending doctor subsequently records a clinical finding in the DocFindings field, the system infers the ground-truth label by comparing the doctor's diagnosis with the AI prediction and sets the IsCorrect flag accordingly in the AiPredictions table.

Chapter 6

SYSTEM INTERFACE AND PHYSICAL DESIGN

6.0.1 System User Interfaces

The platform gives a web-based UI with specific roles dashboards for Admins, Doctors, and Patients, supporting features like AI prediction, image upload, and report viewing. It is designed to be self explanatory for all users regardless of any background, while making sure secure access through authentication and role-based permissions.

6.0.1.1 Homepage Interface

In figure 6.1 it can be seen that the homepage is the main entry point of the system, which gives the overall interface of the cancer detection system and its capabilities. The homepage provides a smooth navigation experience to various pages like services, doctors, and patient login pages.

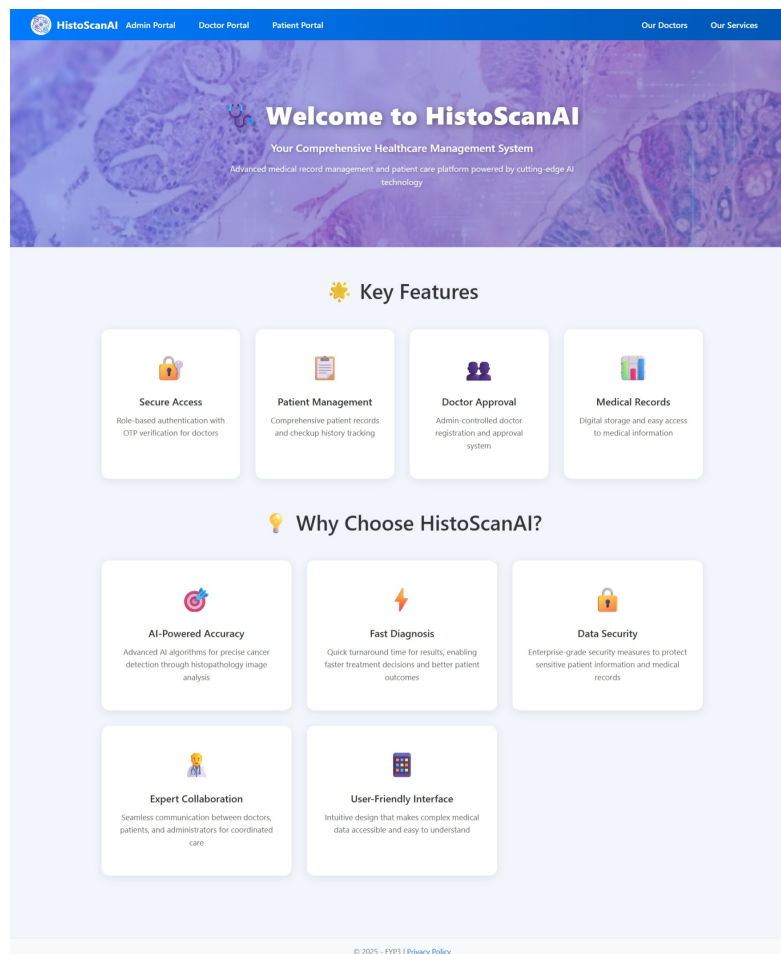


Figure 6.1: Homepage Interface

6.0.1.2 Our Doctors Interface

The doctor’s interface is shown in Figure 6.2 below. When the cursor hovers on a particular doctor, information about that specific doctor is displayed including the name of the doctor, specializations, number of years in practice, and doctorate degree.

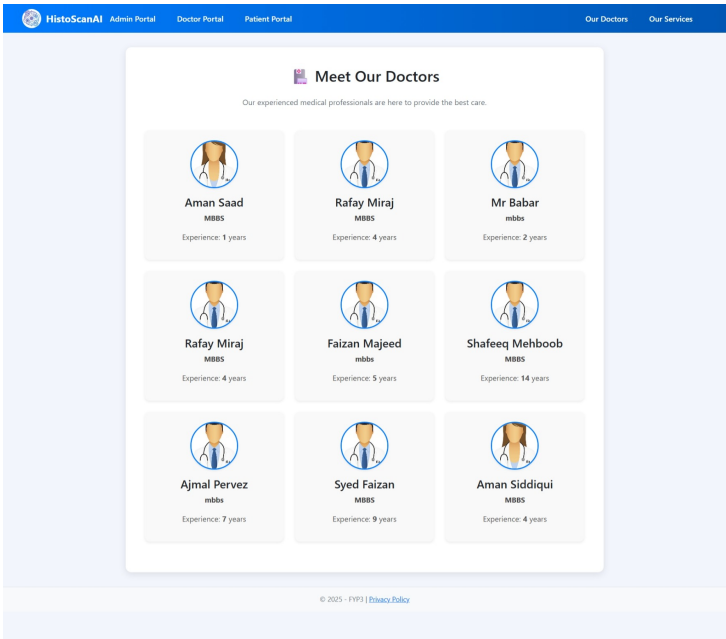


Figure 6.2: Our Doctors Interface

6.0.1.3 Our Services Interface

In the figure 6.3, the services page outlines the key functions provided by the system, which includes analysis of the histopathology images, detection of the cancer using artificial intelligence technology, and generation of the report.

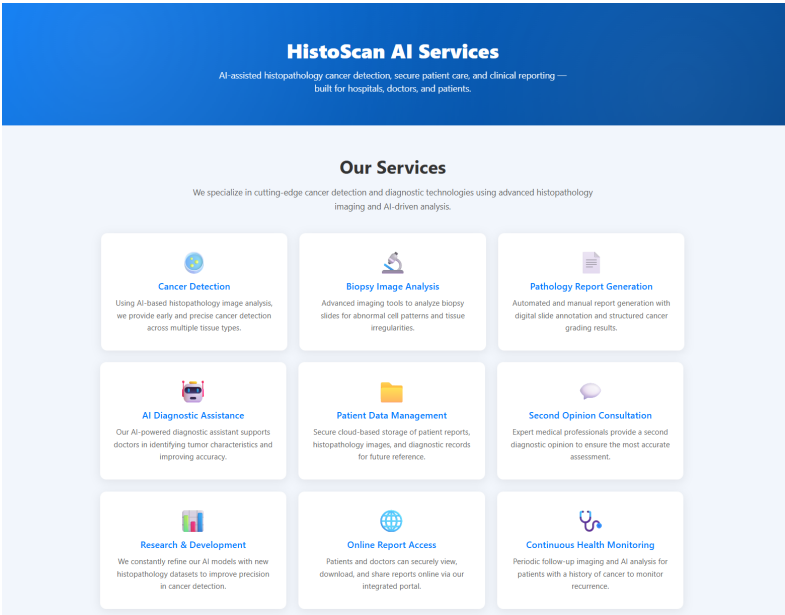


Figure 6.3: Our Services Interface

6.0.1.4 HistoScan AI Detection Success Statistics

Figure 6.4 gives an overview of the efficiency of HistoScan after analyzing 122 scans for 36 patients from five different hospitals by ten experienced radiologists. During its operation, the system performed 120 artificial intelligence analysis tasks and produced 162 reports, diagnosing cancer in 38 patients with an efficiency of 66.7%. There were 82 negative scans reported by HistoScan. The most extensive use occurred in January 2026, with subsequent declining use until a small upsurge occurred in June 2026. This demonstrates that HistoScan is still in the experimental phase. Despite that, HistoScan shows great promise but needs to improve on its efficiency and hospital adoption.

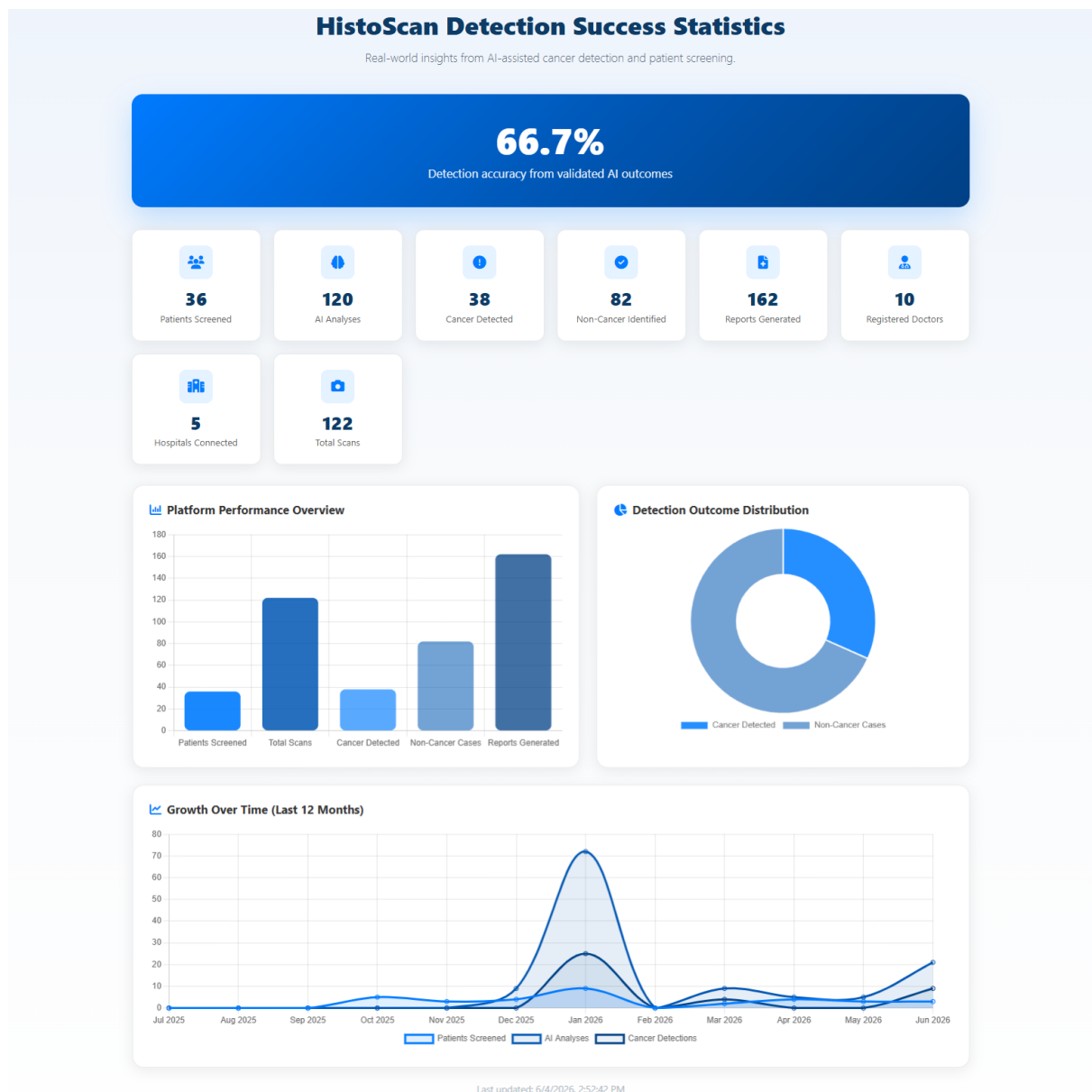


Figure 6.4: Detection Success Statistics

6.0.1.5 Admin Login Interface

It can be seen from Figure 6.5 that the login screen for the administrator helps in making sure that users who have valid credentials authenticate themselves and thus have access to the operations on the system. If the credentials are not valid, the login screen

rejects them.

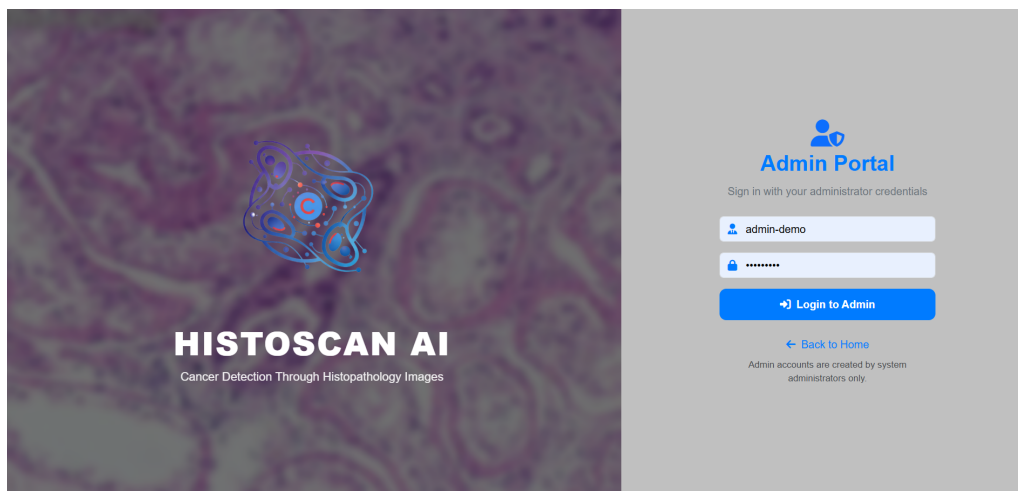


Figure 6.5: Admin Login Interface

6.0.1.6 Admin Portal Interface

The Admin Portal provided in Figure 6.6 enables the administrators to control everything in the portal and approve/reject the registrations of the doctors/patients along with their profiles being displayed. The list of approved doctors and patients is available in tables with options to delete entries or even enter the data of the patient manually. Filters such as Hospital can be used by the administrator and the other facilities available in the portal include: Feedback Management, Analytics Dashboard, and Audit & Activity Logs.

HistoScanAI

Admin Portal

Doctor Portal

Patient Portal

Our Doctors

Our Services

Admin Portal

Manage Feedback

Analytics Dashboard

Audit & Activity Logs

Logout

Filter by hospital

All hospitals

Apply

Clear

Pending Doctor Approvals

Full Name	Username	Email	Hospital	Degree	Experience	Gender	Actions
Sadiq Hussain	Sadiq_28	61raf225@gmail.com	North Oncology Institute	mbbs	8 years	M	Approve Reject

Approved Doctors

Full Name	Username	Email	Hospital	Degree	Experience	Gender	Actions
Aman Saad	Aman Saad	aman16siddiqui@gmail.com	Unassigned	MBBS	1 years	F	Delete
Rafay Miraj	Abdul Rafay	aman16siddiqui@gmail.com	Unassigned	MBBS	4 years	M	Delete
Mr Babar	Babar	aman16siddiqui@gmail.com	Unassigned	mbbs	2 years	M	Delete
Rafay Miraj	Rafay_Miraj22	amanfyp2025@gmail.com	Unassigned	MBBS	4 years	M	Delete

Pending Patient Registrations

Full Name	Username	Email	Hospital	Contact No	Gender	Date of Birth	Blood Group	Actions
rafeeq ali	rafeeq	Unassigned	03345099903	M	2004-12-06	AB +	Approve Reject	
Usman Ali	Usman	Unassigned	0511111111	M	2004-12-06	A-	Approve Reject	
Zubair Ahmed	Zubair_Ahmed	Unassigned	03216549674	M	2001-02-22	O-	Approve Reject	

Approved Patients

Add Patient

Full Name	Username	Email	Hospital	Contact No	Gender	Date of Birth	Blood Group	Address	Actions
Noor Siddiqui	Noor_2006	noor2006siddiqui@gmail.com	North Oncology Institute	03348124805	F	2006-08-09	A-	bahria phase 7	Delete
Abdul Rafay	rafay_2003	siddiquinoor373@gmail.com	HistoScanAI Central Hospital	03405560461	M	2006-08-09	O-	dha phase 2	Delete
Faisal Pervez	Faisal_84	siddiquinoor@gmail.com	City Medical Center	03451233365	M	1984-02-04		dha 6	Delete
Majeed Saglain	Majeed_22	aman16siddiqui@gmail.com	North Oncology Institute	03215621478	M	2024-04-04	B +	dha	Delete

Figure 6.6: Admin Portal Interface

6.0.1.7 Admin Audit and Activity Log Interface

Activity history of all the activities that the user has carried out through the software can be found in Figure 6.7; this covers everything from logging into the system to viewing data, modifying it, and producing reports. These actions are always dated so that they can be traced to their respective user.

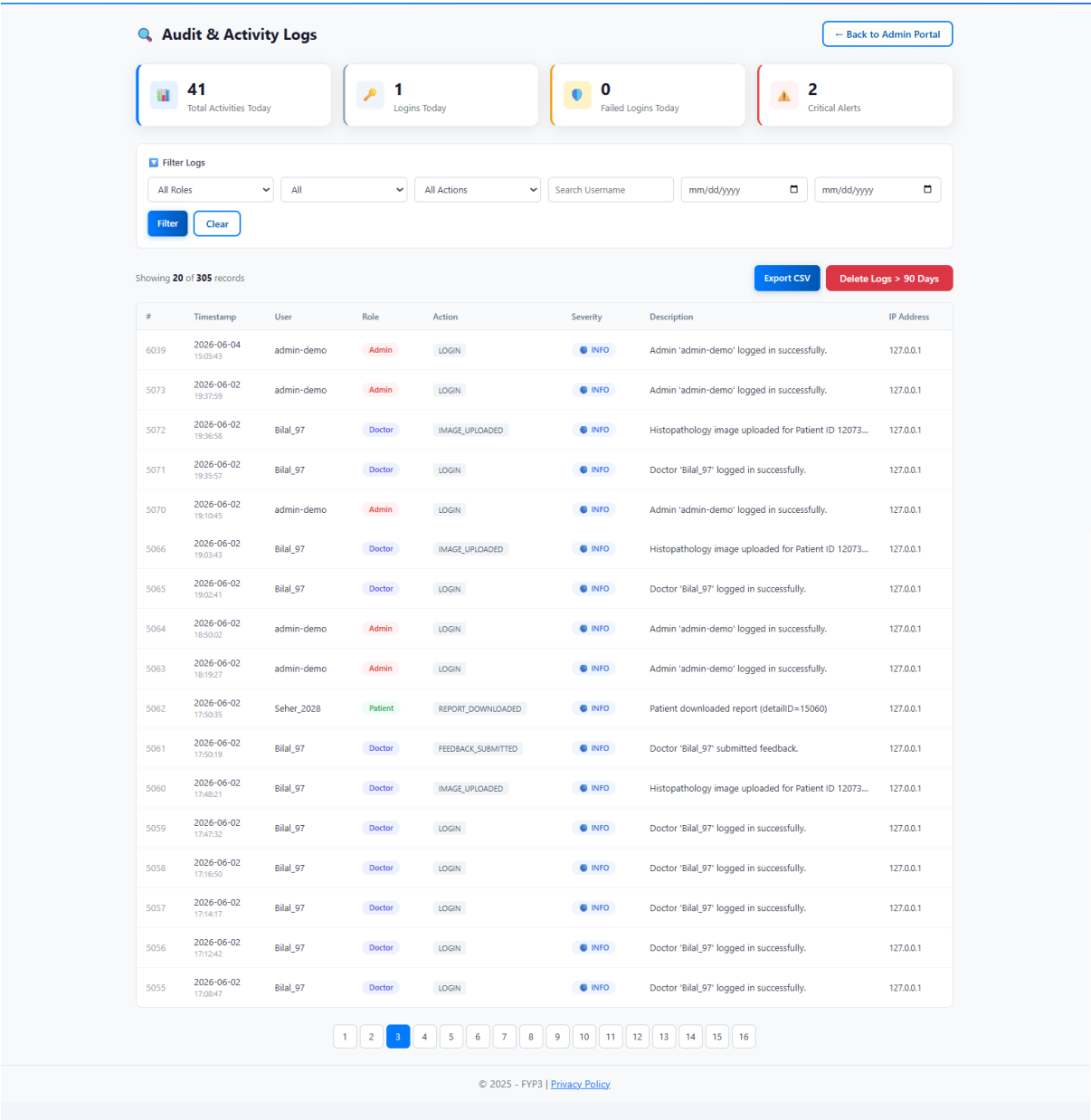


Figure 6.7: Audit and Activity Log Interface

6.0.1.8 Admin Feedback

For instance, in Figure 6.8 the Feedback Management module captures all feedback from both the doctors and patients for the platform totaling 20 feedback items in total and an average score of 4.1/5. As regards categories, it appears that the performance in terms of System Performance and Dashboard Ease stands out scoring 4.8/5 and 4.5/5 respectively. However, it is clear that there are areas that require improvements, such as Overall Performance (3.3/5) and User Management (3.3/5). AI Quality receives an

impressive score of 4.0/5 indicating that there is trust in the functionality of the algorithm. Nevertheless, Report Upload scoring 3.8/5 shows that users had challenges uploading reports.

The Feedback Management module presents the feedback details with ratings, tags showing the user's role and comments for a better understanding of the feedback. Moreover, feedback can be filtered using different criteria like the user's role, period and even key word. Additionally, feedback can be downloaded to Excel in CSV format.

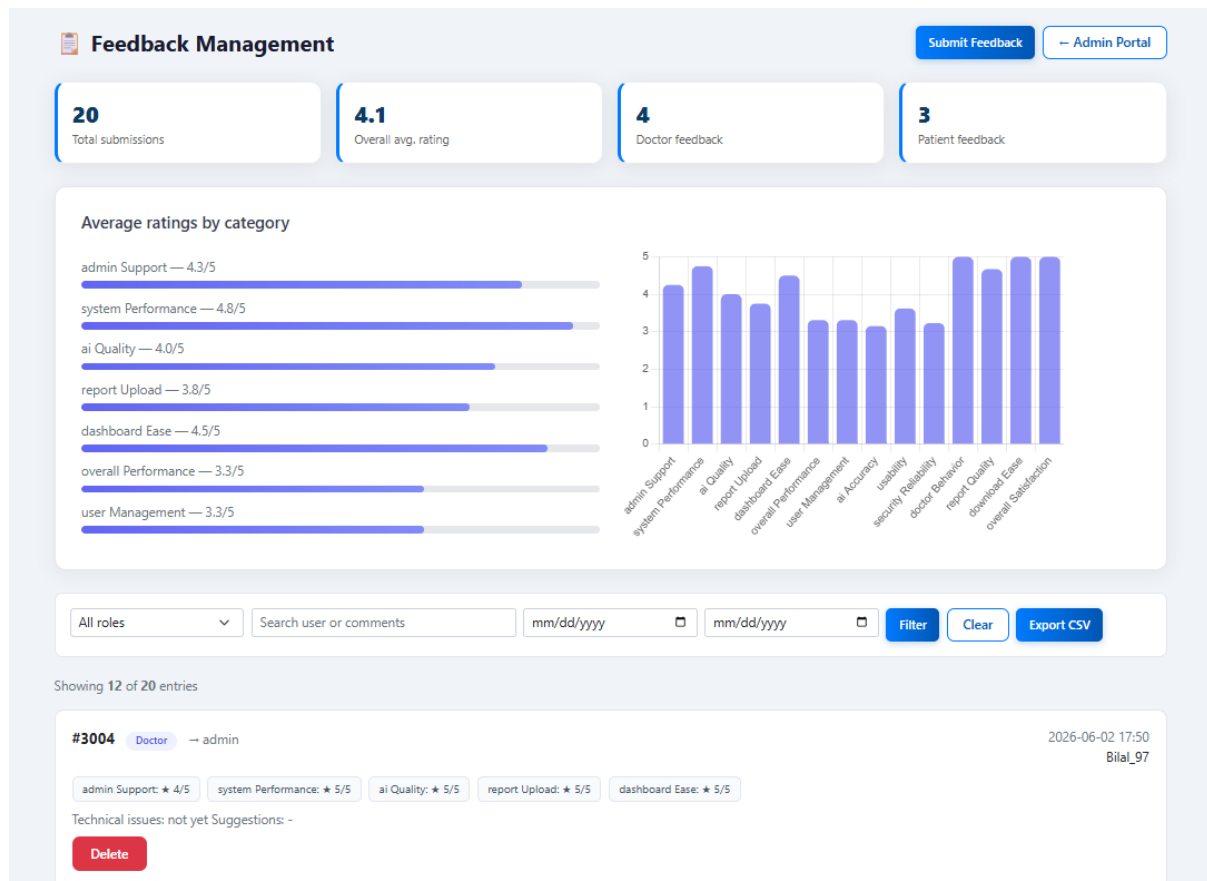


Figure 6.8: Admin Feedback

6.0.1.9 Analytics and Dashboard

From the following figures 6.9 – 6.16, it can be observed that the platform HistoScanAI Analytics offers a highly detailed dashboard that offers the administrators full insight into all the processes associated with the cancer detection system.

It includes various modules focused on the performance of AI model, doctor and patient engagement, hospital comparison, usage analytics, data collection, and user feedback. All the information about these aspects is available within a single filterable interface.

To begin with, there is a filter used by the administrator to select specific statistics. Then it describes the number of patients, doctors, pending doctors, AI scans' total numbers and accuracy rate, the number of generated reports, active users, total hospitals, and average rating feedback. In addition, there is a graph showing AI scan trends and split pie chart indicating the percentage of results showing cancerous vs non-cancerous

conditions.

Moreover, one can see that AI model utilized in the system is highly reliable since it shows very high confidence scores with no false positives and negative results. There is a clearly observable tendency of increasing doctor and patient engagement.

Comparison module helps to determine leading hospitals based on the criteria selected by the administrator. Information on platform usage helps to identify peak times and behavior of sessions on the website.

Data collection analytics provide information regarding a steady increase in the number of images uploaded and analyzed by using both AI model and doctors. User feedback obtained from admins, doctors and patients proves high user satisfaction in terms of system performance and patient experience.

Overview

Figure 6.9 illustrates that the Dashboard Overview serves as the primary starting page for the analytics system, providing an overview of all the important metrics, which include 8 patients, 10 doctors, 31 AI scans, and a 66.7% rate of accuracy for detecting cancer. An AI Insights section provides some pro-active information regarding insights into the busiest hospital, the most active doctor, and recommendations to enhance the user experience.

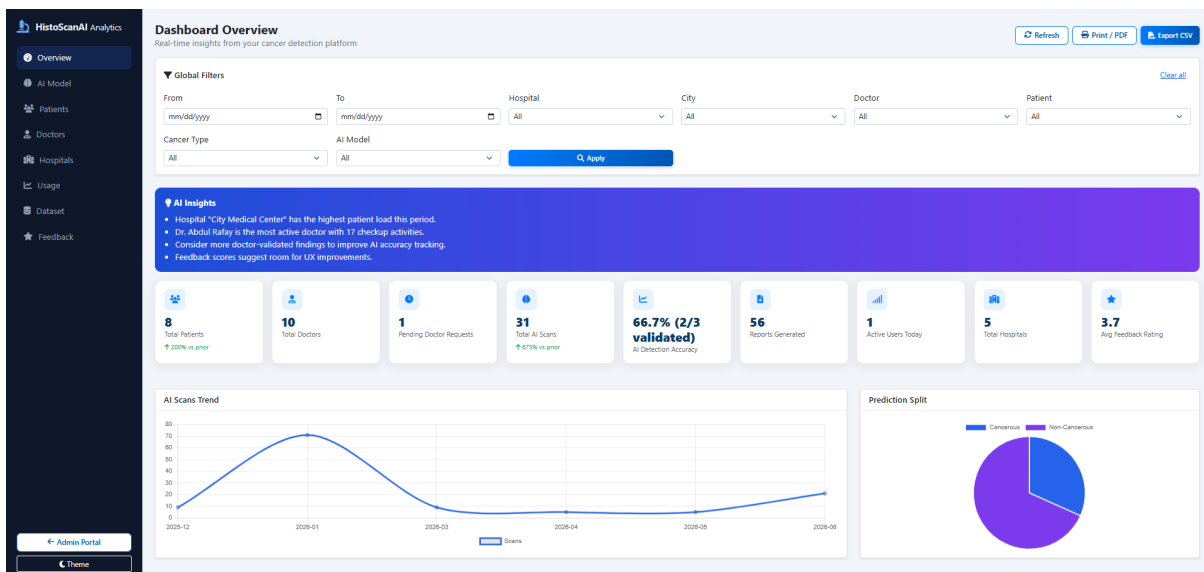


Figure 6.9: Overview In Analytics and Dashboard

AI Model

The AI Model Analytics for Figure 6.10 gives a measure of the efficiency of the EfficientNet-B0 model, which has made a total of 120 detections out of which 38 were identified as cancer patients while 82 were found not to be cancer patients; there is one false positive and no false negative detection. The confidence level stands at 96.4 percent with an overall accuracy rate of 66.7 percent. The performance graphs indicate significant scan activity in January 2026, followed by a gradual improvement.

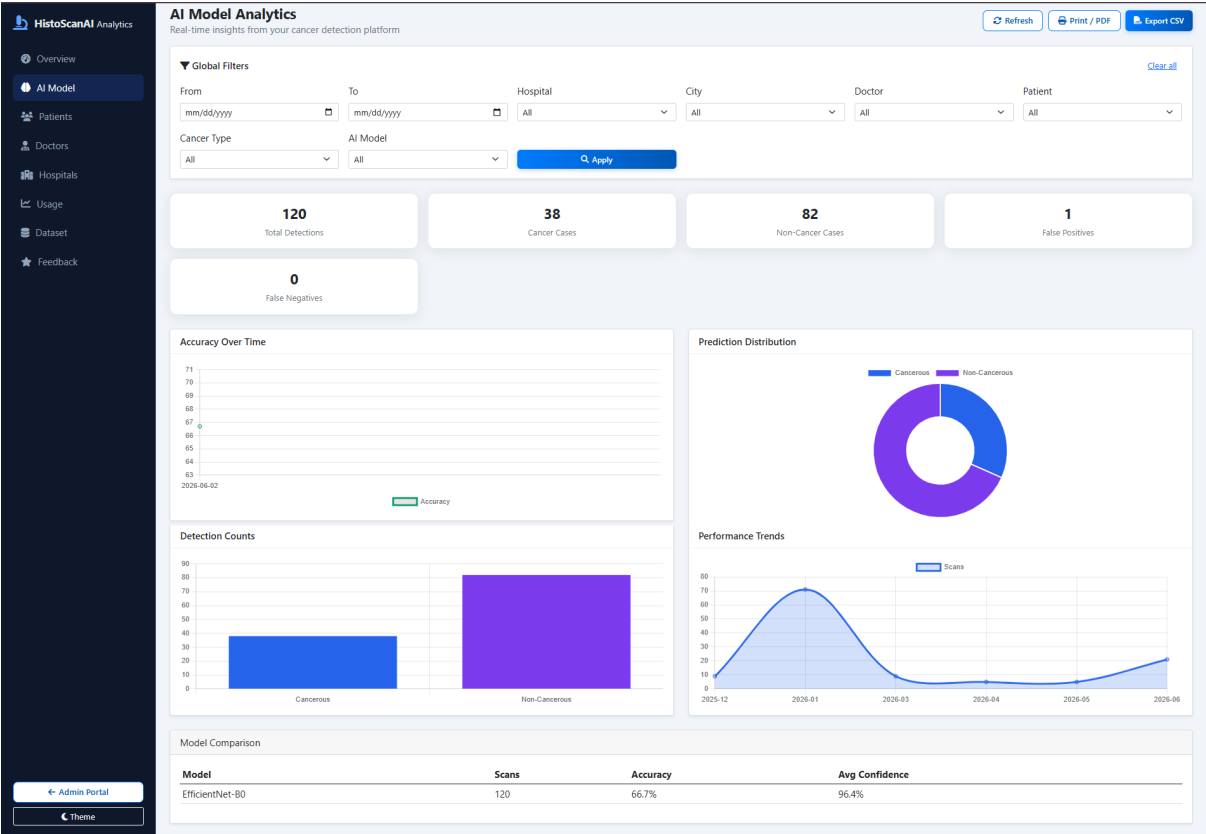


Figure 6.10: AI Model In Analytics and Dashboard

Patients

Figure 6.11 represents Patient Analytics which consists of total eight patients where three were active in the last 24 hours and four were newly registered within the last 30 days, along with 37 patient reports created and 27 AI analyzes performed. Both monthly registration graph and monthly patient activity trend graph show a U-shaped recovery curve where it falls in mid-April and rises to June 2026.

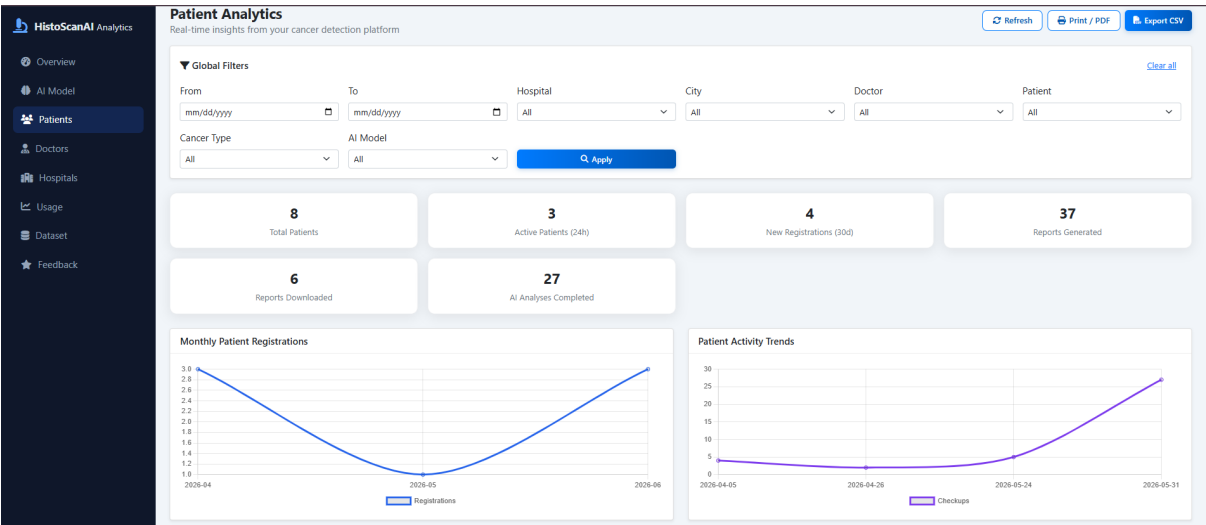


Figure 6.11: Patients In Analytics and Dashboard

Doctors

In Figure 6.12, the Doctor Analytics dashboard tracks 10 approved doctors having 22 AI analyses, 12 reports uploads, and even more impressive is that their average time to diagnose is just 0.3 hours. The Activity Leaderboard highlights engagement levels of doctors; here, Abdul Rafay and Bilal Pervez perform quite impressively while the weekly activity graph displays a massive increase from mid-May 2026. The department breakdown also proves that all active doctors belong to the MBBS category.

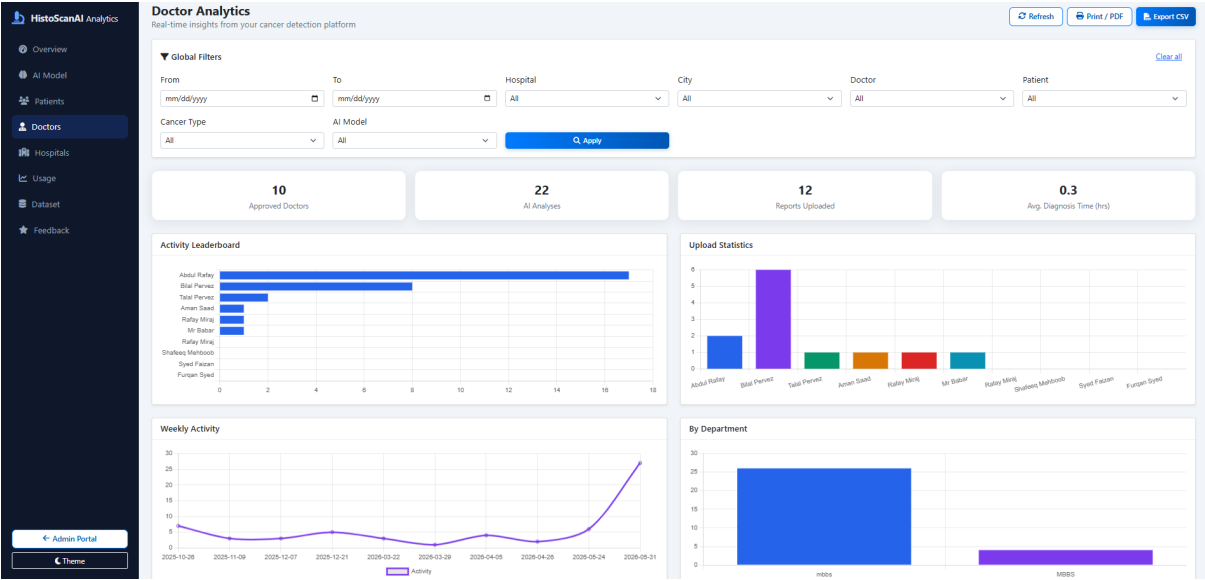


Figure 6.12: Doctor In Analytics and Dashboard

Hospital

Figure 6.13 represents the Hospital Analytics page that depicts the performance of all 5 connected hospitals based on the patient load, accuracy of the AI, and scan distribution. The leading hospital is City Medical Center having 7 scans and an AI accuracy rate of 66.7%, followed by North Oncology Institute in terms of overall performance. Donut chart showing scan distribution also proves the same.

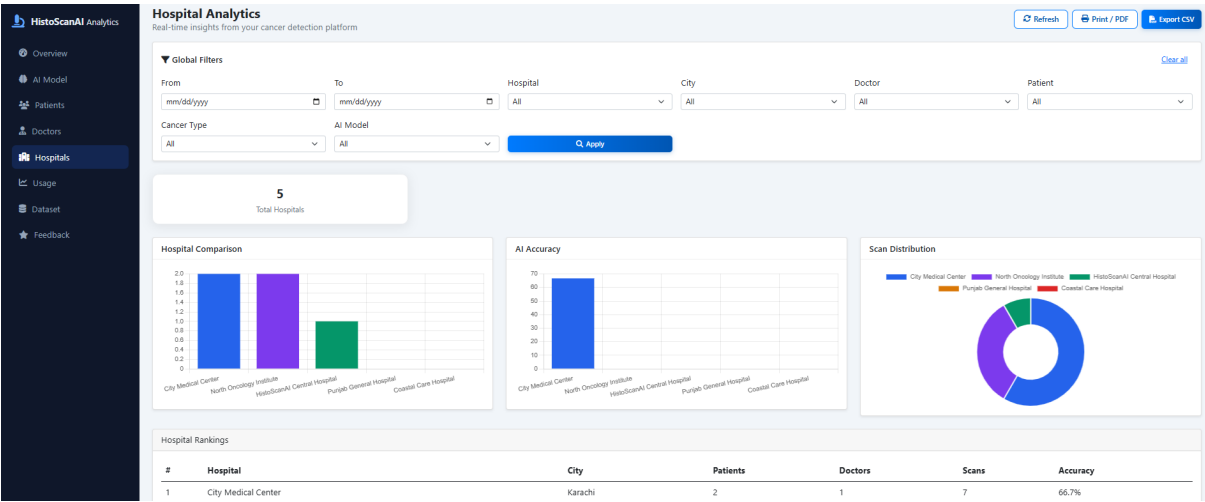


Figure 6.13: Hospitals In Analytics and Dashboard

Usage

In Figure 6.14, the Usage Analytics Module tracks a total of 41 users logging into the system, with 22 sessions completed, 24 active sessions, and an average session time of 2.3 days. The User Activity Timeline reveals that the maximum number of concurrent users logged on was from May 31 to June 2, 2026, after which there is a steep fall until June 4. The Peak Hours graph shows 15:00 hours as the peak hour.

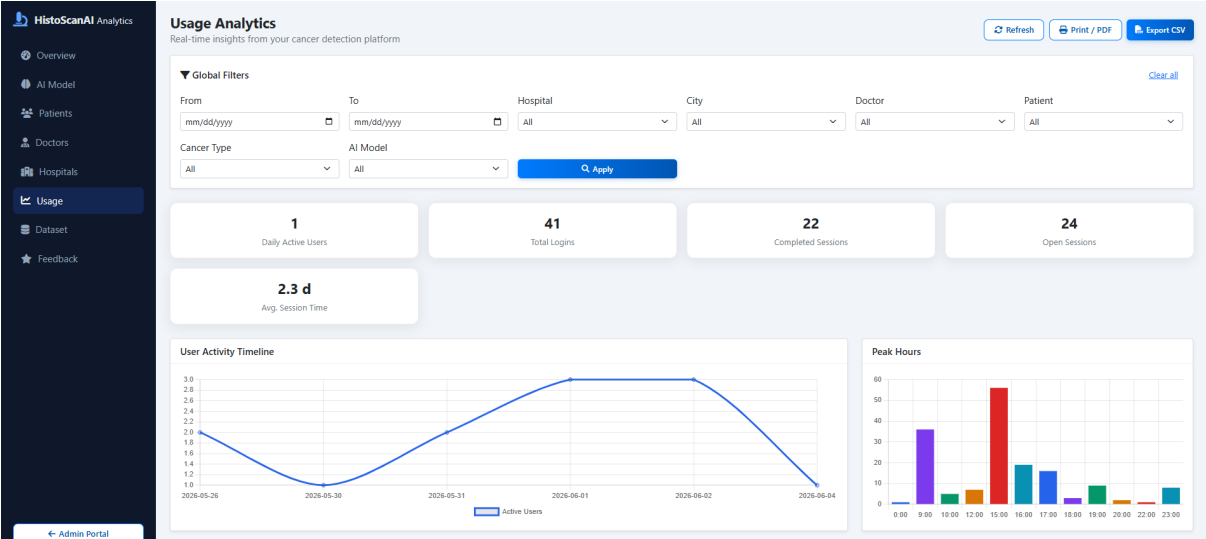


Figure 6.14: Usage In Analytics and Dashboard

Dataset

From figure 6.15, we have a total of 32 images uploaded into the Dataset Growth category, with 31 being analyzed using the AI and 29 having reports made by doctors. From the graph, there is a clear indication that initially the upload rate was slow between December 2025 and April 2026, but a marked increase began to be experienced in May-June 2026.

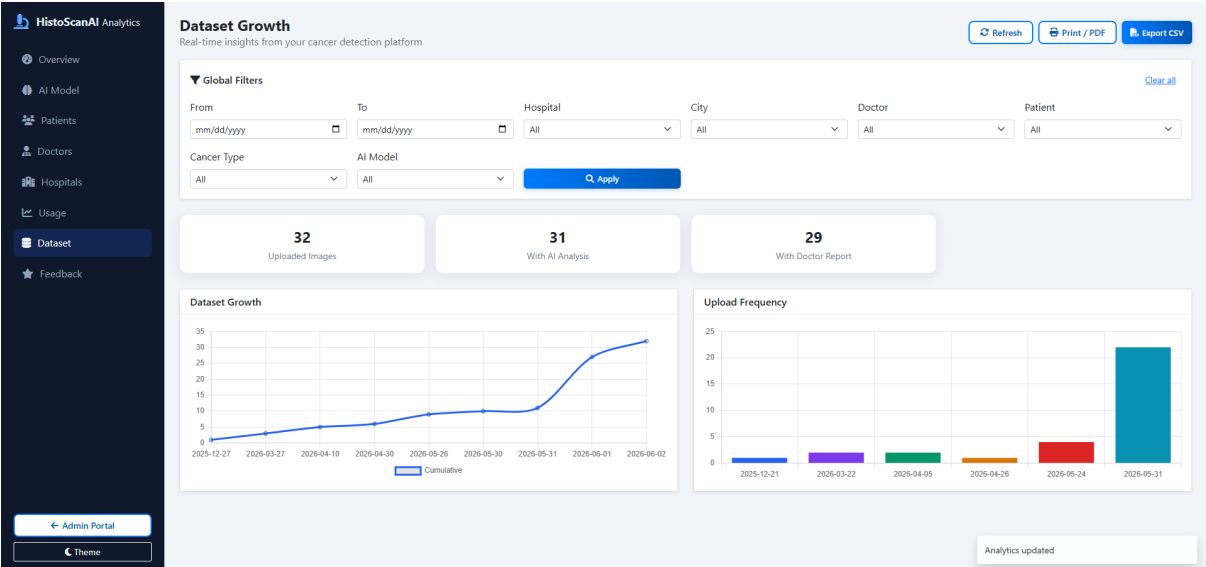


Figure 6.15: Dataset In Analytics and Dashboard

Feedback

The total number of submissions for figure 6.16 is 20, with the average score being 3.7/5, a very high level of patient satisfaction of 4.9/5, doctor satisfaction of 4.3/5, resulting in a usability score of 74.2.

There is a strong bias towards getting 5-stars, although from the trend line for June 2026, there is an evident decline in the average ratings from the month of May.

Below are some recent feedbacks from doctors and admins for further analysis.

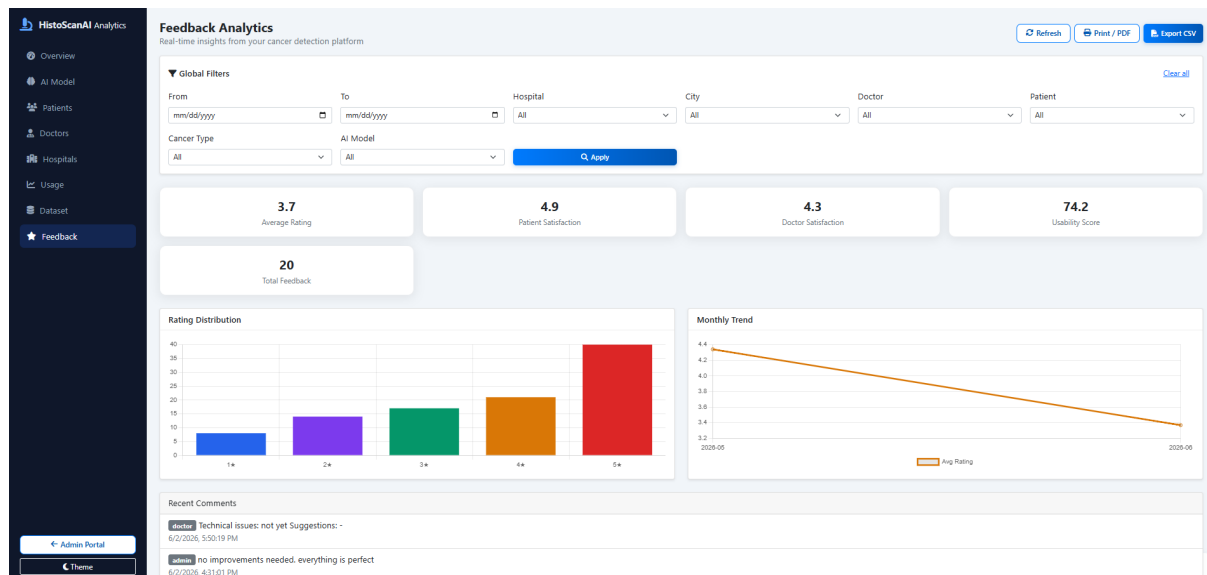


Figure 6.16: Feedback In Analytics and Dashboard

6.0.1.10 Doctor Sign up Interface

In figure 6.17 doctors can register themselves on the interface by supplying their professional as well as personal details. Only legitimate and authentic doctors will be able to access this interface by using otp verification. Following are the details required for signing up:

- Username: enter username with correct format
- Password: enter password with correct format
- Full Name: enter full name
- Contact Number: enter correct number
- Degree: enter the degree doctor has
- Years of Experience: enter the experience years a doctor has
- Gender: select from drop down box the gender of person
- Email: enter active email to get notifications from admin
- Hospital: select hospital from where the doctor belongs

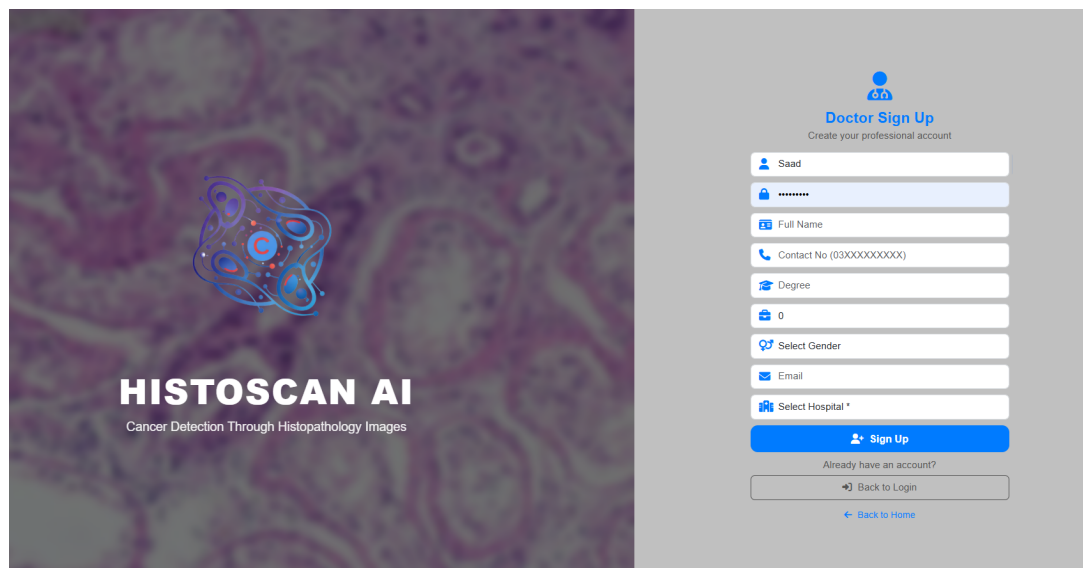


Figure 6.17: Sign up Interface

6.0.1.11 Doctor Login Interface

In figure 6.18 the interface allows doctors to register in the system by providing their professional and personal details. It ensures that only valid medical practitioners gain access to the platform through otp verification.

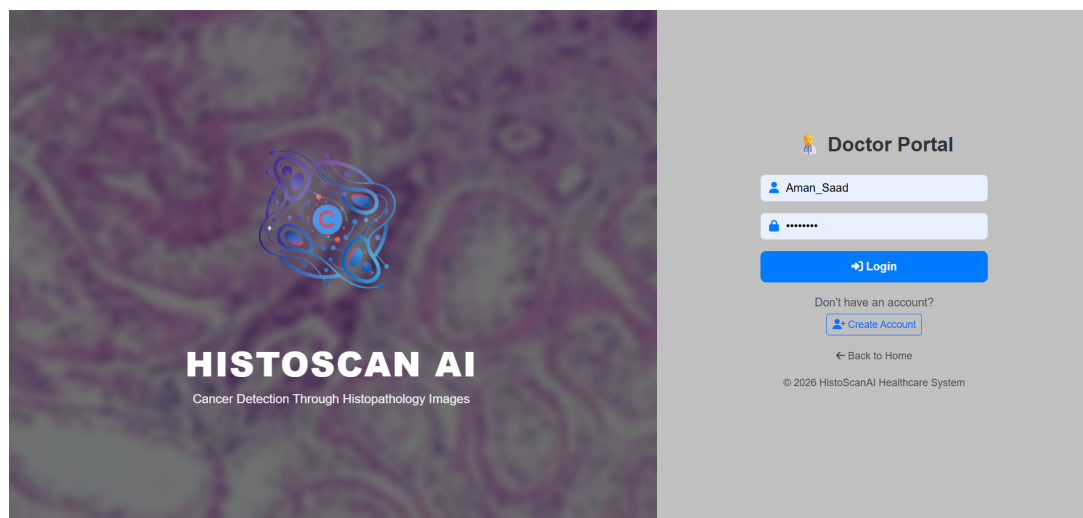


Figure 6.18: Doctor Login Interface

6.0.1.12 Doctor Portal Interface

In figure 6.19 the doctor portal facilitates for entering patient symptoms and doctor findings then viewing the patients history, uploading of histopathology images, carrying out predictions based on artificial intelligence, and viewing the results along with visuals. This platform also offers reporting and patient management capabilities. It also gives proper feedback system for doctor. They give feedback to the system by giving answers to the following questions:

- Admin support / service
- System performance
- AI image analysis quality
- Report upload experience
- Ease of using dashboard
- Technical issues faced
- Suggestions / recommendations

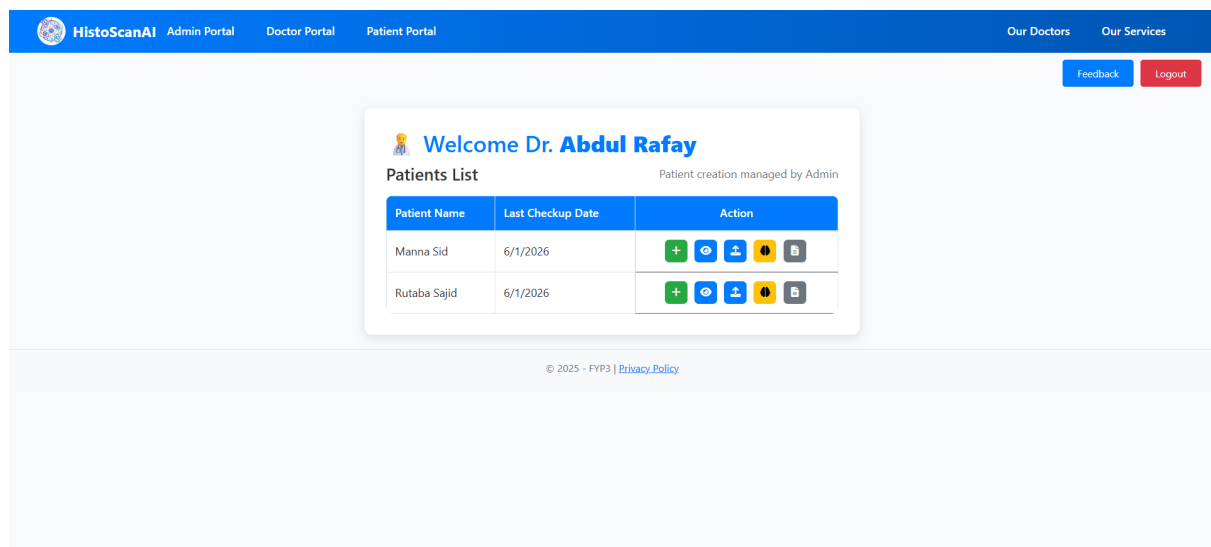


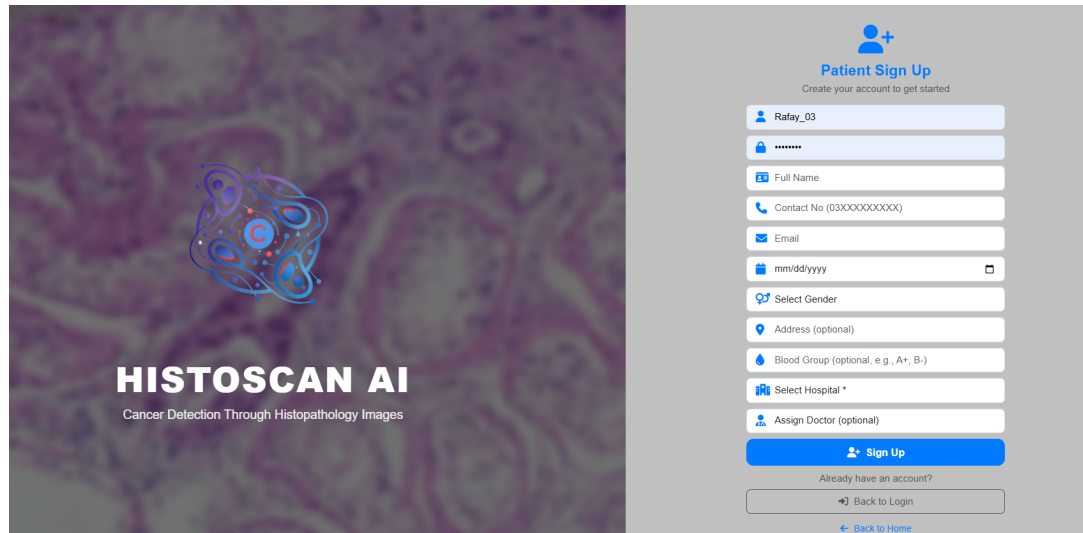
Figure 6.19: Doctor Portal Interface

6.0.1.13 Patient Sign up Interface

In figure 6.20 the interface allows patients to create an account by entering their basic information. It ensures easy registration and access to the system's services. The sign-up form requires the patients to enter the following information:

- Username: enter username with correct format
- Password: enter password with correct format
- Full Name: enter full name
- Contact Number: enter correct number
- Email: enter active email to get notifications from doctor
- Date of Birth: enter correct date of birth
- Gender: select from drop down box the gender of person
- Address: enter correct address (optional)

- Blood Group: enter blood group (optional)
- Assigning Doctor: select the doctor that you want to get treated by
- Hospital: selecting hospital from where the patient is going to be registered

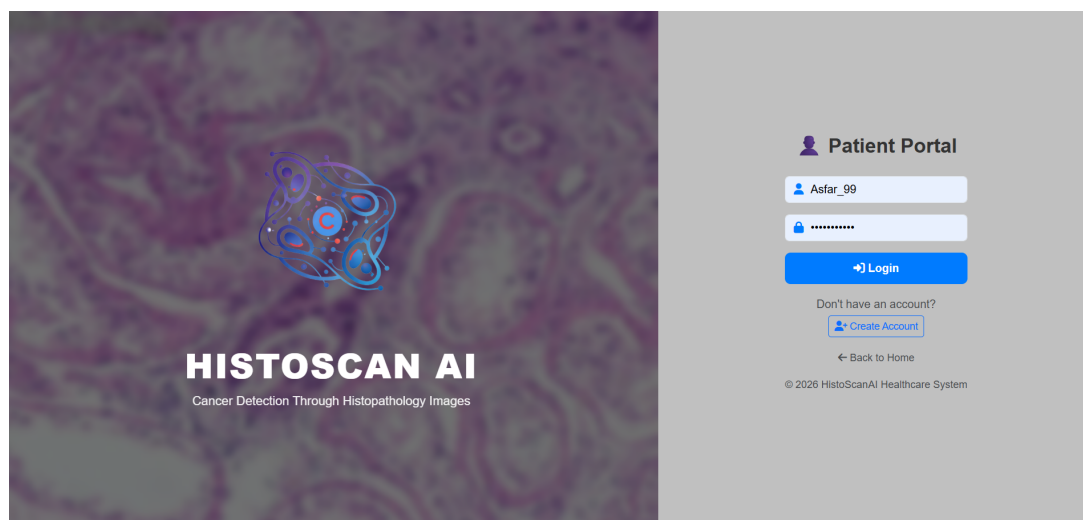


The image shows a mobile application interface for patient sign-up. On the left is a dark purple banner with a microscopic cell image and the text "HISTOSCAN AI" and "Cancer Detection Through Histopathology Images". On the right is a white sign-up form titled "Patient Sign Up" with the subtitle "Create your account to get started". The form contains the following fields: Username (Rafay_03), Password (masked with dots), Full Name, Contact No (03XXXXXXXX), Email, Date of Birth (mm/dd/yyyy), Gender (Select Gender), Address (optional), Blood Group (optional, e.g., A+, B-), Select Hospital *, and Assign Doctor (optional). At the bottom of the form is a blue "Sign Up" button. Below the button are links for "Already have an account? Back to Login" and "Back to Home".

Figure 6.20: Patient Sign up Interface

6.0.1.14 Patient Login Interface

In figure 6.21 the patient login interface provides secure access for registered users to their personal dashboard. It ensures data privacy through authentication.



The image shows a mobile application interface for patient login. On the left is a dark purple banner with a microscopic cell image and the text "HISTOSCAN AI" and "Cancer Detection Through Histopathology Images". On the right is a white login form titled "Patient Portal". The form contains the following fields: Username (Asfar_99) and Password (masked with dots). Below the password field is a blue "Login" button. Below the button is a link for "Don't have an account? Create Account". At the bottom of the form is a link for "Back to Home". At the very bottom of the screen is a copyright notice: "© 2026 HistoScanAI Healthcare System".

Figure 6.21: Patient Login Interface

Correct credentials are supposed to be entered otherwise it will show error.

6.0.1.15 Patient Portal Interface

In figure 6.22 the patient portal enables users to view AI-based diagnosis results, access their reports and download them.

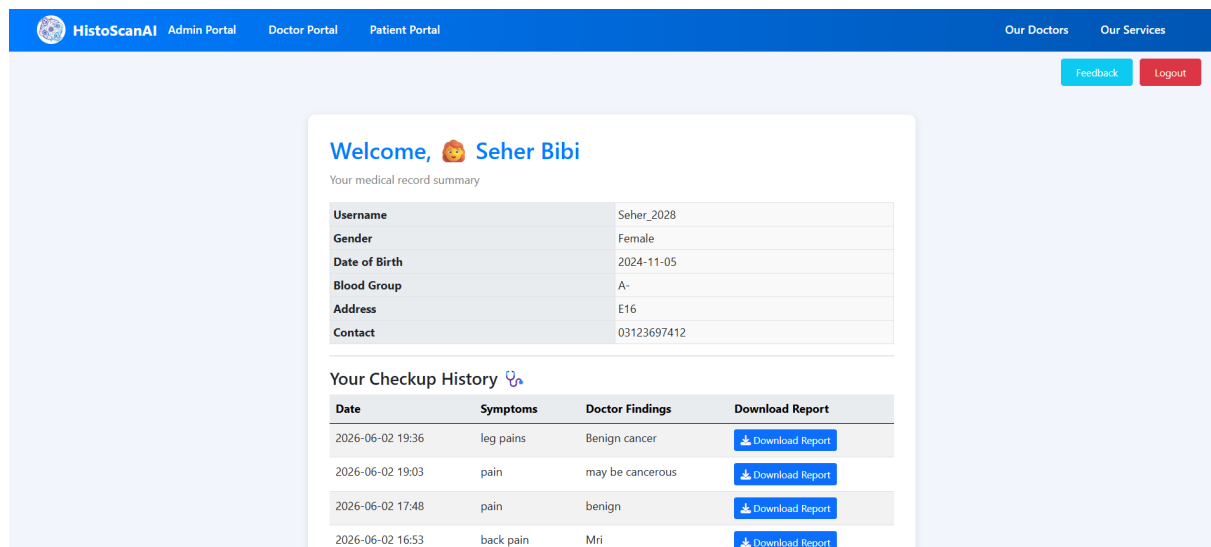


Figure 6.22: Patient Portal Interface

It also allows patients to track their history and receive updates from doctors that their reports are generated and patients can simply download it. In report it has clinical examination, histopathology image, AI results and confidence.

6.0.2 Permanent Doctors Database

In figure 6.23 the doctor table stores information about registered doctors on the platform. Each doctor has a unique DoctorID, along with their username, full name, degree, experience (in years), gender, and email.

	DoctorID	UserName	FullName	Degree	Experience	Gender	Email	Status	HospitalID
16	12066	Rafay_03	Abdul Rafay	mbbs	8	M	aman16siddiqui@gmail.com	approved	3
17	12072	Bilal_97	Bilal Pervez	mbbs	5	M	aman16siddiqui@gmail.com	approved	2
18	13066	Sadiq_28	Sadiq Huss...	mbbs	8	M	61raf225@gmail.com	pending	3

Figure 6.23: Permanent Doctors Database

A Status column indicates whether the doctor's account has been approved, rejected, or is still pending admin review. It also shows from which hospital the doctor is connected. It also gives doctorID, username, fullname, degree, experience, email and gender too.

6.0.3 Patient Details Database

The table in figure 6.24 contains data about patient consultation or appointments, which includes the symptoms observed, observations by doctors, paths to the image files, diagnosis outcomes, and the date on which the records were entered. In some of the columns, NULL values are included since no data is available in those columns yet. This table presents the medical history of one particular patient (PatientID: 12073) who has been seen on June 2, 2026. There are four rows in this table (DetailIDs 15059-15062) and include observations like back pain, general pain, and leg pains.

	DetailID	PatientID	AddDate	Symptoms	DocFindings	ImagePath	AIAnalysisResult	AIAnalysisDate
58	15059	12073	2026-06-02 16:53...	back pain	Mri	/uploads/...	Non-Cancerou...	2026-06-02 ...
59	15060	12073	2026-06-02 17:48...	pain	benign	/uploads/...	Non-Cancerou...	2026-06-02 ...
60	15061	12073	2026-06-02 19:03...	pain	may be can...	/uploads/...	Cancerous (Co...	2026-06-02 ...
61	15062	12073	2026-06-02 19:36...	leg pains	Benign canc...	/uploads/...	Cancerous (Co...	2026-06-02 ...
62	16057	12073	2026-06-06 19:03...	pain	Benign	/uploads/...	Non-Cancerou...	2026-06-06 ...
63	16058	12073	2026-06-06 19:26...	cough wi...	Benign	/uploads/...	Cancerous (Co...	2026-06-06 ...

Figure 6.24: Patient Details Database

It shows the complete data stored in database the DetailsID, PatientID, and dates etc.

6.0.4 Patients Database

In figure 6.25 the patients table holds personal profile information for patients, including date of birth, gender, address, blood group, and approval status.

	PatientID	DateOfBirth	Gender	Address	BloodGroup	DoctorID	Status	Email	HospitalID
1	10067	2006-08-09	M	dha phase 2	O-	10066	approved	siddiquinoor373@gmail.com	1
2	12065	2024-04-04	M	dha	B+	4067	approved	aman16siddiqui@gmail.com	3
3	12073	2024-11-05	F	E16	A-	12072	approved	aman16siddiqui@gmail.com	2

Figure 6.25: Patients Database

A column appears to reference an associated DoctorID, suggesting that each patient is linked to a specific doctor. The status field shows whether the patient profile has been approved by the admin.

6.0.5 Roles Database

In figure 6.26 it can be seen that it is a simple lookup table that defines the three user roles in the system: Admin, Doctor, and Patient.

	RoleID	RoleName
1	1	Admin
2	2	Doctor
3	3	Patient

Figure 6.26: Roles Database

It is referenced by other tables (like the signup table) via RoleID to control access and permissions throughout the application.

6.0.6 Users Database

In figure 6.27 the user table controls the users' authentication and registration in the system. The user table consists of a unique ID, username, password, and RoleID, which is used to connect the Roles table to determine whether a person is an admin, doctor, or patient.

	id	UserName	Password	RoleID	FullName	ContactNo
49	12067	Rutaba_23	rutaba_23	3	Rutaba Sajid	03335129622
50	12069	Manna_04	manna_04	3	Manna Sid	03345035478
51	12070	Sami_2008	sami_2008	3	Sami Din	03124578945
52	12071	rutaba_24	rutaba_24	3	Rutaba Sajid	03335129622
53	12072	Bilal_97	Bilal_97	2	Bilal Pervez	03321478963
54	12073	Seher_2028	seher_2028	3	Seher Bibi	03123697412
55	13066	Sadiq_28	sadiq_28	2	Sadiq Hussain	03314587487

Figure 6.27: Users Database

6.0.7 Activity and Audit Log Database

In Figure 6.28, it shows the system audit log is a database that monitors all activities related to user logon and logoff sessions in the application. The table documents all successful and unsuccessful login attempts by users of various types such as Admin, Patient, and Doctor.

	LogID	UserID	UserName	UserRole	ActionType	Description	IPAddress	Severity	LogTimestamp
1	1	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 19:53:18.800
2	2	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:11:58.190
3	3	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:13:24.440
4	4	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:16:01.347
5	5	1042	Asfar_99	Unknown	LOGIN_FAILED	Failed login attempt for username 'Asfar_99'.	127.0.0.1	WARNING	2026-04-05 20:20:17.773
6	6	1042	Asfar_994	Unknown	LOGIN_FAILED	Failed login attempt for username 'Asfar_994'.	127.0.0.1	WARNING	2026-04-05 20:20:21.557
7	7	7063	Asfar_99	Patient	LOGIN	Patient 'Asfar_99' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:20:25.910
8	8	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:20:31.680
9	9	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:28:40.870
10	10	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:30:28.733
11	11	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:33:44.127
12	12	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:35:03.560
13	13	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:39:11.923
14	14	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 20:56:32.130
15	15	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 21:04:31.180
16	16	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 21:05:08.580
17	17	1042	furqan_90	Unknown	LOGIN_FAILED	Failed login attempt for username 'furqan_90'.	127.0.0.1	WARNING	2026-04-05 21:05:27.863
18	18	1042	furqan_900	Unknown	LOGIN_FAILED	Failed login attempt for username 'furqan_900'.	127.0.0.1	WARNING	2026-04-05 21:05:31.860
19	19	1042	admin-demo	Admin	LOGIN	Admin 'admin-demo' logged in successfully.	127.0.0.1	INFO	2026-04-05 21:05:40.293
20	20	1042	admin-demo	Unknown	LOGIN_FAILED	Failed login attempt for username 'admin-demo'.	127.0.0.1	WARNING	2026-04-05 21:06:56.057

Figure 6.28: Activity and Audit Log Database

The logs show the trend that there are multiple logins by the Admin account, where the username is admin-demo, with User ID 1042 occurring one after another. The logs indicate some significant security incidents, such as unsuccessful login attempts using usernames like Asfar_99, Asfar_994, furqan_90, furqan_900, and even admin-demo itself, with a WARNING status.

6.0.8 AI Predictions Database

In Figure 6.29, AI-based cancer detection predictions made by the EfficientNet-B0 model are recorded. Each row contains a prediction ID, patient/doctor/hospital references, the predicted label (Cancerous or Non-Cancerous), confidence score, and whether the prediction was correct. Specially one prediction (ID 3006) was incorrect the model predicted "Cancerous" with 96.6% confidence but the ground truth was actually "Non-Cancerous."

	PredictionID	DetailID	PatientID	DoctorID	HospitalID	ModelName	PredictedLabel	Confidence	GroundTruthLabel	IsCorrect	PredictedAt
1	3004	15060	12073	12072	2	EfficientNet-B0	Non-Cancerous	0.58	Non-Cancerous	1	2026-06-02 17:49:39.9630000
2	3005	15061	12073	12072	2	EfficientNet-B0	Cancerous	1	Cancerous	1	2026-06-02 19:04:59.7200000
3	3006	15062	12073	12072	2	EfficientNet-B0	Cancerous	0.996	Non-Cancerous	0	2026-06-02 19:37:39.8830000

Figure 6.29: AI Predictions Database

6.0.9 Feedback Database

In figure 6.30, the table represents user feedback provided by various users such as admin, doctors. Ratings have been saved in JSON format including parameters such as performance rating, user management rating, and AI rating. Perfect scores were given by one admin without any feedback, and one doctor (Bilal_97, 12072) gave suggestions regarding technical problems.

	FeedbackID	UserID	UserRole	TargetType	Ratings	Comments	UserName	CreatedAt
1	3003	1042	admin	system	{"overallPerforma...	no improve...	admin-demo	2026-06-02 1...
2	3004	12072	doctor	admin	{"adminSupport":4...	Technical is...	Bilal_97	2026-06-02 1...

Figure 6.30: Feedback Database

6.0.10 Hospital Database

From Figure 6.31, one will notice that there are 3 hospitals listed in the table registered within the database in different cities of Pakistan, including Lahore, Karachi, and Islamabad. The information in the table contains a hospital number, hospital name, city, and the address of the hospitals. For instance, HistoScanAI Central Hospital and North Oncology Institute.

	HospitalID	Name	City	Address
1	1	HistoScanAI Central Hospital	Lahore	Main Boulevard
2	2	City Medical Center	Karachi	Clifton Block 5
3	3	North Oncology Institute	Islamabad	Sector G-8

Figure 6.31: Hospital Database

6.0.11 User Session Database

In Figure 6.32 the table logs user login sessions, tracking session ID, user ID, role, login/logout times, last activity, IP address, and browser info. User admin-demo (ID 1042) has multiple sessions, indicating frequent or repeated logins. All users appear to be accessing from a Windows NT 10.0 browser (likely Windows 10/11), and all IP addresses show ::1, suggesting local/internal network access.

	SessionID	UserID	RoleID	UserName	LoginAt	LastActivityAt	LogoutAt	IPAddress	UserAgent
41	2016	1042	1	admin-demo	2026-06-01 10:20:45.2308989	2026-06-01 23:01:27.4715247	2026-06-01 23:01:27.4715247	::1	Mozilla/5.0 (Windows NT 10.0;...
42	3008	1042	1	admin-demo	2026-06-02 16:12:36.2934838	2026-06-02 18:19:27.0974162	2026-06-02 18:19:27.0974162	::1	Mozilla/5.0 (Windows NT 10.0;...
43	3009	12073	3	Seher_2028	2026-06-02 17:02:52.2000925	2026-06-02 19:05:22.9845194	2026-06-02 19:05:22.9845194	::1	Mozilla/5.0 (Windows NT 10.0;...
44	3010	1042	1	admin-demo	2026-06-02 18:50:02.5762798	2026-06-02 19:05:42.5820755	2026-06-02 19:05:42.5820755	::1	Mozilla/5.0 (Windows NT 10.0;...
45	3011	1042	1	admin-demo	2026-06-02 19:10:45.5883765	2026-06-02 19:37:59.7044672	2026-06-02 19:37:59.7044672	::1	Mozilla/5.0 (Windows NT 10.0;...
46	4008	1042	1	admin-demo	2026-06-04 15:05:43.3313899	2026-06-04 19:07:05.6442692	2026-06-04 19:07:05.6442692	::1	Mozilla/5.0 (Windows NT 10.0;...

Figure 6.32: User Session Database

Chapter 7

TEST PLAN

A test plan[36] is a comprehensive document that lists all the test processes, objects, time frames, costs, approximations, time limits, and resources required to complete a project. It serves as a model for doing the test required to ensure the software is operating correctly. A test plan is essential because it will briefly touch on all the crucial elements required to produce interactive and comprehend how the application will function in the real world. It will let programmers and other team members see and test their application in a real-world environment.

Every test of software starts with a test plan. It is the most crucial task that guarantees the availability of the lists of all scheduled activities in the correct order. A test plan helps us to define test objectives, levels of testing, designing test strategy, defining test criteria, resource planning, planing test environment, and schedule and estimation for our project.

7.1 Objective of the Testing Phase

Our objective is to assess the quality of our software product throughout the testing phase and then to fix any defects that are found. Software testing primary goal is to identify and fix all current issues and defects while also maintaining control over the quality of the final software output. In addition, testing serves two other fundamental reasons, the first of which involves confirming purchase specifications, and the second of which involves risk management. The testing phase will assist us in confirming and ensuring that our software product satisfies all of the functional, design, implementation, and performance requirements. Following are the objectives of the Testing phase:

- To define the scope of what will be tested.
- To plan the activities and deadlines.
- To describe the strategy used for testing. Specify the desired deliverable.
- To indicate how the test results will be assessed.
- To determine ways to mitigate the testing plan's hazards.
- To assess the needs of stakeholders, including design and code.
- To analyze performance. To eliminate errors and defects.
- To ensure that the system will fulfill his desired requirements.
- To help doctors to diagnose cancer with efficiency and accuracy.
- To ensure that the tools, design, and code are current and adhere to the standard to identify product issues prior to deployment.

7.2 Levels of Tests for Testing Software

Before our software product is deployed, it has to undergo a testing process in which we ensure that our product is working in the manner in which it was intended. There are three main stages of testing that we have to complete before our system is ready to deploy. The three levels of testing are:

1. Unit Testing
2. Integration Testing
3. System Testing

7.2.1 Unit Testing

Unit testing[37] is the very first stage of testing in which we tested each component of the system in isolation. This is done to make sure that each individual component is performing its intended functionality. The goal of this stage is to determine whether the components of system are functioning as they were designed to.

7.2.2 Integration testing

Integration testing[38] looks at component integration or interfaces, interactions with various system components. This type of testing is carried out when more than 02 modules are joined or integrated. In this testing phase, we combine different modules or components of our system as a group in order to check the data flow from one to other module.

7.2.3 System Testing

System testing[39] is a type of testing that verifies a fully integrated and finished piece of software. A system test's objective is to evaluate every component of system requirements. In system testing, we tested all components of our system as a whole to make sure that the overall product meets the specific requirements. this phase is very important because it allows us to verify whether our application or system meets all the functional, and business requirements that is demanded by our customer.

7.3 Test Management Process

In order to create a high-quality software product, we must manage all testing operations within the test management process. We need to plan and manage the testing procedure in this process. To create a high-quality system, we must simultaneously assure the traceability and visibility of the testing process. For the purpose of efficiently and effectively completing all testing tasks, an appropriate testing strategy is created. A test plan is useful for outlining the testing strategy, its scope, the resources needed, and the timeline for all testing activities[40].

7.3.1 Design the Test Strategy

It is a very important step in the testing process to establish a set of rules or principles that will govern the test design and how we test the product and accomplish our objectives. The Test Strategy Phase's primary goal is to streamline the testing process so that we can guarantee quality, traceability, reliability, and better planning[41]. Listed below are a few important sections of the test strategy design:

1. **Scope and Overview:**

We must first specify the scope of testing tasks that must be performed with respect to the project's time frames as specified in the test plan.

2. **The Test Approach:**

The next step is to outline the scope of the product testing as well as the duties and responsibilities of each team member.

3. **Testing Tools:**

Then we have to specify the tools needed for performance, load, and security testing as well as for test execution.

4. **Risk Analysis:**

The next step is to list any potential dangers that can arise while doing the test. To reduce these risks as well as provide a backup plan in case one of these dangers materializes, a risk mitigation strategy must be established.

7.3.2 Test Objectives

The goal of testing is to find as many software flaws as possible and to guarantee that the product is bug-free before it is made available to the customer. Enhancing software quality is one of the most important goals of software testing. High-quality software has fewer defects than lower-quality software.

7.3.3 Test Criteria

Test criteria[42] is defined to assist us (testers) in organizing the testing process. The available test effort influences the selection of the test criteria. In order to meet the requirements, test coverage measures must also be defined (as a ratio between the necessary test cases). These test coverage measures will be used to gather data on how comprehensive integration tests are. The goal is to make it possible for us (testers) to define integration tests and assess the effectiveness of the findings.

7.3.4 Resource Planning

Resource planning[43] is the comprehensive description of all the resources needed to complete a project. We have tested our module with different tools. Table 7.1 shows the resource planning of our project. It contains serial no., module name, tester name, tools and time required to test modules.

Table 7.1: Resource Planning

Sr. No.	Module	Tester Name	Tools	Time Required
01	Sign Up	Rafay, Aman & Rutaba	Visual Studio Code, Test Link	2 Hour
02	Login	Rafay, Aman & Rutaba	Visual Studio Code, Test Link	2 Hour
03	Admin Portal	Rafay, Aman & Rutaba	Visual Studio Code, Test Link	2 Hour
04	Doctor Portal	Rafay, Aman & Rutaba	Visual Studio Code, Test Link	2 Hour
05	Patient Portal	Rafay, Aman & Rutaba	Visual Studio Code, Test Link	2 Hour
06	AI Model	Rafay, Aman & Rutaba	Visual Studio Code, Test Link	3 Hour
07	Report Generation	Rafay, Aman & Rutaba	Visual Studio Code	2 Hour
08	Audit Log	Rafay, Aman & Rutaba	Visual Studio Code	2 Days
09	Notification	Rafay, Aman & Rutaba	Visual Studio Code	1 Hour

The tables shows the modules being worked upon of sign up, login, admin portal, doctor and patient portal, ai model, report generation, audit logs and notifications. Also displaying the tester names and the tools we used to test along with the time.

7.3.5 Plan Test Environment

We know that a testing environment[44] is a configuration of hardware and software that enables testers to successfully carry out test cases. However, the test environment can be set up to meet the needs of the system that is being tested. The success of the software testing phase will be ensured by setting up the proper test environment. But we must also exercise caution because any errors or defects in this process could result in a client cost and time increase.

- **Visual Studio:**

We will be using Visual Studio[45] to test our front-end and back-end code.

- **SQL Server Management Studio 21:**

We have used SQL Server Management Studio 21[46] to test our database.

- **TestLink:**

In order to test our system test cases, we used the TestLink tool[47].

7.3.6 Schedule and Estimation

A testing schedule and estimation[48] is to determine how much time, money, and resources will be required to execute tests in a given environment. We divide the project into smaller tasks for test estimating and scheduling, allocating the time and effort needed for each. Table 7.2 show the schedule and estimation of our project. It contain serial no., module name, tester name, start and end date, and time required to test each module.

Table 7.2: Schedule and Estimation

Sr. No.	Module	Tester Name	Start Date	End Date	Time Required
01	Sign Up	Rafay, Aman Rutaba	March 30, 2026	March 30, 2026	2 Hour
02	Login	Rafay, Aman Rutaba	March 30, 2026	March 30, 2026	2 Hour
03	Admin Portal	Rafay, Aman Rutaba	March 31, 2026	March 31, 2026	2 Hour
04	Doctor Portal	Rafay, Aman Rutaba	March 31, 2026	March 31, 2026	2 Hour
05	Patient Portal	Rafay, Aman Rutaba	April 1, 2026	April 1, 2026	2 Hour
06	AI Model	Rafay, Aman Rutaba	April 1, 2026	April 1, 2026	3 Hour
07	Report Generation	Rafay, Aman Rutaba	April 2, 2026	April 2, 2026	2 Hour
08	Audit Log	Rafay, Aman Rutaba	April 3, 2026	April 3, 2026	2 Days
09	Notification	Rafay, Aman Rutaba	April 5, 2026	April 5, 2026	1 Hour

The tables shows the modules being worked upon of sign up, login, admin portal, doctor and patient portal, ai model, report generation, audit logs and notifications. Along with tester names, starting and ending dates with the time required.

7.4 Test Cases

The steps necessary to validate a particular feature or functionality in software testing are referred to as test cases[49]. The test case outlines the procedures, information, requirements, and post conditions needed to validate a feature. Test cases are essential for ensuring that each module of the system functions as intended under different conditions. They help in identifying defects early in the development life cycle and improve overall software quality. Each test case typically includes a unique identifier, input data, expected output, and actual results for comparison.

7.4.1 Admin Login

The login module is responsible for authenticating users who wish to gain access to the system. As seen from Table 7.3 below, the test cases will be used to test the reliability and functionality of the login module when presented with various inputs.

The standard operations carried out during testing are as follows: (1) Opening the Admin Login screen, (2) Inputting the username, (3) Inputting the password, and (4) Pressing the Login button.

Table 7.3: Admin Login Test Cases

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
01	Login with valid details	admin-demo admin@123	Login successful	Login successful	Pass
02	Login with incorrect username	admin- DEMO admin@123	Login failed due to incorrect username	Login failed	Pass
03	Login with incorrect password	admin-demo demo@11123	Login failed due to incorrect password	Login failed	Pass
04	Login with both incorrect fields	demoooooooo admin@8347p	Login failed	Login failed	Pass
05	Login with empty fields	empty fields	Login failed	Login failed	Pass

These include total of 5 scenarios of login with valid, invalid, correct and incorrect credentials and with empty fields. Each test case evaluates whether the system correctly authenticates users, displays appropriate error messages for invalid attempts, and maintains security by preventing unauthorized access.

7.4.2 Doctor Sign Up

The doctor sign up component facilitates the creation of an account through the provision of the appropriate information. Test cases have been created to ensure that the sign up process works well with regard to validation and functionality. Table 7.4 shows test cases with successful signing up using valid data and dealing with the situation of duplicate or erroneous data.

For each of the following test cases, the following actions are carried out: (1) Navigate to the Sign Up page, (2) Input the username, (3) Input password, (4) Input full name, (5) Input contact number, (6) Input degree, (7) Input years of experience, (8) Input gender, (9) Input email address, (10) Click on Sign Up button, and (11) Input OTP.

Table 7.4: Doctor Sign up Test Cases

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
01	Register with all correct data	Aman_123 aman12345 Aman Saad 03345099903 MBBS 03 years Female aman16sid@gmail.com entered and then waits for approval from admin	Registration Successful	Registration Successful	Pass
02	Register with already used username	Aman_123 alii3214f Ali Syed 03045136403 MBBS 07 years Male alisyed@gmail.com	Username already exists	Already in use username error	Pass

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
03	Register with already used email	rutaba_123 rutaba123 Rutaba Sajid 03140326478 MBBS 09 years Female aman16sid@gmail.com	Email already in use	Registration successful	Fail
04	Register without entering full name	Rafay_123 rafay1234 03125869745 MBBS 19 years Male mirajdin23@gmail.com	Registration fails due to missing full name	Full name is required	Pass
05	Register with invalid contact #	Rafay_123 rafay1234 Rafay Miraj 0314565865345 MBBS 11 years Male raf16miraj@gmail.com	Invalid contact number	Contact number must start with 03 and contain 11 digits	Pass
06	Register without entering degree	Fayyaz_123 fayyaz1234 Fayyaz Mir 03084441515 01 years Male	Registration fails due to missing degree	Degree is required	Pass

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
		fayyaz123@gmail.com			
07	Register without selecting gender	Ayesha_123 ayeshaa2345 Ayesha Aley 031371296555 MBBS 03 years ayeshaaley@gmail.com	Registration fails due to missing gender	Gender is required	Pass
08	Register without entering email	Noor_123 noorsaad345 Noor Saad 03455559455 MBBS 06 years Female	Registration fails due to missing email	Email is required	Pass
09	Register without entering any details	empty fields	Registration fails due to missing details	All details are required to signup	Pass

The following table lists out test cases for the functionality of Doctor Sign Up. There are 9 test cases in total. 8 test cases passed and most of them returned correct error messages when they were using incorrect or missing information to sign up as a Doctor. Registering with correct data, then only incorrect username, email, full name, contact, degree, gender, email and then empty fields. However, only the Test Case 03; Duplicate email failed because multiple doctors can sign up using the same email.

7.4.3 Doctor Login

The doctor login module ensures that only registered doctors can access their accounts securely, by entering login credentials and then verifying through an otp which has a five minutes timer. After the timer expired the doctor has to re-generate the otp for verification again. The test cases in table 7.5 shows that the test cases are designed to verify the functionality and reliability of the login feature under different conditions.

Doctor login has proper test steps that are followed through out; first open the doctors' login page then enter valid credentials, press login button and page will be redirected to otp page, otp will be mailed to the doctor and then doctor will use that otp to enter in the login dashboard.

The following standard steps are performed in each test case: (1) Open the Doctor Login page, (2) Enter username, (3) Enter password, (4) Press the Login button and (5) Enter otp. These steps remain consistent across all scenarios to evaluate how the system behaves with different input conditions.

Table 7.5: Doctor Login Test Cases

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
01	Login with correct details	Rafay_23 rafay12345 entered and then logs in	Login successful	Login successful	Pass
02	Login with incorrect username	rafayyyy23 rafay12345	Login failed due to incorrect username	Login failed	Pass
03	Login with incorrect password	Rafay_23 ali28746	Login failed due to incorrect password	Login failed	Pass
04	Login with both wrong fields	raf_8347 hf832fuh8	Login failed	Login failed	Pass
05	Login with empty fields	empty fields	Login failed	Login failed	Pass

These cover scenarios such as logging in with correct credentials, incorrect username,

incorrect password, and empty fields. Each case verifies whether the system properly authenticates doctors, displays appropriate error messages for invalid attempts, and maintains secure access or not.

7.4.4 Doctor Dashboard

Doctor Dashboard helps the doctor effectively monitor the management of patients through interactive elements. These functions include checking, adding checkups, viewing patients' information, adding medical imaging, AI analysis request, and report generation, among others, as shown in Table 7.6 below.

Table 7.6: Doctor Dashboard Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass or Fail
01	Add Checkup	1. Click "Add Checkup" button 2. Enter patient symptoms 3. Enter doctor findings 4. Click "Save"	Symptoms: Fever, cough Findings: Mild infection	Checkup added successfully	Added and saved in patient details	Pass
02	View Patient Details	1. Click "View Details" button 2. Check personal info and history	Patient: Rutaba Sajid	Details displayed correctly	Information properly displayed	Pass
03	Upload Image	1. Click "Upload Image" button 2. Choose image file 3. Enter description 4. Click "Upload"	File: img567.png Description: Suspected tumor	Image uploaded successfully	Saved in patient details	Pass
04	Request AI Analysis	1. Click "Request AI Analysis" 2. Select checkup record 3. Select analysis type 4. Click "Analyze"	Select checkup + analysis type	AI generates report	Report generated successfully	Pass
05	Generate Report	1. Click "Generate Report" 2. Select checkup record	Patient: Rutaba Date: 28-03-2026	PDF generated with XAI	Report ready and downloadable	Pass

Every test case ensures that the system does what is expected to be done right, the information about patients is stored and retrieved in a proper way, there is no problem with using AI to analyze data, and everything is reflected in the report generated.

7.4.5 Patient Sign Up

The Patient Sign up module lets the user register an account by giving the necessary personal information. From Table 7.7 it can be noticed that the features of the module involve the successful registration with valid information, dealing with duplicate usernames and email IDs and missing input validation like full name, contact number, gender and email ID.

Another feature that is tested within the system is OTP verification which makes sure that the account creation process is safe and reliable. Based on the test results we can notice that the functionality is performing well and enforcing validation requirements and showing correct response messages.

The following standard steps are performed in each test case: (1) Open the Sign Up page, (2) Enter username, (3) Enter password, (4) Enter full name, (5) Enter contact number, (6) Enter email, (7) Enter dob(optional), (8) Select gender, (9) Enter address,(10) Add blood group (optional), (11) Select doctor, (12) Press the Sign Up button, and (13) Enter OTP for verification and wait for admin approval. These steps remain consistent across all test cases to evaluate system behavior under different input conditions.

Table 7.7: Patient Sign up Test Cases

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
01	Register with all correct data	Aman_123 aman12345 Aman Saad 03345099903 aman16sid@gmail.com 01/01/2001 Female Pakistan Town B+ Aman Saad entered and then waits for approval from admin	Registration successful	Registration successful	Pass

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
02	Register with already used user-name	aliey_123 aliey12345 Syed Ali 03341122203 alisyed@gmail.com 07/06/2004 Male Bahria Town AB+	Username already exists	Already in use username error. Please choose another	Pass
03	Register with already used email	rutaba_123 rutaba123 Rutaba Sajid 03140326478 aman16sid@gmail.com 04/16/2003 Female Ghourri Town AB-	Email already in use	Registration successful	Fail
04	Register without entering full name	Rafay_123 rafay1234 (empty) 03125869745 mirajdin23@gmail.com 10/06/2003 Male Fauji Colony O-	Registration fails due to missing full name	Full name is required	Pass
05	Register with invalid contact #	Rafay_123 rafay1234 Rafay Miraj 0314565865345 raf16miraj@gmail.com 10/06/2003 Male	Invalid contact number	Contact number must start with 03 and contain 11 digits	Pass

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
06	Register without entering email	Fayyaz_123 fayyaz1234 Fayyaz Mir 03084441515 (empty) 10/06/2003 Male Westridge A-	Registration fails due to missing email	Email is required	Pass
07	Register without selecting gender	Ayesha_123 ayeshaa2345 Ayesha Aley 03137129655 ayeshaaleey@gmail.com (not selected) Soan Garden A+	Registration fails due to missing gender	Gender is required	Pass
08	Register without entering any details	empty fields	Registration fails due to missing details	All non-optional details are required to signup	Pass

This table outlines eight test cases for the Patient Sign Up module. It includes a positive test case and negative test cases such as duplicate username and email, missing full name, invalid contact number, missing email, missing gender, and registering with no details. However, Test Case 03 failed because the system allowed successful registration instead of showing an error.

7.4.6 Patient Login

The patient login module allows registered patients to securely access their accounts. The test cases in Table 7.8 also cover multiple scenarios. Each case ensures that the system correctly authenticates patients, shows correct error messages for invalid attempts, and prevents unauthorized access. These tests confirm that the login functionality is reliable and behaves as expected in all situations.

These tests cover the various possibilities for logging in to the portal, such as logging in successfully using proper credentials and failures due to improper or incorrect user name and password combination. There are also other possibilities included, such as

entering incorrect credentials altogether and emptying fields.

Moreover, the patient portal has been tested against repetitive attempts to log in using the wrong credentials so that the security system may be enabled upon identifying any suspicious behavior on the part of the user. The portal has mechanisms in place that prevent any further action from being taken once security measures have been initiated. Moreover, this also adds security to the system. Also, it ensures security for the process. One such process is session management, which permits the user to stay active within a certain period of time.

In every test case, the following procedures will be followed: (1) Open the Patient Login page, (2) Provide username, (3) Provide password, and (4) Click the Login button. This is consistent among all test cases to check various possibilities.

Table 7.8: Patient Login Test Cases

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Pass or Fail
01	Login with valid details	rutaba_23 03345996687	Login successful	Login successful	Pass
02	Login with incorrect username	rubl8430 03345996687	Login failed due to incorrect username	Login failed	Pass
03	Login with incorrect password	rutaba_23 fvcn894099	Login failed due to incorrect password	Login failed	Pass
04	Login with both invalid details	ruTRSX_23 rutab5834	Login failed	Login failed	Pass
05	Login with empty fields	empty fields	Login failed	Login failed	Pass

The system correctly grants access when valid details are provided, ensuring that only registered users can enter the application. This confirms that the authentication mechanism is properly integrated and functioning as intended. In addition, the system effectively handles negative test scenarios such as invalid credentials and empty input

fields.

7.4.7 Patient Dashboard

With the patient dashboard module, the patients can be able to handle their medical data conveniently. This functionality is tested using the test cases in table 7.9 such as viewing the dashboard, downloading medical data, and reviewing the downloaded files.

Table 7.9: Patient Dashboard Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass or Fail
01	Login with incorrect username	1. Open Patient dashboard 2. Press download report button 3. Preview the downloaded report	MedicalReport_Aman_Asfar_9052202603	Report downloaded successfully	Report downloaded successfully	Pass

In each of these cases, there is guarantee that the system provides proper access to patients' information, download of reports is easily possible, and that information presentation is reliable. All these tests show that patient dashboard application works as expected.

7.4.7.1 Test Method

We use unit testing method to test our system. The goal of unit testing is to isolate each component of the software and test each one separately to show that it is right in terms of meeting the requirements and functioning as intended[50].

7.4.7.2 Test Case Description

The test suite mentioned above has sufficient details regarding the testing of each functionality. Every functionality is put to the test in three different ways: with all the right inputs, with the wrong ones, and even with some of them missing. The test case's status indicates whether the system was able to carry out the needed tasks correctly or not. The serial number indicates the order in which the test cases were run. The tables above identify every prerequisite functionality that must be completed or run before testing any capability, and they also include the date on which that functionality is tested, making it simple to track changes. If the actual output of the system for any feature matches the needed output exactly, only then it is considered to be pass; otherwise, if the actual output for any functionality does not match the required output, the system is considered to be failed[51].

7.4.7.3 Test Case Data

Test data[52] is information needed to carry out test cases correctly and confirm the desired results in any software program that is being tested. While executing test cases, we need some data to input in order to get the expected output. A Test Data is an input

that is required to run our software application to complete the software test execution. Data required for every test case differs for example the data required to perform the testing process on login functionality is only the email and password. For Registration it will be data about user personal information.

7.4.7.4 Test Case Report

The system successfully completed all test cases that were run on it and are listed above in the test case table. The system is acting exactly as stated in the system requirement specification document and does not deviate from the specified behavior. After careful examination and system monitoring, it is concluded that all the system's components are functioning properly after doing unit, integration, and system testing[53].

7.5 Bugs Report

A bug report[54] is a record that contains all the information required to identify, track down, and resolve issues that have arisen with software or websites. Device logs, stack traces, and other diagnostic data are included in a bug report to assist you in identifying and resolving errors in your website. In a bug report, the number of errors and their solutions are discussed. Table 7.10 shows the bug report of our system. it contain Bug ID, Module name, Bug Detail, Reporter name, expected and actual result along with its severity.

Table 7.10: Bug Report

Bug ID	Module Name	Bug Detail	Reporter Name	OS	Browser	Expected Result	Actual Result	Severity
BG1	Doctor Sign Up	Register with already used email	Aman Siddiqui	Windows 11	Chrome	Registration Failed	Registration successful	High
BG1	Patient Sign Up	Register with already used email	Aman Siddiqui	Windows 11	Chrome	Registration Failed	Registration successful	High

The above table shows the bug that was found during testing of the system. The details include Bug ID, affected module, bug description, reporter details, Operating System, Browser, Expected Result, Actual Result, and Severity. The purpose of reporting this type of bug is to help document the functional problems and make sure that the system is reliable, through analysis and tracking of bugs in the various modules.

According to the above table, the system had bugs in the doctor sign-up and patient sign-up modules, which allowed for users to register their details even though there were existing accounts with the same email addresses. This bug was supposed to reject any registration of duplicate email addresses, but instead allowed for a successful sign-up process in both modules. The above bugs have a severity level of 'High' since they impact on user registration and hence user authentication.

Chapter 8

Conclusion

This project set out to address a critical challenge in modern oncology: the need for fast, accurate, privacy-preserving, and clinically explainable analysis of histopathology tissue slides. The resulting system, HistoScanAI, is a full-stack web platform built on ASP.NET Core MVC (.NET 8) and Microsoft SQL Server that brings together deep learning inference, Explainable AI, and Federated Learning under a single, role-governed clinical workflow.

Summary of Achievements

- **End-to-End Clinical Workflow:** A three-tier role architecture (Admin, Doctor, Patient) with dedicated dashboards, server-side session management, and a tamper-evident audit log recording every sensitive operation with IP address, timestamp, and severity level.
- **AI-Powered Classification:** An optimized EfficientNet-B0 CNN classifies the histo-pathology slide as Cancerous or Non-Cancerous using the output from the Softmax layer. A decoupled inference module coded in Python runs on both CPU and GPU platforms.
- **Explainable AI via Grad-CAM:** The Gradient-weighted Class Activation Mapping is used for explaining the predictions made by EfficientNet-B0 for any test data point in real time. There is analysis of the activation intensities for nine regions of the last convolutional block in the model, and the result is captured into a simple English language statement that is stored along with each prediction.
- **Privacy-Preserving Federated Learning:** The FedAvg algorithm allows each hospital to train locally; only encrypted weight updates are transmitted to the central server. No raw patient image ever leaves the hospital's premises, satisfying HIPAA and GDPR requirements.
- **Supporting Infrastructure:** OTP-verified patient registration (SMTP/TLS), automated email notifications, downloadable self-contained HTML reports with embedded Grad-CAM, an Analytics Dashboard (Chart.js), a structured Feedback Module, server-side XSS sanitization, and brute-force login detection.

Overall Assessment

The project titled HistoScanAI shows how an explainable and privacy-preserving clinical decision support system powered by AI can be developed for real-world use, even while adhering to the limited scope of a final-year academic project. This project utilizes an integrated AI system incorporating EfficientNet-B0 model, Grad-CAM, and Federated Learning which tackle the core challenges of accuracy, explainability, and privacy in healthcare, respectively.

8.1 Future Work

Future scope of the CDHI framework is concerned with making the system more accurate, scalable, secure, and clinically usable. Future improvements planned for the system include extending the current capabilities of the framework with advanced methods of artificial intelligence, more efficient federated learning methodologies, and practical application within the domain of health care for automating the process of cancer detection.

- **Multi-Class Classification:** From binary classification towards detection of different types of cancers such as Invasive Ductal Carcinoma or Adenocarcinoma using EfficientNet-B0 with N -class output layer.
- **Advanced Federated Learning:** Use FedProx (heterogeneous datasets), Differential Privacy (DP-SGD), Secure Aggregation through homomorphic encryption, Personalized FL, and Asynchronous FL when dealing with hospitals with varying capacities.
- **Whole-Slide Image (WSI) Support:** Use OpenSlide to extract patches and generate a spatial heatmap at slide level for processing slides up to gigabytes with magnification of $40\times$.
- **Visual Grad-CAM Overlay:** Use spatial heatmap (colored blue $\rightarrow$ red) instead of textual explanations on the image of histopathology slide itself.
- **Mobile App & HIS Integration:** Build an iOS/Android front-end (.NET MAUI or React Native) and integrate with Hospital Information Systems via HL7 FHIR to automate patient demographics and record synchronization.
- **Continuous Learning & Security:** Implement an automated retraining pipeline on pathologist-confirmed cases, enforce HTTPS/HSTS, apply SQL Server TDE column-level encryption, and add a formal GDPR Right-to-Erasure workflow.
- **Clinical Validation & LLM Reporting:** Benchmark EfficientNet-B0 against ResNet-50 and DenseNet-121 on PatchCamelyon datasets, conduct a prospective clinical trial, and integrate an LLM (e.g., GPT-4) to generate context-aware clinical narrative reports.

The framework of HistoScanAI has laid down a robust platform for the use of artificial intelligence in cancer diagnostics. This is because each future direction will develop on the current framework and database. As such, it can be observed that the platform was created in such a way that its future scalability was anticipated. If further developed, the platform may become instrumental in the fight against cancer.

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